1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	APPD DATE
8	0006400877	ENGINEERING RELEASED	2016-06-14

D10 MLB - DVT

LAST_MODIFICATION=Tue Jun 14 15:20:28 2016

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8	<CSA_PAGE8>	SOC:PCIE	<SYNC_MASTER8>	<SYNC_DATE8>
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19	<CSA_PAGE19>	SYSTEM POWER:PMU (2/3)	<SYNC_MASTER19>	<SYNC_DATE19>
20	<CSA_PAGE20>	SYSTEM POWER:PMU (3/3)	<SYNC_MASTER20>	<SYNC_DATE20>
21	<CSA_PAGE21>	SYSTEM POWER:CHARGER	<SYNC_MASTER21>	<SYNC_DATE21>
22	<CSA_PAGE22>	SYSTEM POWER:BATTERY CONN	<SYNC_MASTER22>	<SYNC_DATE22>
23	<CSA_PAGE23>	SYSTEM POWER:BOOST	<SYNC_MASTER23>	<SYNC_DATE23>
24	<CSA_PAGE24>	SENSORS	<SYNC_MASTER24>	<SYNC_DATE24>
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26	<CSA_PAGE26>	CAMERA:STROBE DRIVER	<SYNC_MASTER26>	<SYNC_DATE26>
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Schematic & PCB Callouts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-00419	1	SCH,MLB,D10	SCH	CRITICAL	?
820-00188	1	PCBF,MLB,D10	PCB	CRITICAL	?

System Block Diagram:

<rdar://problem/16684269>

SCH 051-00419

BRD 820-00188

MCO 056-01342

NAND BOM Options

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00169	1	NAND,H,32GB,16nm,MLC	U1701	CRITICAL	NAND_32G
335S00182	1	NAND,H,128GB,16nm,TLC	U1701	CRITICAL	NAND_128G
335S00156	1	NAND,H,256GB,3Dv3,TLC	U1701	CRITICAL	NAND_256G
138S0867	5	CAP,X5R,10UF,20%,6.3V,0.65MM,HRZTL,0402	C1748,C1713,C1716,C1721,C1733	CRITICAL	NAND_32G
138S00003	5	CAP,X5R,15UF,20%,,6.3V,0.65MM,HRZTL,0402	C1748,C1713,C1716,C1721,C1733	CRITICAL	NAND_128G
138S00003	5	CAP,X5R,15UF,20%,,6.3V,0.65MM,HRZTL,0402	C1748,C1713,C1716,C1721,C1733	CRITICAL	NAND_256G

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00201	335S00169	ALTERNATE	U1701	T,15nm,MLC,32GB
335S00209	335S00169	ALTERNATE	U1701	S,16nm,MLC,32GB
335S00195	335S00182	ALTERNATE	U1701	SS,1Ynm,TLC,128GB
335S00180	335S00182	ALTERNATE	U1701	T,15nm,TLC,128GB
335S00179	335S00182	ALTERNATE	U1701	SD,15nm,TLC,128GB
335S00148	335S00183	ALTERNATE	U1701	SD,3Dv2,TLC,256GB
335S00190	335S00183	ALTERNATE	U1701	SS,3Dv3,TLC,256GB

#22686038:See Radar

Magnesium Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00173	338S00203	ALTERNATE	U2402	Larger Wafer (-29 flow) Magnesium

Carbon Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
338S00087	338S00226	ALTERNATE	U2401,U2404	Updated version of Carbon

UT LDO Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00889	353S00015	ALTERNATE	U2501	ST, LDO REG, 2.925V, CSP 0.65x0.65

Mamba LDO Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S00932	353S00576	ALTERNATE	U3801	ST, LDO REG, 2.75V

I2C5 Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00234	335S00233	ALTERNATE	U1101	I2C5 ALTERNATE

Active Diode Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S00106	376S00047	ALTERNATE	Q2101	DIODES INC. ACT DIODE

DDR PLL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00095	155S00068	ALTERNATE	FL1501	FERR BD,100OHM,25%,100MA,2OHM,01005

Power Inductor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
152S00118	152S00075	ALTERNATE	ALL	IND,PWR,SHLD,1.2 UH,3.0A,0.080 OHM,2016	
152S00077	152S00397	ALTERNATE	ALL	IND,PWR,SHLD,1.0 UH,2.25A,0.150 OHM,2016	updated 11/12
152S00121	152S00081	ALTERNATE	ALL	IND,PWR,SHLD,0.47 UH,3.8A,0.048 OHM,2012	
152S00123	152S1936	ALTERNATE	ALL	IND,PWR,SHLD,15 UH,0.72A,0.900 OHM,3225	
152S00402	152S00366	ALTERNATE	ALL	IND,MULT,1UH,1.2A,0.320 OHM,0603	updated 11/12
152S00297	152S1843	ALTERNATE	ALL	CYNTEC 2012 1UH	reverted 11/13
152S00365	152S00297	ALTERNATE	ALL	CYNTEC 2012 1UH	
152S00398	152S00204	ALTERNATE	ALL	IND,PWR,0.22UH,20%,6.7a,23MOHM,2012	For Chestnut inductor; so it doesn't interfere with PMU inducotr Buck 7 alts
152S00120	152S00077	ALTERNATE	ALL	For Chestnut inductor only	Except BUCK5 LX (BUCK5 LX is Taiyo only)
152S00117	152S00074	ALTERNATE	L1806,L1810,L1814,L1816,L1817	IND,PWR,SHLD,1.0 UH,3.0A,0.060 OHM,2016	

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
118S0764	118S0717	ALTERNATE	ALL	RES, 3.92K, 0.1%, 0201
138S0702	138S0657	ALTERNATE	ALL	CAP, X5R, 4.3UF, 4V, 0610
138S00006	138S0835	ALTERNATE	ALL	CAP, 3-TERM, 4.3UF, 4V, 0402
138S00005	138S00003	ALTERNATE	ALL	CAP,X5R,15UF,6.3V,0.65MM,0402,TAIYO
138S00048	138S00003	ALTERNATE	ALL	CAP,X5R,15UF,6.3V,0.65MM,0402,KYOCERA
138S0648	138S0652	ALTERNATE	ALL	CAP,X5R,4.7UF,6.3V,0.65MM,0402,TAIYO
132S0400	132S0436	ALTERNATE	ALL	CAP,X5R,0.22UF,6.3V,01005,TDK
138S00024	138S0986	ALTERNATE	ALL	CAP,CER,3-TERM,7.5UF,20%,4V,0402,TAIYO/TDK
138S0706	138S0739	ALTERNATE	ALL	CAP,CER,1UF,20%,10V,X5R,0201,MURATA
138S0945	138S0739	ALTERNATE	ALL	CAP,CER,1UF,20%,10V,X5R,0201,KYOCERA
132S0436	132S0400	ALTERNATE	ALL	CAP,CER,X5R,0.22UF,20%,6.3V,20%

Global Ferrite Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00067	155S0581	ALTERNATE	ALL	FERR, 240OHM, 0.38OHM DCR, 0201
155S0581	155S00067	ALTERNATE	ALL	FERR, 240OHM, 0.38OHM DCR, 0201
155S00012	155S00168	ALTERNATE	ALL	FLTR, 65 OHMS, 0605
155S00194	155S0610	ALTERNATE	ALL	FERR BD, 150OHM, TDK
155S00200	155S0610	ALTERNATE	ALL	FERR BD, 150OHM, TY
152S00489	152S00456	ALTERNATE	ALL	FERR BD, 0.47UH, TY

Global Varistor Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0168	377S0140	ALTERNATE	ALL	VARISTOR, 6.8V, 100PF, 01005

ACC BUCK CIRCUIT Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S00087	371S00064	ALTERNATE	D2700	DIODE,SHOTTKY,30V,200MA,0201
152S00558	152S00557	ALTERNATE	L2700	IND,MLD,0.47UH,2.5A,80Mohm,1608
376S00166	376S00164	ALTERNATE	Q2700,Q2701	PFET,12V,CSP4
353S01007	353S01039	ALTERNATE	U2710	IC,LOAD SWITCH,WLCSP4

D10 EEEE CALLOUTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-6838	1	EEEE CODE FOR 639-01754	EEEE_GXD5	CRITICAL	EEEE_BEST
825-6838	1	EEEE CODE FOR 639-01755	EEEE_GXD6	CRITICAL	EEEE_SUPREME
825-6838	1	EEEE CODE FOR 639-01756	EEEE_GXD7	CRITICAL	EEEE_EXTREME
825-6838	1	EEEE CODE FOR 639-02372	EEEE_H6TF	CRITICAL	EEEE_BEST_ROW
825-6838	1	EEEE CODE FOR 639-02373	EEEE_H6TG	CRITICAL	EEEE_SUPREME_ROW
825-6838	1	EEEE CODE FOR 639-02374	EEEE_H6TH	CRITICAL	EEEE_EXTREME_ROW

CAYMAN DDR Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00254	339S00253	ALTERNATE	ALL	DDR-H, 2G, B1
339S00255	339S00253	ALTERNATE	ALL	DDR-S, 2G, B1

Cap 2.2UF Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00049	138S00032	ALTERNATE	(C2507,C2531)	CAP,CER,X5R,2,2UF,20%6.3V,20%, KYOCERA
138S0831	138S00032	ALTERNATE	ALL	CAP,CER,X5R,2,2UF,20%,6.3V,20%,MURATA

#25634778: Exclude Kyocera as 2.2UF alt at only C2507/C2531 REFDES (other refdes no impact)

D10x Specific BOM Callouts

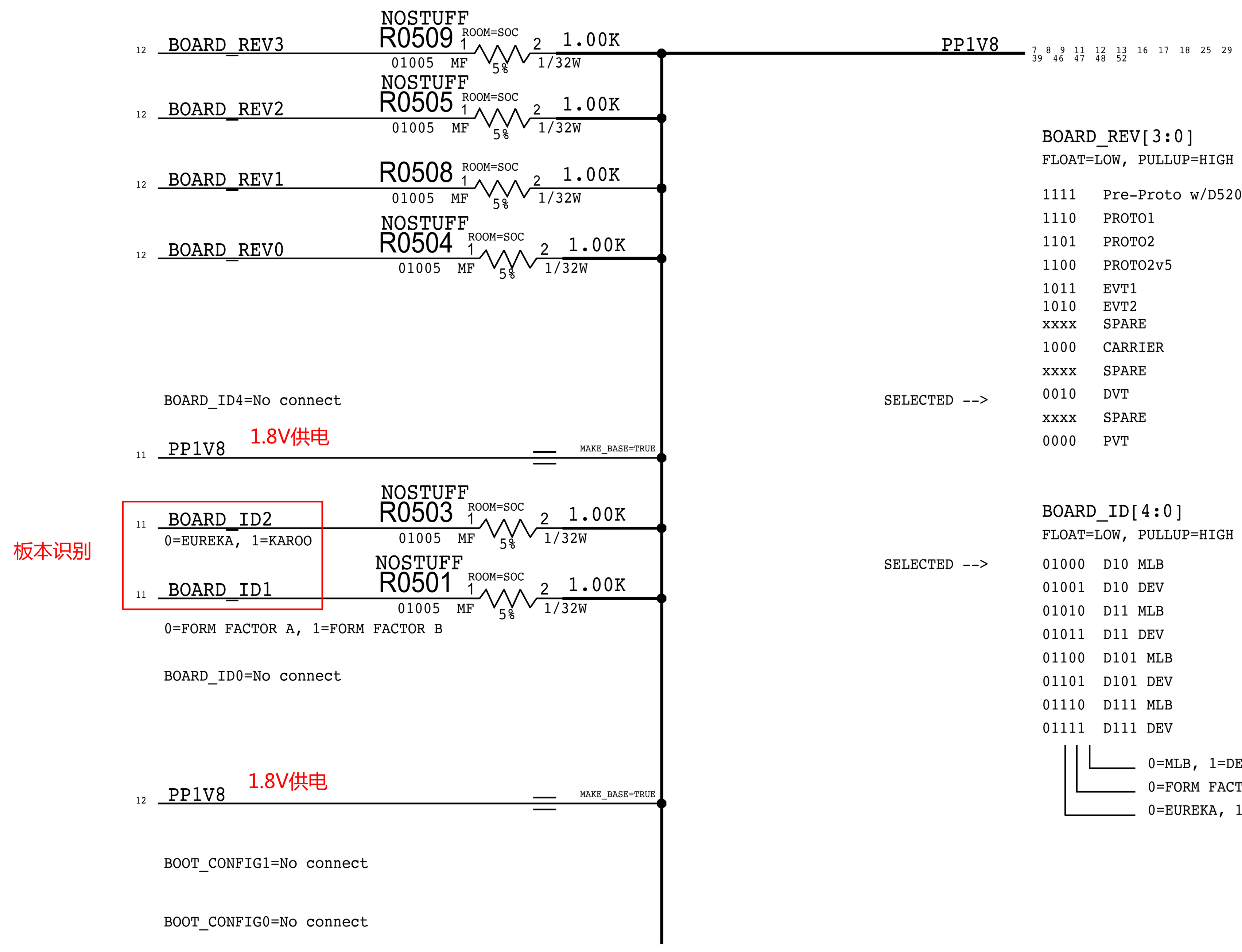
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00117	1	TAIYO,IND,PWR,SHLD,1UH,3.6A,0.060OHM,2016	L1803	CRITICAL	B5LX_TAIYO
117S0156	2	RES,MF,1K OHM, 5%, 1/32W, 01005	R4808,R4809	CRITICAL	UTAH_C

#24681501

#24629229

Current as of D10 MCO 056-01342-78

BOOTSTRAPPING:BOARD REV
BOARD ID
BOOT CONFIG



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	1110	PROTO1
	1101	PROTO2
	1100	PROTO2v5
	1011	EVT1
	1010	EVT2
	xxxx	SPARE
	1000	CARRIER
	xxxx	SPARE
SELECTED -->	0010	DVT
	xxxx	SPARE
	0000	PVT

BOARD_ID[4:0]
FLOAT=LOW, PULLUP=HIGH

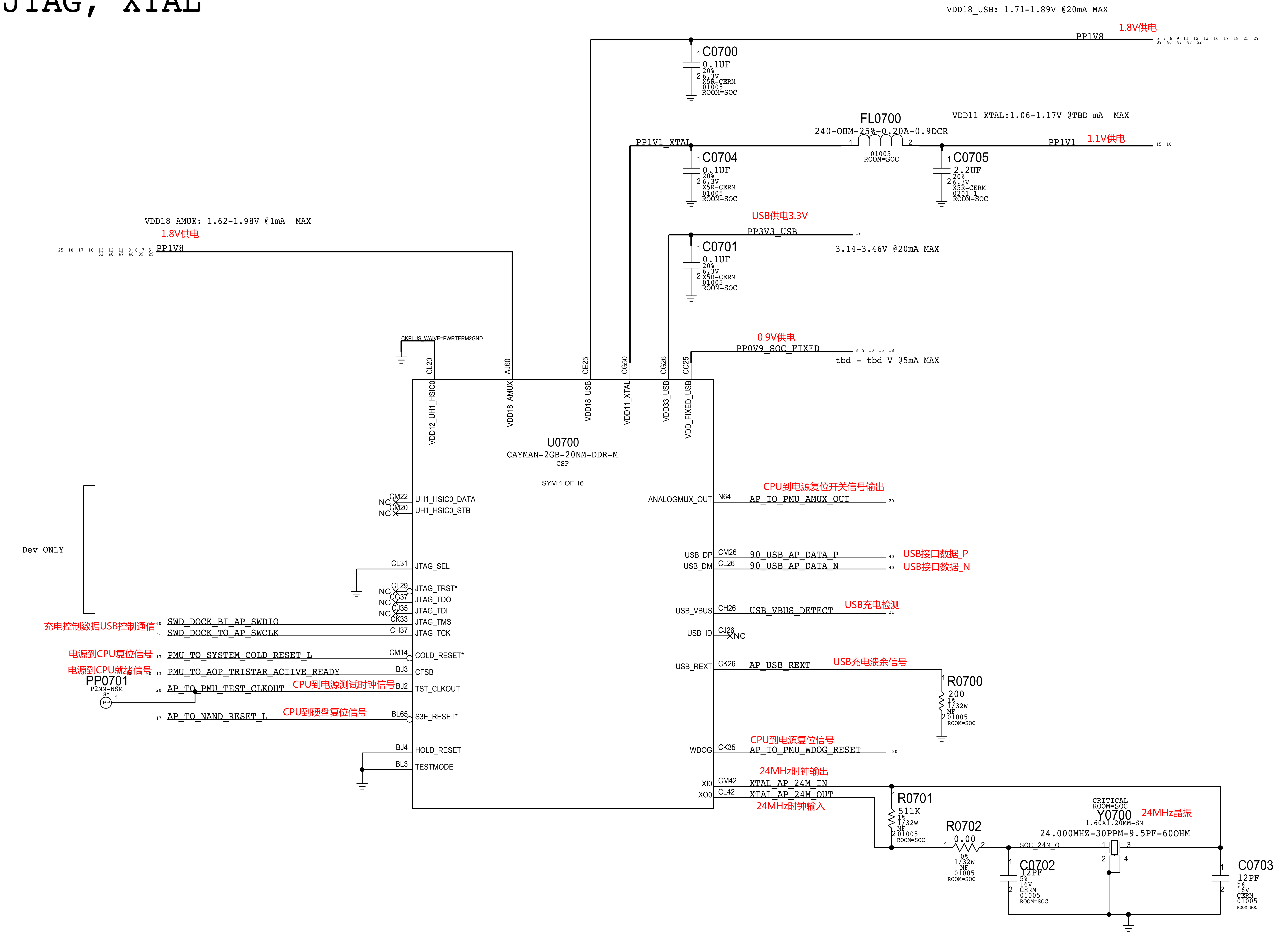
SELECTED -->	01000	D10 MLB
	01001	D10 DEV
	01010	D11 MLB
	01011	D11 DEV
	01100	D101 MLB
	01101	D101 DEV
	01110	D111 MLB
	01111	D111 DEV

0=MLB, 1=DEV
0=FORM FACTOR A, 1=FORM FACTOR B
0=EUREKA, 1=KAROO

BOOT_CONFIG[2:0]
FLOAT=LOW, PULLUP=HIGH

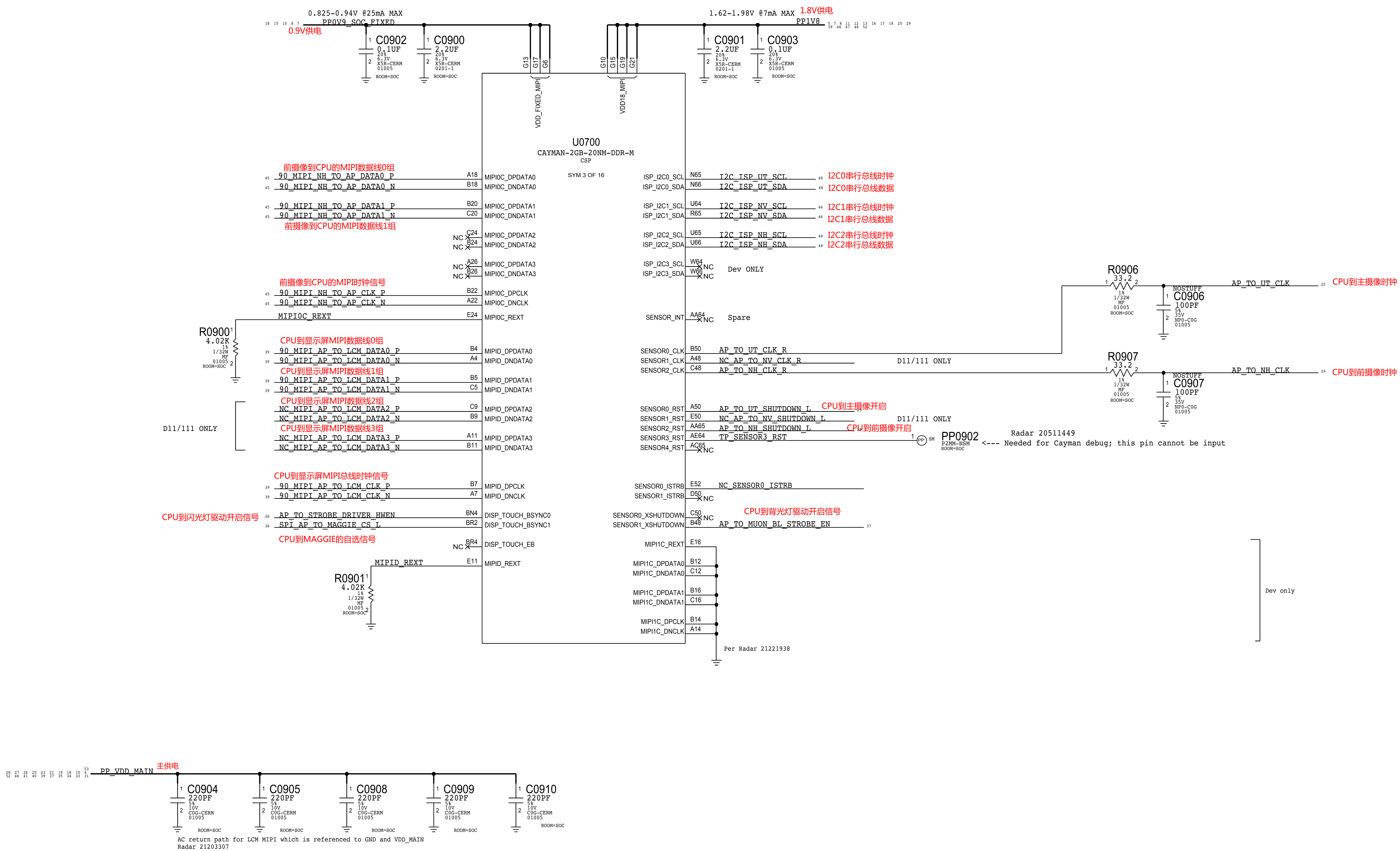
	000	SPI0
	001	SPI0 TEST MODE
	010	NVME0_X2
	011	NVME0 X2 TEST
SELECTED -->	100	NVME0 X1
	101	NVME0 X1 TEST
	110	SLOW SPI0 TEST
	111	FAST SPI0 TEST

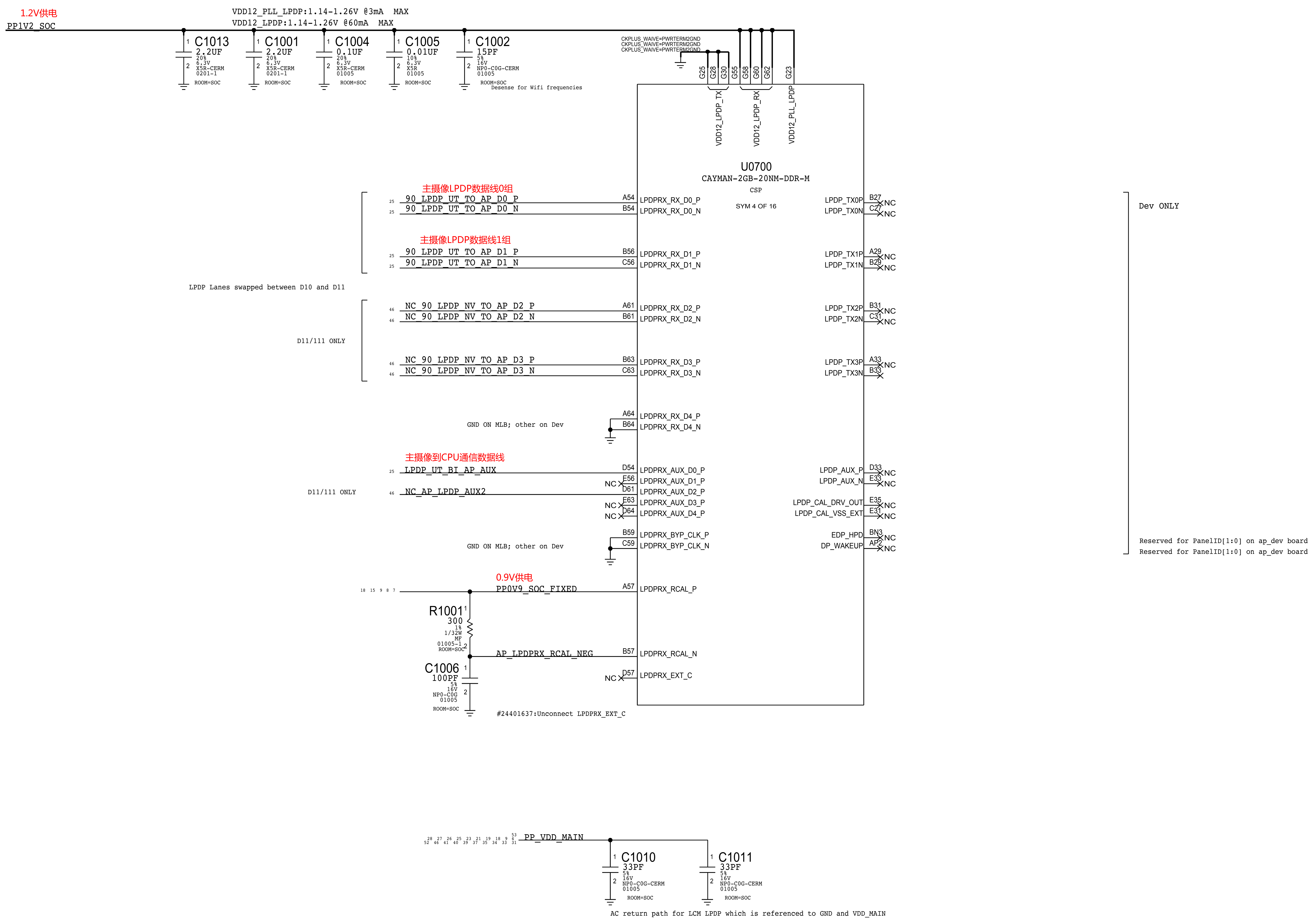
SOC - USB, JTAG, XTAL



SOC - PCIE INTERFACES

SOC - MIPI & ISP INTERFACES





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PP1V2_SOC
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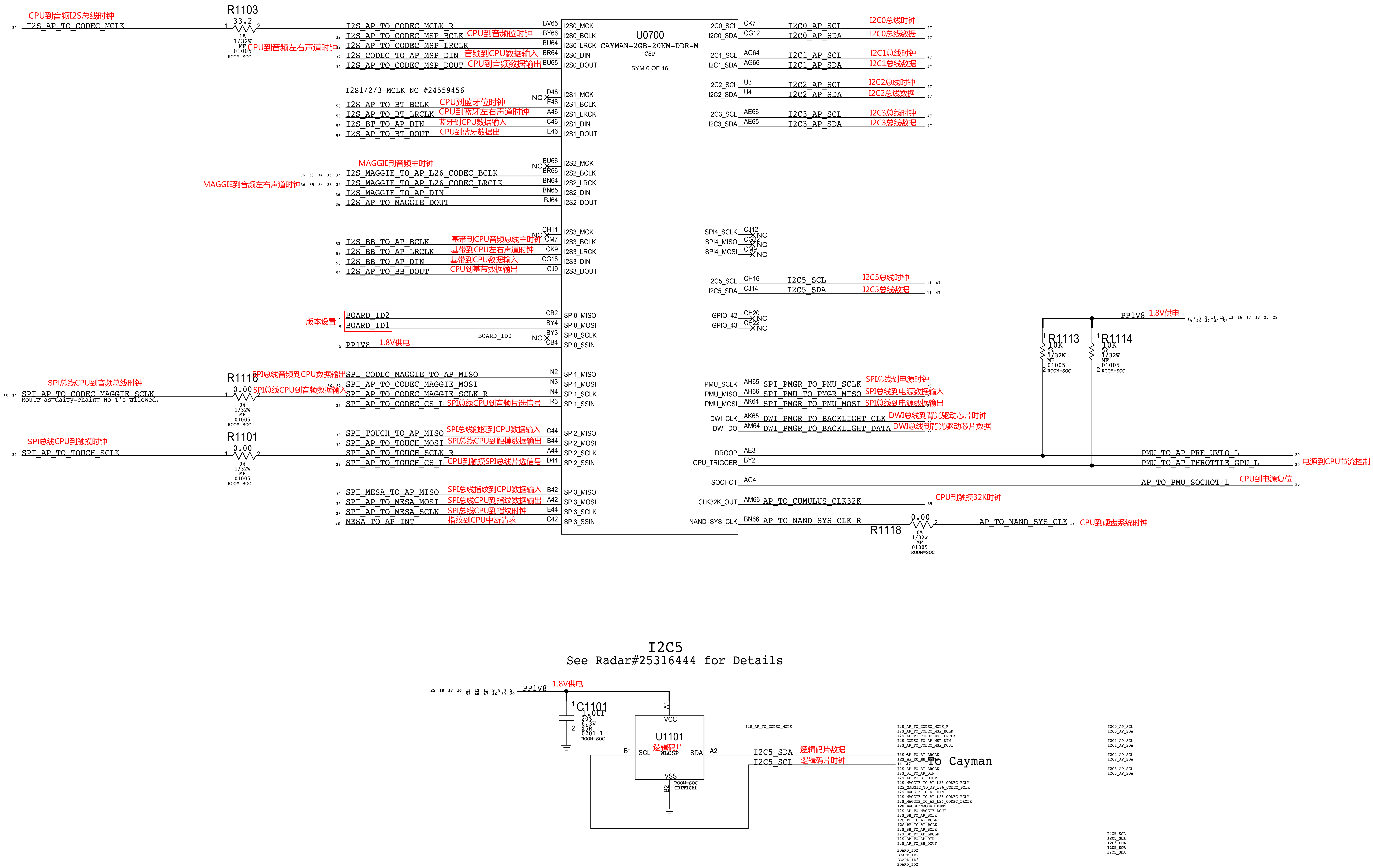
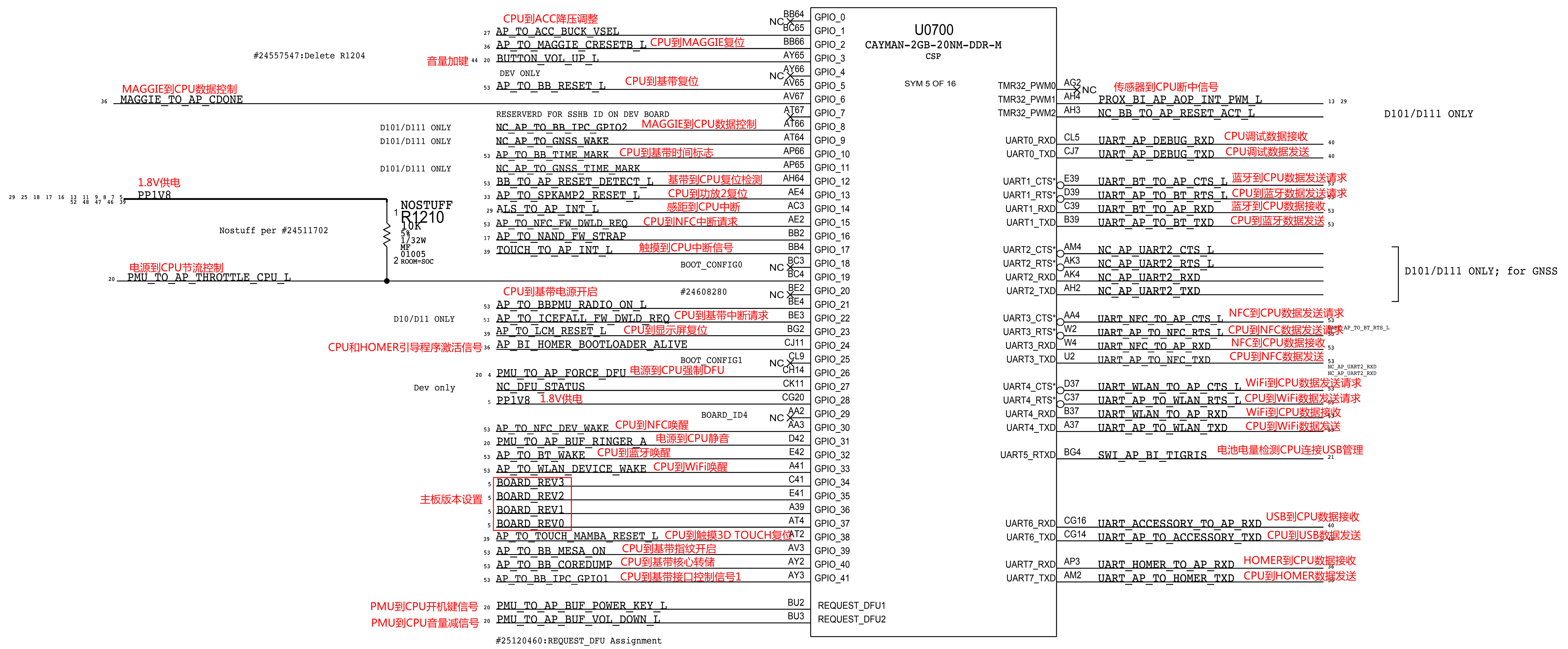
VDD12_LPDP:1.14-1.26V @60mA MAX
C1013 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=SOC
C1001 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=SOC
C1004 0.1UF 20% 6.3V X5R-CERM 01005 ROOM=SOC
C1005 0.01UF 10% 6.3V X5R 01005 ROOM=SOC
C1002 15PF 5% 16V NP0-C0G-CERM 01005 ROOM=SOC
Desense for Wifi frequencies
CKPLUS_WAIVE=PWRTERM2GND
CKPLUS_WAIVE=PWRTERM2GND
CKPLUS_WAIVE=PWRTERM2GND
G25 G28 G30 G55 G58 G60 G62 G23
VDD12_LPDP_TX
VDD12_LPDP_RX
VDD12_PLL_LPDP
U0700
CAYMAN-2GB-20NM-DDR-M
CSP
SYM 4 OF 16
主摄像LPDP数据线0组
90_LPDP_UT_TO_AP_D0_P
90_LPDP_UT_TO_AP_D0_N
主摄像LPDP数据线1组
90_LPDP_UT_TO_AP_D1_P
90_LPDP_UT_TO_AP_D1_N
LPDP Lanes swapped between D10 and D11
NC_90_LPDP_NV_TO_AP_D2_P
NC_90_LPDP_NV_TO_AP_D2_N
D11/111 ONLY
NC_90_LPDP_NV_TO_AP_D3_P
NC_90_LPDP_NV_TO_AP_D3_N
A54 LPDPRX_RX_D0_P
B54 LPDPRX_RX_D0_N
B56 LPDPRX_RX_D1_P
C56 LPDPRX_RX_D1_N
A61 LPDPRX_RX_D2_P
B61 LPDPRX_RX_D2_N
B63 LPDPRX_RX_D3_P
C63 LPDPRX_RX_D3_N
A64 LPDPRX_RX_D4_P
B64 LPDPRX_RX_D4_N
GND ON MLB; other on Dev
主摄像到CPU通信数据线
LPDP_UT_BI_AP_AUX
D11/111 ONLY
NC_AP_LPDP_AUX2
D54 LPDPRX_AUX_D0_P
E56 LPDPRX_AUX_D1_P
D61 LPDPRX_AUX_D2_P
E63 LPDPRX_AUX_D3_P
D64 LPDPRX_AUX_D4_P
B59 LPDPRX_BYP_CLK_P
C59 LPDPRX_BYP_CLK_N
GND ON MLB; other on Dev
0.9V供电
PP0V9_SOC_FIXED
A57 LPDPRX_RCAL_P
R1001 300 1% 1/32W MF 01005-1 ROOM=SOC
AP_LPDPRX_RCAL_NEG
B57 LPDPRX_RCAL_N
C1006 100PF 5% 16V NP0-C0G 01005 ROOM=SOC
D57 LPDPRX_EXT_C
#24401637:Unconnect LPDPRX_EXT_C
LPDP_TX0P B27 NC
LPDP_TX0N C27 NC
LPDP_TX1P A29 NC
LPDP_TX1N B29 NC
LPDP_TX2P B31 NC
LPDP_TX2N C31 NC
LPDP_TX3P A33 NC
LPDP_TX3N B33
LPDP_AUX_P D33 NC
LPDP_AUX_N E33 NC
LPDP_CAL_DRV_OUT E35 NC
LPDP_CAL_VSS_EXT E31 NC
EDP_HPD BN3 NC
DP_WAKEUP AP2 NC
Dev ONLY
Reserved for PanelID[1:0] on ap_dev board
Reserved for PanelID[1:0] on ap_dev board
PP_VDD_MAIN
C1010 33PF 5% 16V NP0-C0G-CERM 01005 ROOM=SOC
C1011 33PF 5% 16V NP0-C0G-CERM 01005 ROOM=SOC
AC return path for LCM LPDP which is referenced to GND and VDD_MAIN

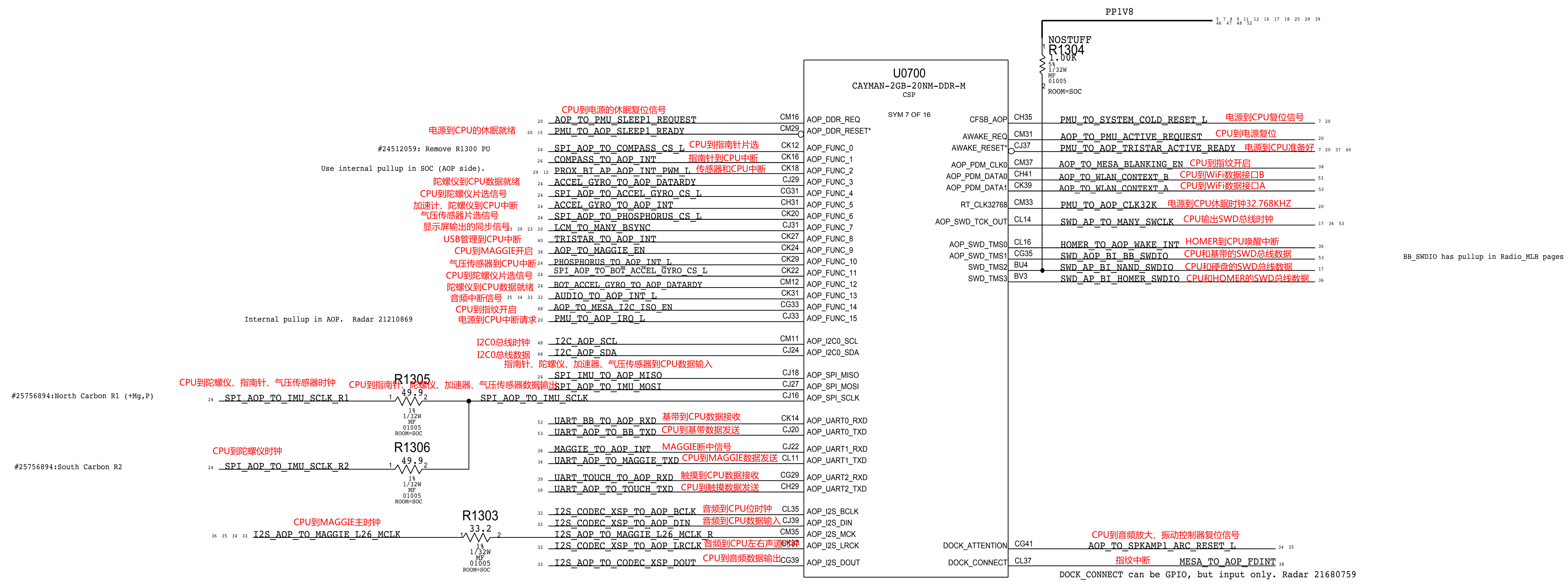
SOC - SERIAL INTERFACES



SOC - GPIO INTERFACES

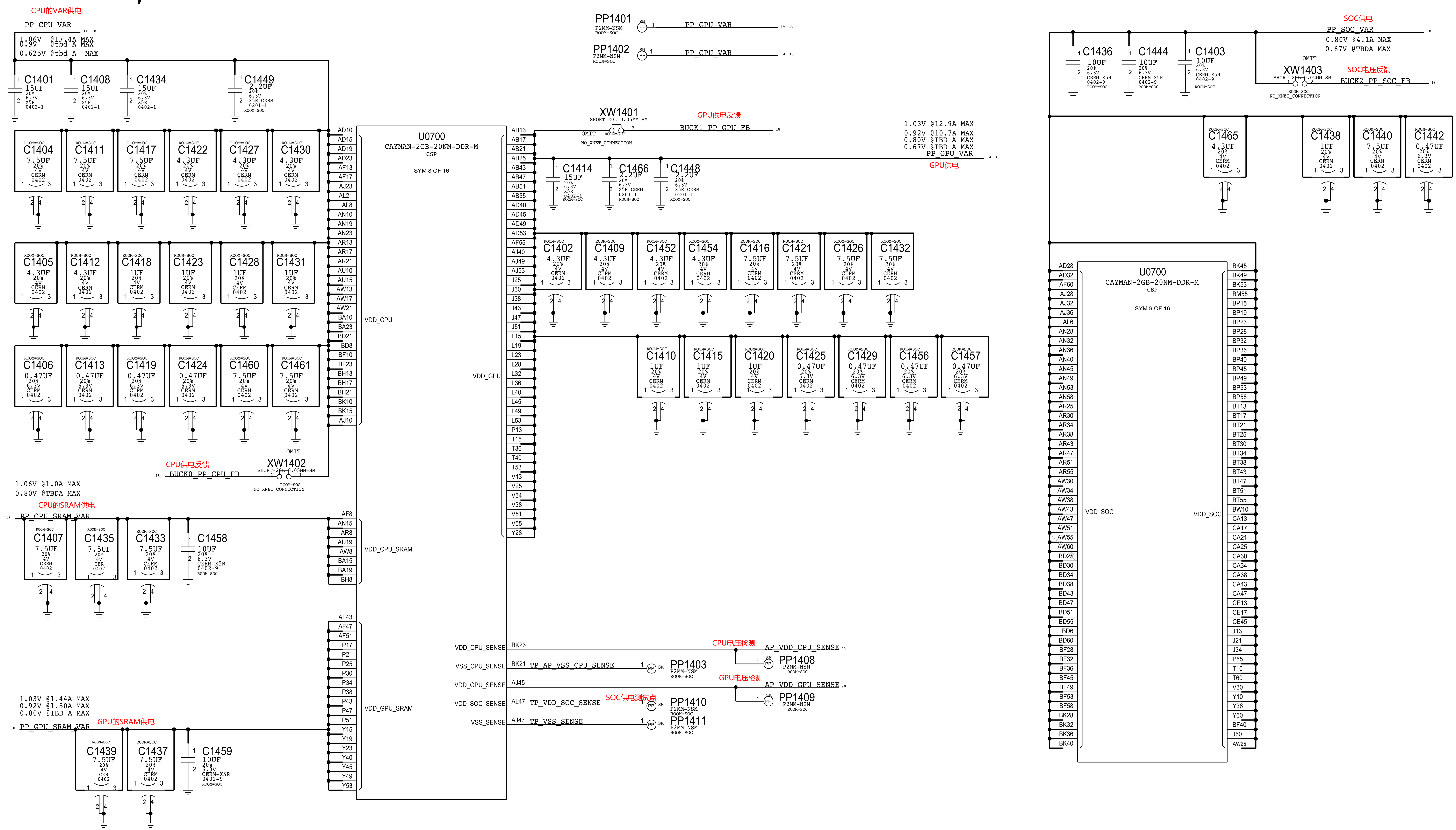


SOC - AOP



PP1V8
NOSTUFF
R1304
1.00K
5%
1/32W
MF
01005
ROOM=SOC
U0700
CAYMAN-2GB-20NM-DDR-M
CSP
SYM 7 OF 16
CPU到电源的休眠复位信号
AOP_TO_PMU_SLEEP1_REQUEST CM16 AOP_DDR_REQ
电源到CPU的休眠就绪
PMU_TO_AOP_SLEEP1_READY CM29 AOP_DDR_RESET*
#24512059: Remove R1300 PU
Use internal pullup in SOC (AOP side).
SPI_AOP_TO_COMPASS_CS_L CPU到指南针片选 CK12 AOP_FUNC_0
COMPASS_TO_AOP_INT 指南针到CPU中断 CK16 AOP_FUNC_1
PROX_BI_AP_AOP_INT_PWM_L 传感器和CPU中断 CK18 AOP_FUNC_2
陀螺仪到CPU数据就绪
ACCEL_GYRO_TO_AOP_DATARDY CJ29 AOP_FUNC_3
CPU到陀螺仪片选信号
SPI_AOP_TO_ACCEL_GYRO_CS_L CG31 AOP_FUNC_4
加速计、陀螺仪到CPU中断
ACCEL_GYRO_TO_AOP_INT CH31 AOP_FUNC_5
气压传感器片选信号
SPI_AOP_TO_PHOSPHORUS_CS_L CK20 AOP_FUNC_6
显示屏输出的同步信号
LCM_TO_MANY_BSYNC CJ31 AOP_FUNC_7
USB管理到CPU中断
TRISTAR_TO_AOP_INT CK27 AOP_FUNC_8
CPU到MAGGIE开启
AOP_TO_MAGGIE_EN CK24 AOP_FUNC_9
气压传感器到CPU中断
PHOSPHORUS_TO_AOP_INT_L CK29 AOP_FUNC_10
CPU到陀螺仪片选信号
SPI_AOP_TO_BOT_ACCEL_GYRO_CS_L CK22 AOP_FUNC_11
陀螺仪到CPU数据就绪
BOT_ACCEL_GYRO_TO_AOP_DATARDY CM12 AOP_FUNC_12
音频中断信号
AUDIO_TO_AOP_INT_L CK31 AOP_FUNC_13
CPU到指纹开启
AOP_TO_MESA_I2C_ISO_EN CG33 AOP_FUNC_14
Internal pullup in AOP. Radar 21210869
电源到CPU中断请求
PMU_TO_AOP_IRQ_L CJ33 AOP_FUNC_15
I2C0总线时钟
I2C_AOP_SCL CM11 AOP_I2C0_SCL
I2C0总线数据
I2C_AOP_SDA CJ24 AOP_I2C0_SDA
指南针、陀螺仪、加速器、气压传感器到CPU数据输入
SPI_IMU_TO_AOP_MISO CJ18 AOP_SPI_MISO
CPU到指南针、陀螺仪、加速器、气压传感器数据输出
SPI_AOP_TO_IMU_MOSI CJ27 AOP_SPI_MOSI
CPU到陀螺仪、指南针、气压传感器时钟
#25756894:North Carbon R1 (+Mg,P)
SPI_AOP_TO_IMU_SCLK_R1
R1305
49.9
1%
1/32W
MF
01005
ROOM=SOC
SPI_AOP_TO_IMU_SCLK CJ16 AOP_SPI_SCLK
UART_BB_TO_AOP_RXD 基带到CPU数据接收 CK14 AOP_UART0_RXD
UART_AOP_TO_BB_TXD CPU到基带数据发送 CJ20 AOP_UART0_TXD
CPU到陀螺仪时钟
R1306
#25756894:South Carbon R2
SPI_AOP_TO_IMU_SCLK_R2
49.9
1%
1/32W
MF
01005
ROOM=SOC
MAGGIE_TO_AOP_INT MAGGIE断中信号 CJ22 AOP_UART1_RXD
UART_AOP_TO_MAGGIE_TXD CPU到MAGGIE数据发送 CL11 AOP_UART1_TXD
UART_TOUCH_TO_AOP_RXD 触摸到CPU数据接收 CG29 AOP_UART2_RXD
UART_AOP_TO_TOUCH_TXD CPU到触摸数据发送 CH29 AOP_UART2_TXD
I2S_CODEC_XSP_TO_AOP_BCLK 音频到CPU位时钟 CL35 AOP_I2S_BCLK
I2S_CODEC_XSP_TO_AOP_DIN 音频到CPU数据输入 CJ39 AOP_I2S_DIN
CPU到MAGGIE主时钟
R1303
33.2
I2S_AOP_TO_MAGGIE_L26_MCLK
1%
1/32W
MF
01005
ROOM=SOC
I2S_AOP_TO_MAGGIE_L26_MCLK_R CM35 AOP_I2S_MCK
I2S_CODEC_XSP_TO_AOP_LRCLK 音频到CPU左右声道时钟 CK37 AOP_I2S_LRCK
I2S_AOP_TO_CODEC_XSP_DOUT CPU到音频数据输出 CG39 AOP_I2S_DOUT
CFSB_AOP CH35 PMU_TO_SYSTEM_COLD_RESET_L 电源到CPU复位信号
AWAKE_REQ CM31 AOP_TO_PMU_ACTIVE_REQUEST CPU到电源复位
AWAKE_RESET* CJ37 PMU_TO_AOP_TRISTAR_ACTIVE_READY 电源到CPU准备好
AOP_PDM_CLK0 CM37 AOP_TO_MESA_BLANKING_EN CPU到指纹开启
AOP_PDM_DATA0 CH41 AOP_TO_WLAN_CONTEXT_B CPU到WiFi数据接口B
AOP_PDM_DATA1 CK39 AOP_TO_WLAN_CONTEXT_A CPU到WiFi数据接口A
RT_CLK32768 CM33 PMU_TO_AOP_CLK32K 电源到CPU休眠时钟32.768KHZ
AOP_SWD_TCK_OUT CL14 SWD_AP_TO_MANY_SWCLK CPU输出SWD总线时钟
AOP_SWD_TMS0 CL16 HOMER_TO_AOP_WAKE_INT HOMER到CPU唤醒中断
AOP_SWD_TMS1 CG35 SWD_AOP_BI_BB_SWDIO CPU和基带的SWD总线数据
SWD_TMS2 BU4 SWD_AP_BI_NAND_SWDIO CPU和硬盘的SWD总线数据
SWD_TMS3 BV3 SWD_AP_BI_HOMER_SWDIO CPU和HOMER的SWD总线数据
BB_SWDIO has pullup in Radio_MLB pages
CPU到音频放大、振动控制器复位信号
DOCK_ATTENTION CG41 AOP_TO_SPKAMP1_ARC_RESET_L
DOCK_CONNECT CL37 指纹中断 MESA_TO_AOP_FDINT
DOCK_CONNECT can be GPIO, but input only. Radar 21680759

SOC - CPU, GPU & SOC RAILS



SOC - POWER SUPPLIES

SOC - POWER SUPPLIES


PROBE POINTS
90_PCIE_AP_TO_NAND_REFCLK_P
PP1701
P2MM-NSM
ROOM=NAND
90_PCIE_AP_TO_NAND_REFCLK_N
PP1702
P2MM-NSM
ROOM=NAND
1.8V供电
315mA MAX
PP1V8
R1703
24.9
1%
1/32W
MF
01005
ROOM=NAND
PP1V8_NAND_AVDD
C1726
2.2UF
C1710
0.1UF
NAND_AGND
C1701
C1707
C1729
C1730
15UF
C1739
C1741
C1743
C1745
C1747
1.0UF
C1708
220PF
C1711
22PF
C1717
68PF
C1723
39PF
C1712
100PF
硬盘0.9V供电
PP0V9_NAND
1007mA MAX
C1704
C1702
C1705
C1722
C1727
C1737
C1738
C1740
C1742
C1744
C1746
C1703
C1706
C1709
C1714
C1720
C1728
1.8V供电
PP1V8
C1724
0.01UF
C1725
0.01UF
硬盘参考电压
NAND_VREF
#24543147:10uF for 32GB
#26326159:10uF for C1719
1230mA MAX (1us peak power)
硬盘3V供电
PP3V0_NAND
C1748
C1713
C1716
C1719
10UF
C1721
C1733
C1749
C1750
C1751
C1752
C1753
C1754
C1715
C1718
C1731
C1732
C1734
C1735
C1736
U1701
THGBX6T1T82LFXF
VLGA
VER-1
ROOM=NAND
BOMOPTION=OMIT_TABLE
CRITICAL
CPU到硬盘系统控制时钟
AP_TO_NAND_SYS_CLK
CPU到硬盘的PCIE总线参考时钟
90_PCIE_AP_TO_NAND_REFCLK_P
90_PCIE_AP_TO_NAND_REFCLK_N
硬盘到CPU的PCIE时钟请求
PCIE_NAND_BI_AP_CLKREQ_L
硬盘PCIE总线复位参考设置
PCIE_NAND_RESREF
R1704
3.01K
90_PCIE_AP_TO_NAND_TXD_P
90_PCIE_AP_TO_NAND_TXD_N
CPU到硬盘PCIE总线发送
90_PCIE_NAND_TO_AP_RXD_P
90_PCIE_NAND_TO_AP_RXD_N
硬盘到CPU的PCIE总线接收
CPU到硬盘复位信号
AP_TO_NAND_RESET_L
NAND_ZQ
R1701
34.8
电源到硬盘的电池电压低启动信号
PMU_TO_NAND_LOW_BATT_BOOT_L
AP_TO_NAND_FW_STRAP
CPU到硬盘调频链接信号
系统激活
SYSTEM_ALIVE
CPU到硬盘的PCIE总线复位
PCIE_AP_TO_NAND_RESET_L
CPU和硬盘SWD总线数据
SWD_AP_BI_NAND_SWDIO_R
R1702
0.00
SWD_AP_BI_NAND_SWDIO
SWD_AP_TO_NAND_SWCLK_R
R1707
0.00
SWD_AP_TO_MANY_SWCLK
CPU和硬盘SWD总线时钟
CLK_IN
PCIE_REFCLK_P
PCIE_REFCLK_M
PCIE_CLKREQ*
PCI_RESREF
PCIE_RX0_P
PCIE_RX0_M
PCIE_RX1_P
PCIE_RX1_M
PCIE_TX0_P
PCIE_TX0_M
PCIE_TX1_P
PCIE_TX1_M
RESET*
TRST*
ZQ
EXT_D0
EXT_D1
EXT_D2
EXT_D3
EXT_D4
EXT_D5
EXT_D6
EXT_D7
EXT_NCE
EXT_NRE
EXT_NWE
EXT_RNB
EXT_CLE
EXT_ALE
NC
NAND_AGND


U1801
D2333A1
WLCSP
SYM 2 OF 4
Voltages per Cayman Power Spec, Sec. 2.2, rev 0.9.2, #24557869
主供电电压检测
VDD_MAIN_SNS
主供电
PP_VDD_MAIN
BAT/USB
BUCK INPUT
SWITCH OUTPUTS
CPU供电
PP_CPU_VAR
0.625V - 1.06V
BUCK0 13.4A MAX
GPU供电
PP_GPU_VAR
0.67V - 0.92V
1.03V for overdrive only
BUCK1 13.4A MAX
CPU供电
PP_SOC_VAR
0.67V/0.80V
BUCK2 4.7A MAX
暂存供电1.8V
PP1V8_SDRAM
BUCK3 1.7A MAX
暂存供电1.1V
PP1V1_SDRAM
BUCK4 4.7A MAX
CPU 0.9V供电
PP0V9_SOC_FIXED
BUCK5 3.2A MAX
1.25V供电
PP1V25_BUCK
BUCK6 1.5A MAX
CPU的SRAM供电
PP_CPU_SRAM_VAR
0.80V - 1.06V
BUCK7 1.5A MAX
GPU的SRAM供电
PP_GPU_SRAM_VAR
0.80V - 0.92V
BUCK8 1.5A MAX
主摄像2.8V聚焦供电
PP2V8_UT_AF_VAR
BUCK9 0.75A MAX
1.8V供电
PP1V8
触摸供电1.8V
PP1V8_TOUCH
PP1V8_MAGGIE_IMU
1.1V供电
PP1V1
D10/D101:#24681501,TY ONLY, D11/D11: Both Vendor
L1806 1.0UH-20%-3.6A-0.060OHM
L1807 0.22UH-20%-6.7A-0.023OHM
L1808 0.22UH-20%-6.7A-0.023OHM
L1809 0.22UH-20%-6.7A-0.023OHM
L1810 1.0UH-20%-3.6A-0.060OHM
L1811 0.22UH-20%-6.7A-0.023OHM
L1812 0.22UH-20%-6.7A-0.023OHM
L1813 0.22UH-20%-6.7A-0.023OHM
L1814 1.0UH-20%-3.6A-0.060OHM
L1815 0.47UH-20%-3.8A-0.048OHM
L1816 1.0UH-20%-3.6A-0.060OHM
L1817 1.0UH-20%-3.6A-0.060OHM
L1818 0.47UH-20%-3.8A-0.048OHM
L1803 1.0UH-20%-3.6A-0.060OHM
L1804 1UH-20%-2.1A-0.120OHM
L1805 1.0UH-20%-2.25A-0.086OHM
L1801 1UH-20%-2.1A-0.120OHM
L1802 1.0UH-20%-1.5A-0.161OHM
XW1802
XW1807
XW1803
XW1801
XW1806
XW1804
XW1805
CRITICAL
VDD_MAIN_SNS
VDD_MAIN
VDD_MAIN_E
VDD_MAIN_N
VDD_MAIN_SW
VDD_MAIN_W
VDD_BUCK0_01
VDD_BUCK0_23
VDD_BUCK1_01
VDD_BUCK1_23
VDD_BUCK2
VDD_BUCK3
VDD_BUCK4
VDD_BUCK5
VDD_BUCK6
VDD_BUCK7
VDD_BUCK8
VDD_BUCK9
BUCK0_LX0
BUCK0_LX1
BUCK0_LX2
BUCK0_LX3
BUCK0_FB
BUCK0_PP_CPU_FB
BUCK1_LX0
BUCK1_LX1
BUCK1_LX2
BUCK1_LX3
BUCK1_FB
BUCK1_PP_GPU_FB
BUCK2_LX0
BUCK2_LX1
BUCK2_FB
BUCK2_PP_SOC_FB
BUCK3_LX0
BUCK3_FB
VBUCK3_SW
BUCK4_LX0
BUCK4_LX1
BUCK4_FB
VBUCK4_SW
BUCK5_LX0
BUCK5_FB
BUCK6_LX0
BUCK6_FB
BUCK7_LX0
BUCK7_FB
BUCK8_LX0
BUCK8_FB
BUCK9_LX0
BUCK9_FB
BUCK3_SW1
BUCK3_SW2
BUCK3_SW3
BUCK4_SW1
C1846 10UF
C1850 10UF
C1853 10UF
C1857 10UF
C1875 2.2UF
C1847 2.2UF
C1851 2.2UF
C1854 2.2UF
C1858 2.2UF
C1876 2.2UF
C1848 2.2UF
C1852 2.2UF
C1855 2.2UF
C1859 2.2UF
C1877 2.2UF
C1849 2.2UF
C1856 2.2UF
C1818 15UF
C1825 15UF
C1831 15UF
C1837 15UF
C1842 15UF
C1844 15UF
C1819 15UF
C1826 15UF
C1832 15UF
C1838 15UF
C1843 15UF
C1845 15UF
C1872 220PF
C1813 15UF
C1820 15UF
C1827 15UF
C1833 15UF
C1839 15UF
C1865 15UF
C1814 15UF
C1821 15UF
C1828 15UF
C1834 15UF
C1866 15UF
C1873 220PF
C1822 15UF
C1829 15UF
C1835 15UF
C1841 15UF
C1864 15UF
C1871 220PF
C1816 15UF
C1823 15UF
C1860 220PF
C1830 15UF
C1836 15UF
C1802 15UF
C1874 15UF
C1861 220PF
C1867 220PF
C1840 15UF
C1801 15UF
C1803 15UF
C1807 15UF
C1811 220PF
C1804 15UF
C1808 15UF
C1868 220PF
C1805 15UF
C1809 15UF
C1869 220PF
C1806 15UF
C1810 15UF
C1870 220PF
C1862 15UF
C1863 15UF

ADELYN LDO SPECS

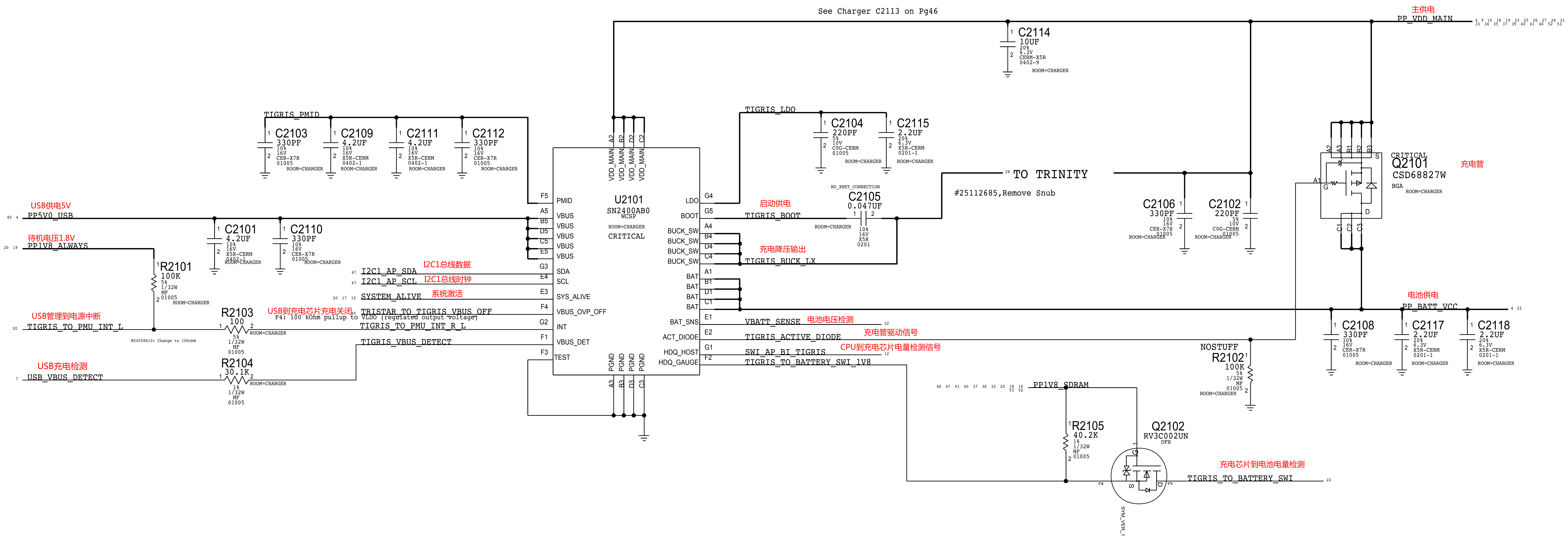
LDO#	ADJ.RANGE, LOW	ADJ.RANGE, HI	ACCURACY	MAX.CURRENT
LDO1 (Ca)	1.2-2.475V	2.4-3.675V	+/-1.4%	50mA
LDO2 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LDO3 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LDO4 (D)	0.7-1.2V		+/-2.5%	60mA
LDO5 (F)	2.5-3.6V(tbc)		+/-75mV	1000mA
LDO6 (Cb)	1.2-2.475V	2.4-3.675V	+/-2.5%	250mA (500/100mA in bypass)
LDO7 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LDO8 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LDO9 (Cb)	1.2-2.475V	2.4-3.675V	+/-25mV	250mA
LDO10 (Ga)	0.7-1.2V		+/-4.5%	1150mA

LDO#	ADJ.RANGE, LOW	ADJ.RANGE, HI	ACCURACY	MAX.CURRENT
LDO11 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LDO12 (E)	1.8V		+/-5%	10mA
LDO13 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LDO14 (Gb)	0.7-1.4V		+/-3.0%	400mA
LDO15 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LDO16 (Cb)	1.2-2.475V	2.4-3.675V	+/-30mV	250mA
LDO17 (Ca)	1.2-2.475V	2.4-3.675V	+/-2.5%	50mA
LDO18 (Gb)	0.7-1.4V		+/-3.0%	400mA
LDO19 (Gb)	0.7-1.4V		+/-3.0%	400mA
LDO_RTC	2.5V		+/-2.0%	10mA
BUF_1V2	1.2V		+/-5.0%	10mA

BUTTON PULL-UP RESISTORS

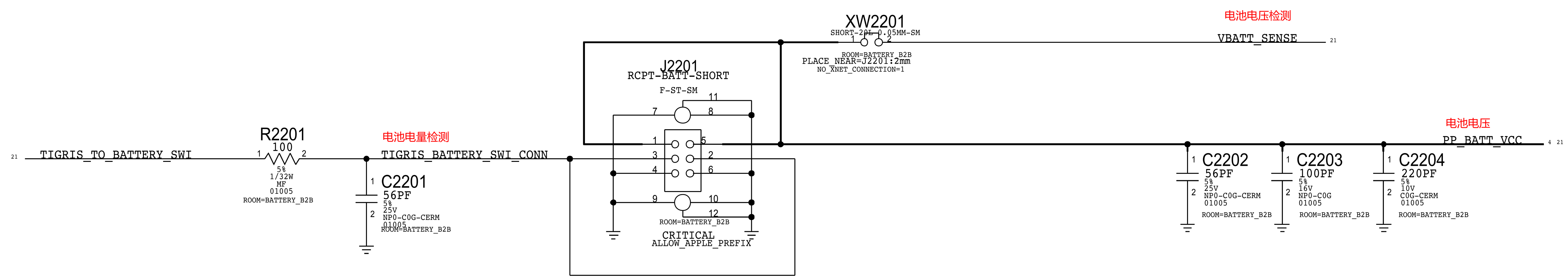
NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC

TIGRIS CHARGER



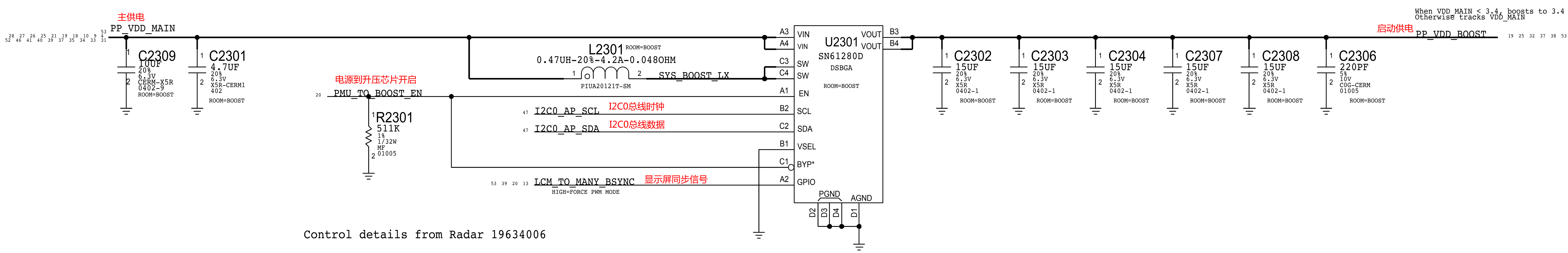
BATTERY CONNECTOR

THIS ONE ON MLB ---> 516S00172 (matches d10 mlb MCO rev 27)



BOOST

升压器



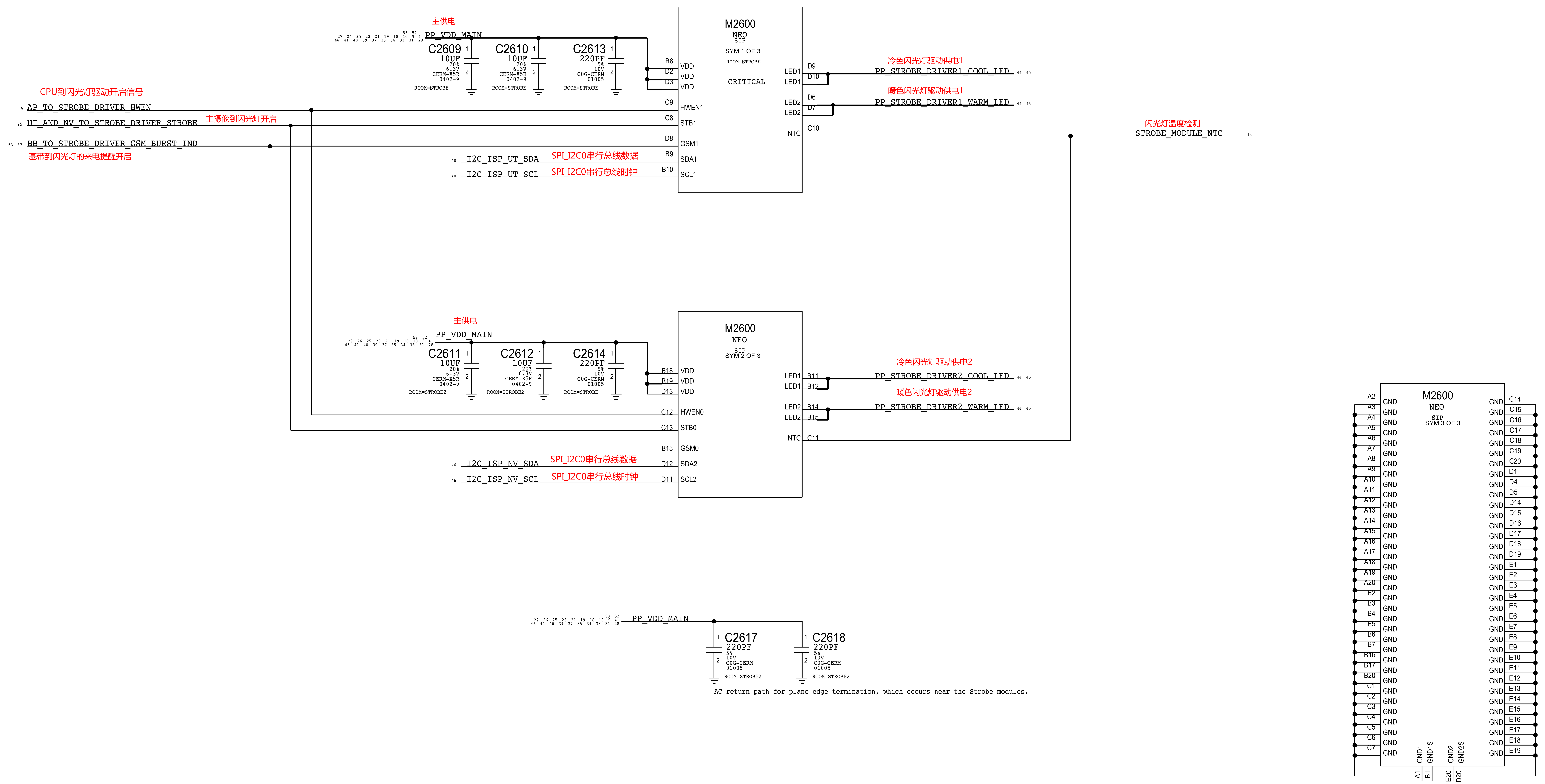
主供电
PP_VDD_MAIN
C2309
10UF
20%
6.3V
CERM-X5R
0402-9
ROOM=BOOST
C2301
4.7UF
20%
6.3V
X5R-CERM1
402
ROOM=BOOST
L2301
ROOM=BOOST
0.47UH-20%-4.2A-0.048OHM
PIUA20121T-SM
SYS_BOOST_LX
电源到升压芯片开启
PMU_TO_BOOST_EN
R2301
511K
1%
1/32W
MF
01005
I2C0_AP_SCL
I2C0总线时钟
I2C0_AP_SDA
I2C0总线数据
LCM_TO_MANY_BSYNC
显示屏同步信号
HIGH=FORCE PWM MODE
U2301
SN61280D
DSBGA
ROOM=BOOST
VIN
VIN
SW
SW
EN
SCL
SDA
VSEL
BYP*
GPIO
VOUT
VOUT
PGND
AGND
C2302
15UF
C2303
15UF
C2304
15UF
C2307
15UF
C2308
15UF
C2306
220PF
5%
10V
C0G-CERM
01005
ROOM=BOOST
When VDD_MAIN < 3.4, boosts to 3.4
Otherwise tracks VDD_MAIN
启动供电
PP_VDD_BOOST
Control details from Radar 19634006

CARBON - ACCEL & GYRO
INVENSENSE, MPU-6800: C2403=0.1UF
陀螺仪
供电电压1.8V
PP1V8_MAGGIE_IMU
C2418 2.2UF
C2402 0.1UF
C2415 0.1UF
R2401 100K
BOMOPTION: CARBON_1
#25765850:Update Carbon APN
U2401
MPU-6900-21
LGA
CPU到陀螺仪片选信号
SPI_AOP_TO_ACCEL_GYRO_CS_L
GYRO_CHARGE_PUMP
ACCEL_GYRO_TO_AOP_INT
加速计、陀螺仪到CPU中断
CPU到陀螺仪、指南针、气压传感器时钟
SPI_AOP_TO_IMU_SCLK_R1
SPI_AOP_TO_IMU_MOSI
CPU到指南针、陀螺仪、加速器、气压传感器数据输出
SPI_IMU_TO_AOP_MISO
指南针、陀螺仪、加速器、气压传感器到CPU数据输入
ACCEL_GYRO_TO_AOP_DATARDY
陀螺仪到CPU数据就绪
C2419 5PF
CRITICAL
C2403 0.1UF
#25782019:Add 0ohm
R2404 0.00
PP1V8_MAGGIE_IMU_R
C2448 2.2UF
C2442 0.1UF
C2445 0.1UF
R2441 100K
U2404
MPU-6900-21
SPI_AOP_TO_BOT_ACCEL_GYRO_CS_L
CPU到陀螺仪时钟
SPI_AOP_TO_IMU_SCLK_R2
BOT_GYRO_CHARGE_PUMP
XW2404
BOT_ACCEL_GYRO_TO_AOP_DATARDY
C2449 5PF
C2443 0.1UF
#25740540:PP for South Carbon MOSI
PP2440
MAGNESIUM - COMPASS
指南针
PP1V8_MAGGIE_IMU_FILT
C2401 0.1UF
C2408 2.2UF
U2402
HSCDTD601A-19A
SPI_IMU_TO_AOP_MISO
SPI_AOP_TO_IMU_MOSI
SPI_AOP_TO_IMU_SCLK_R1
SPI_AOP_TO_COMPASS_CS_L
CPU到指南针片选
指南针到CPU中断
COMPASS_TO_AOP_INT
PP2404
PP2401
PP2402
PP2403
气压计
PHOSPHORUS
#24593845, #25691124
BOSCH (APN:338S00188): nostuff C2420/C2421/C2422/C2423 and R2403 PU
R2422 0.00
C2413 0.1UF
C2405 0.1UF
C2420 4PF
C2421 20PF
C2422 20PF
C2423 5.6PF
C2414 2.2UF
BOSCH: Internal PU
R2403 100K
U2403
BMP284AA
SPI_AOP_TO_IMU_MOSI
SPI_AOP_TO_IMU_SCLK_R1
SPI_AOP_TO_PHOSPHORUS_CS_L
气压传感器片选信号
PHOSPHORUS_TO_AOP_INT_L
气压传感器到CPU中断
6 OF 81


UTAH POWER
摄像供电
NOTE: OUTPUT IMPEDANCE MUST BE >0.005-OHM
IN ORDER TO MEET CAP ESR REQUIREMENT PER LDO SPEC.
VENDOR ALSO RECOMMENDS CIN = COUT FOR STABILITY
TI:353S00015
ST:353S00889
U2501
LP5907UVX2.925-S
DSBGA
VIN
VOUT
VEN
GND
VOLTAGE=2.925V
See Page46: D10x C2531/C2507 are 2.2UF
启动供电
PP_VDD_BOOST
C2527
2.2UF
摄像供电2.9V
PP2V9_UT_AVDD_CONN
C2502
0.22UF
C2504
220PF
摄像聚焦供电2.8V
PP2V8_UT_AF_VAR
FL2500
33-OHM-25%-1500MA
PP2V8_UT_AF_VAR_CONN
C2505
2.2UF
C2501
220PF
距离感应、感光供电3V
PP3V0_ALS_APS_CONVOY
FL2502
33-OHM-25%-1500MA
PP3V0_UT_SVDD_CONN
C2519
2.2UF
C2518
220PF
摄像供电1.2
PP1V2_UT_DVDD
FL2505
33-OHM-25%-1500MA
PP1V2_UT_VDD_CONN
C2506
2.2UF
C2508
2.2UF
C2510
2.2UF
C2503
220PF
C2521
15PF
Desense for Wifi frequencies
供电1.8
PP1V8
FL2506
33-OHM-25%-1500MA
PP1V8_UT_CONN
C2509
1.0UF
C2511
220PF
IO FILTERS
CPU到主摄像时钟
AP_TO_UT_CLK
FL2504
150OHM-25%-200MA-0.7DCR
NOSTUFF
C2512
100PF
C2513
56PF
AP_TO_UT_CLK_CONN
CPU到主摄像开启
AP_TO_UT_SHUTDOWN_L
FL2501
150OHM-25%-200MA-0.7DCR
C2514
220PF
AP_TO_UT_SHUTDOWN_CONN_L
主摄像到闪光灯开启
UT_AND_NV_TO_STROBE_DRIVER_STROBE
FL2503
150OHM-25%-200MA-0.7DCR
C2515
220PF
UT_AND_NV_TO_LED_DRIVER_STROBE_EN_CONN
LPDP FILTERS
PP_VDD_MAIN
C2522
33PF
C2528
33PF
C2529
33PF
AC return path for LPDP which is referenced to GND and VDD_MAIN
主摄像LPDP数据线0组
90_LPDP_UT_TO_AP_D0_P
90_LPDP_UT_TO_AP_D0_N
C2523
C2524
0.1UF
90_LPDP_UT_TO_AP_D0_CONN_P
90_LPDP_UT_TO_AP_D0_CONN_N
主摄像LPDP数据线1组
90_LPDP_UT_TO_AP_D1_P
90_LPDP_UT_TO_AP_D1_N
C2525
C2526
90_LPDP_UT_TO_AP_D1_CONN_P
90_LPDP_UT_TO_AP_D1_CONN_N
主摄像到CPU通信数据线
LPDP_UT_BI_AP_AUX
C2530
0.1UF
LPDP_UT_BI_AP_AUX_CONN
C2520
56PF
#24543254: Need to Scrub C2520 Value

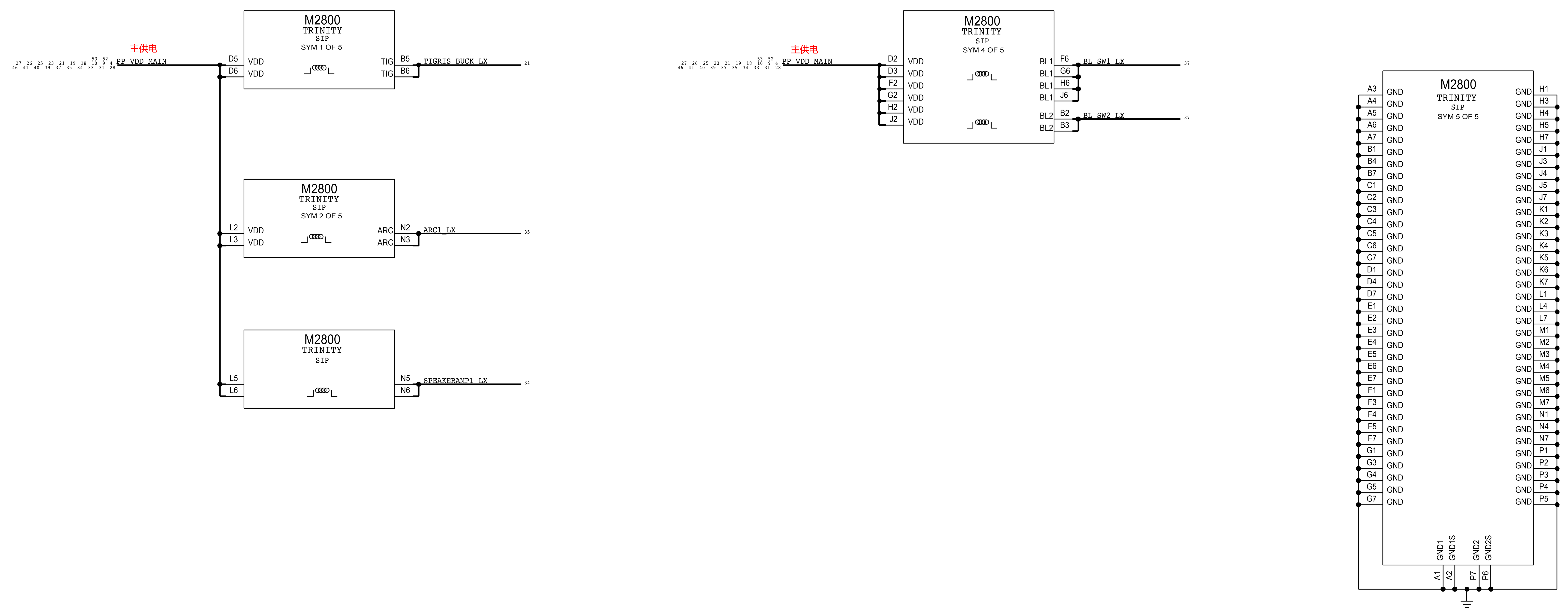
STROBE DRIVERS INSIDE NEO SIP MODULE

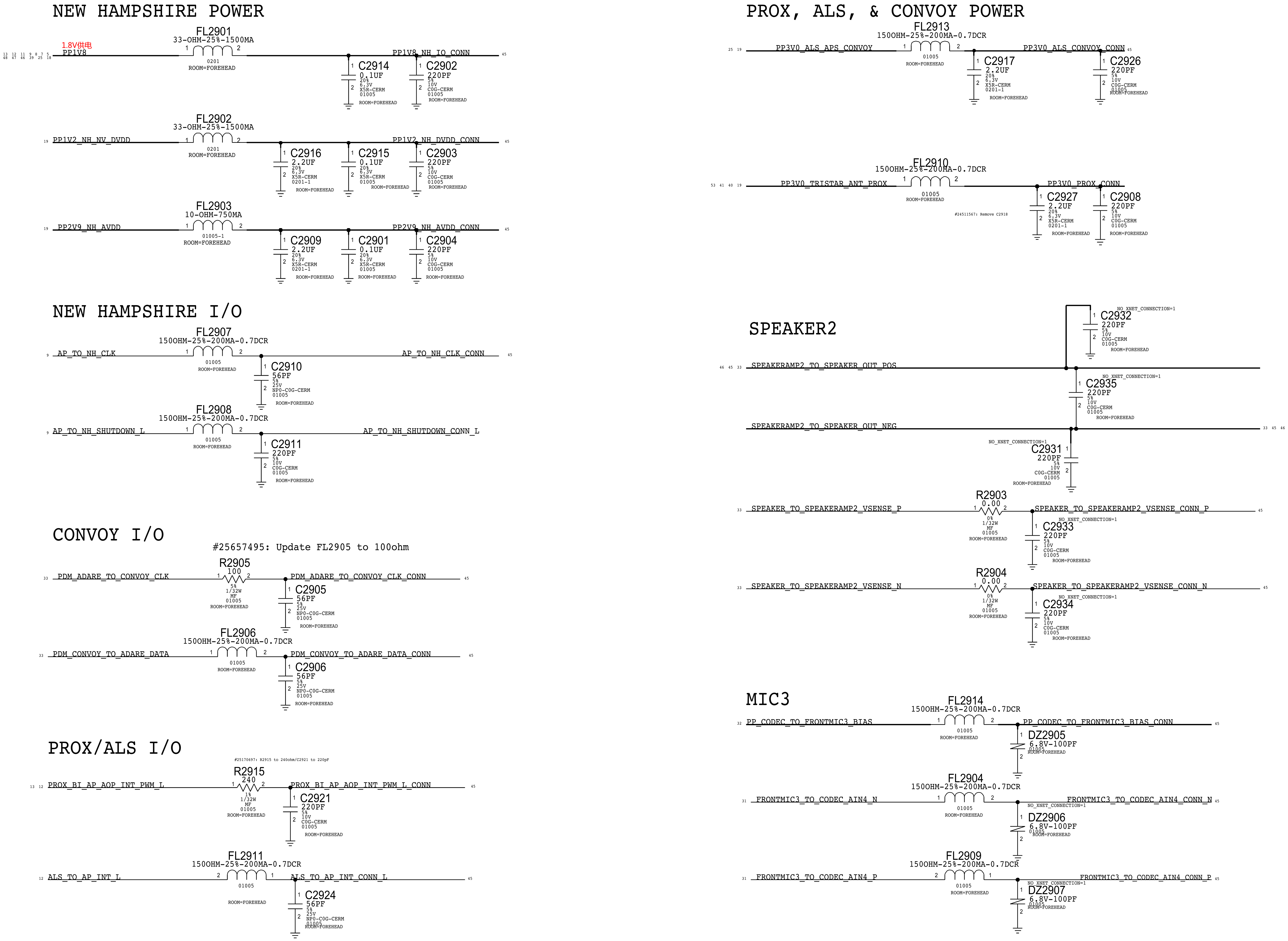
D10/sip_neo



ACCESSORY BUCK

THIS PAGE UNIQUE TO SMALL FORM FACTOR

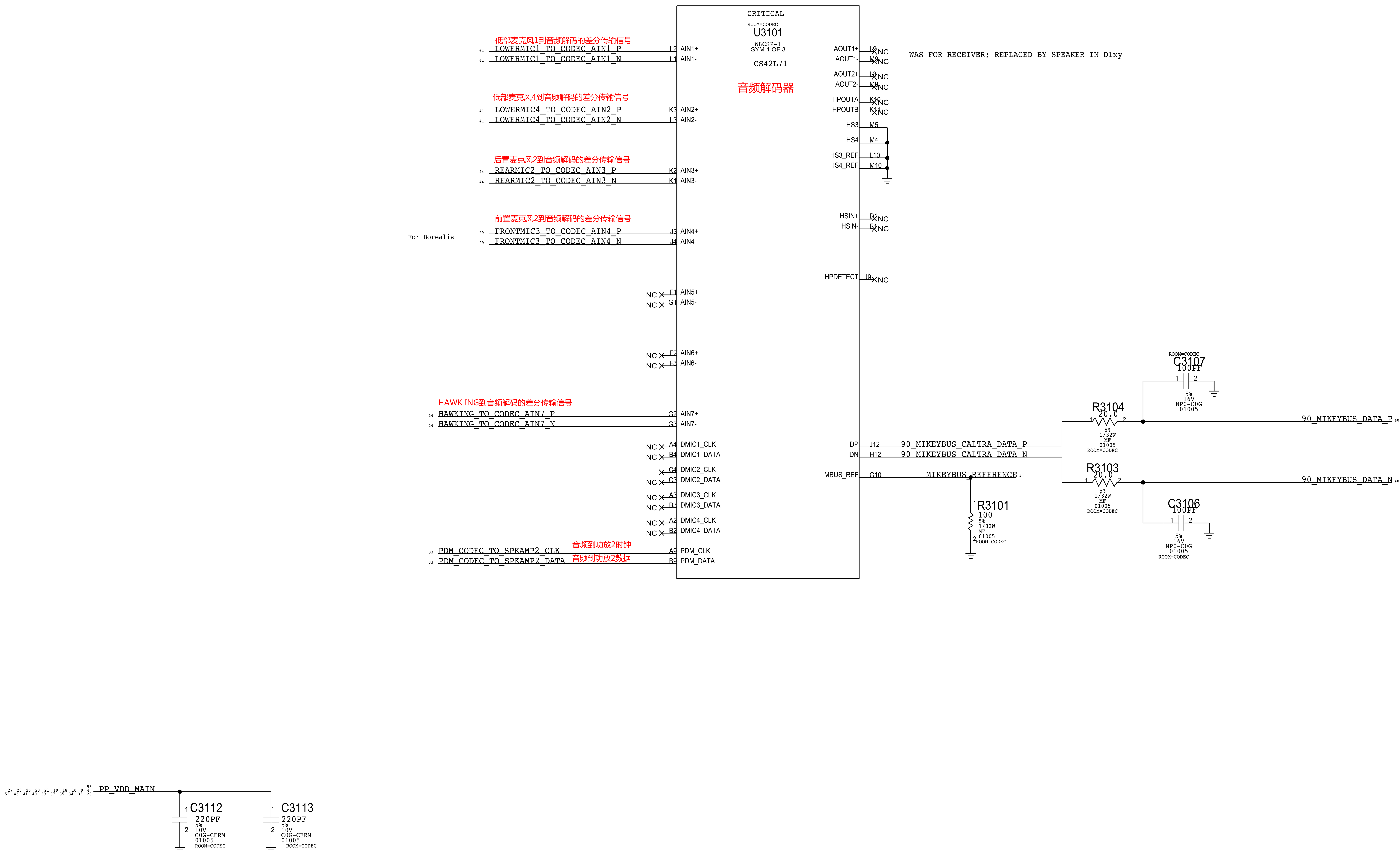




NEW HAMPSHIRE POWER
1.8V供电
PP1V8
FL2901
33-OHM-25%-1500MA
0201
ROOM=FOREHEAD
PP1V8_NH_IO_CONN
C2914
0.1UF
C2902
220PF
PP1V2_NH_NV_DVDD
FL2902
33-OHM-25%-1500MA
PP1V2_NH_DVDD_CONN
C2916
2.2UF
C2915
0.1UF
C2903
220PF
PP2V9_NH_AVDD
FL2903
10-OHM-750MA
01005-1
PP2V9_NH_AVDD_CONN
C2909
2.2UF
C2901
0.1UF
C2904
220PF
NEW HAMPSHIRE I/O
FL2907
150OHM-25%-200MA-0.7DCR
AP_TO_NH_CLK
AP_TO_NH_CLK_CONN
C2910
56PF
FL2908
150OHM-25%-200MA-0.7DCR
AP_TO_NH_SHUTDOWN_L
AP_TO_NH_SHUTDOWN_CONN_L
C2911
220PF
CONVOY I/O
#25657495: Update FL2905 to 100ohm
R2905
100
PDM_ADARE_TO_CONVOY_CLK
PDM_ADARE_TO_CONVOY_CLK_CONN
C2905
56PF
FL2906
150OHM-25%-200MA-0.7DCR
PDM_CONVOY_TO_ADARE_DATA
PDM_CONVOY_TO_ADARE_DATA_CONN
C2906
56PF
PROX/ALS I/O
R2915
240
PROX_BI_AP_AOP_INT_PWM_L
PROX_BI_AP_AOP_INT_PWM_L_CONN
C2921
220PF
FL2911
150OHM-25%-200MA-0.7DCR
ALS_TO_AP_INT_L
ALS_TO_AP_INT_CONN_L
C2924
56PF
PROX, ALS, & CONVOY POWER
FL2913
150OHM-25%-200MA-0.7DCR
PP3V0_ALS_APS_CONVOY
PP3V0_ALS_CONVOY_CONN
C2917
2.2UF
C2926
220PF
FL2910
150OHM-25%-200MA-0.7DCR
PP3V0_TRISTAR_ANT_PROX
PP3V0_PROX_CONN
C2927
2.2UF
C2908
220PF
SPEAKER2
C2932
220PF
SPEAKERAMP2_TO_SPEAKER_OUT_POS
C2935
220PF
SPEAKERAMP2_TO_SPEAKER_OUT_NEG
C2931
220PF
R2903
0.00
SPEAKER_TO_SPEAKERAMP2_VSENSE_P
SPEAKER_TO_SPEAKERAMP2_VSENSE_CONN_P
C2933
220PF
R2904
0.00
SPEAKER_TO_SPEAKERAMP2_VSENSE_N
SPEAKER_TO_SPEAKERAMP2_VSENSE_CONN_N
C2934
220PF
MIC3
FL2914
150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_FRONTMIC3_BIAS
PP_CODEC_TO_FRONTMIC3_BIAS_CONN
DZ2905
6.8V-100PF
FL2904
150OHM-25%-200MA-0.7DCR
FRONTMIC3_TO_CODEC_AIN4_N
FRONTMIC3_TO_CODEC_AIN4_CONN_N
DZ2906
6.8V-100PF
FL2909
150OHM-25%-200MA-0.7DCR
FRONTMIC3_TO_CODEC_AIN4_P
FRONTMIC3_TO_CODEC_AIN4_CONN_P
DZ2907
6.8V-100PF

D C B A

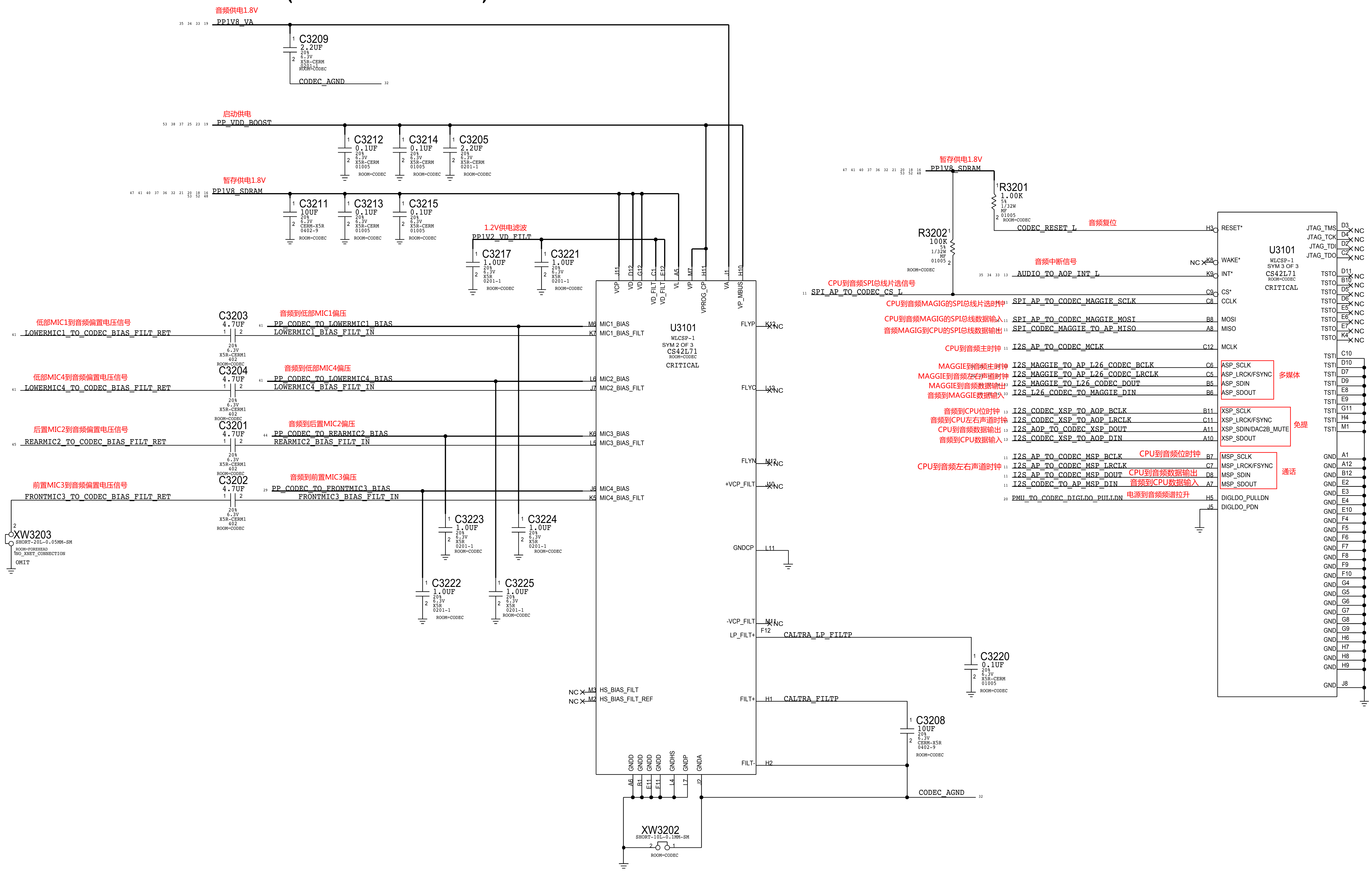
CALTRA AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



AC return path for Mikeybus which is referenced to GND and VDD_MAIN

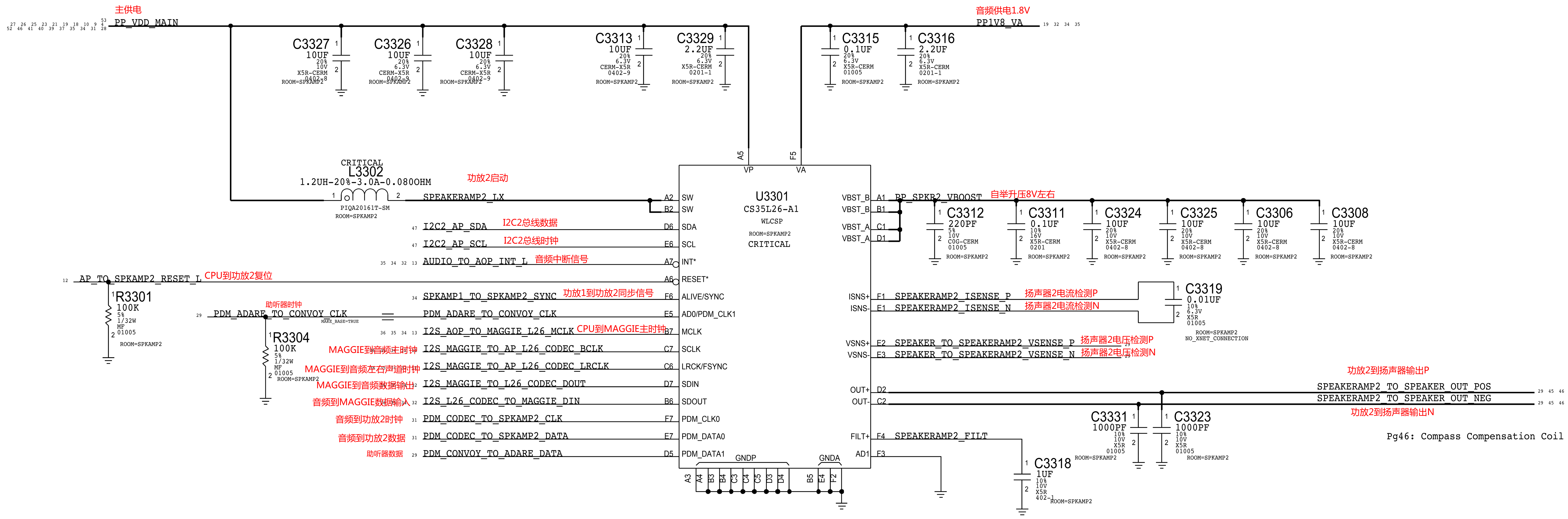
Radar 21203307

CALTRA AUDIO CODEC (POWER & I/O)

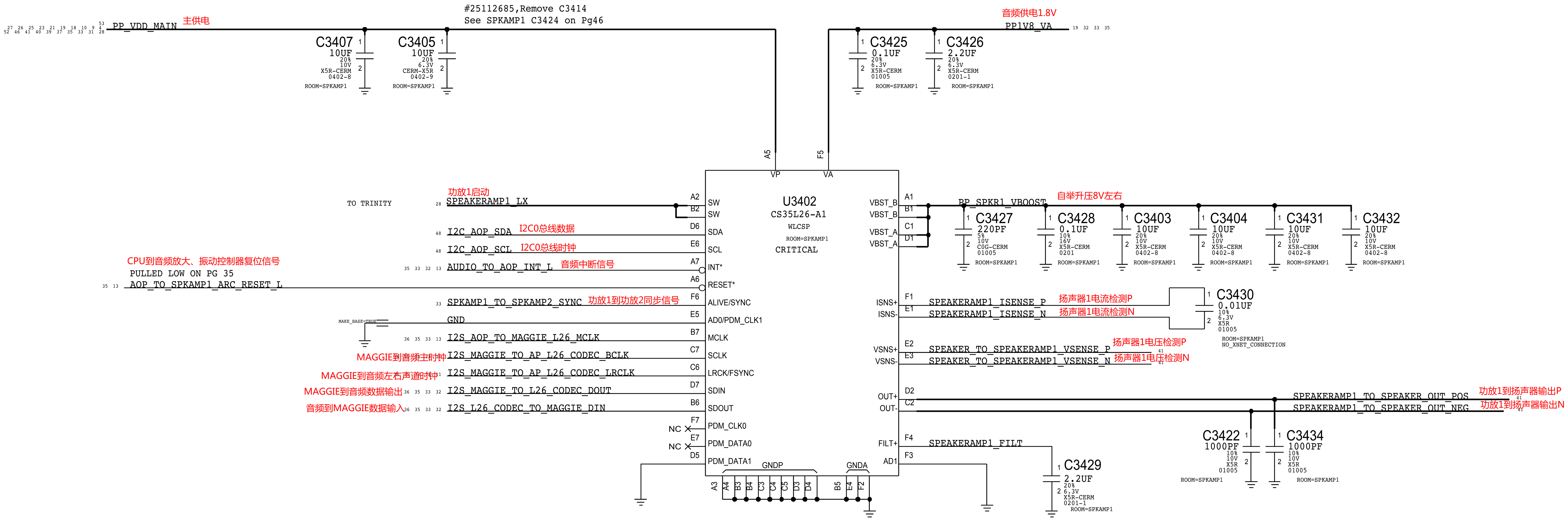


SPEAKER AMPLIFIER 2 (North)

扬声器放大器2

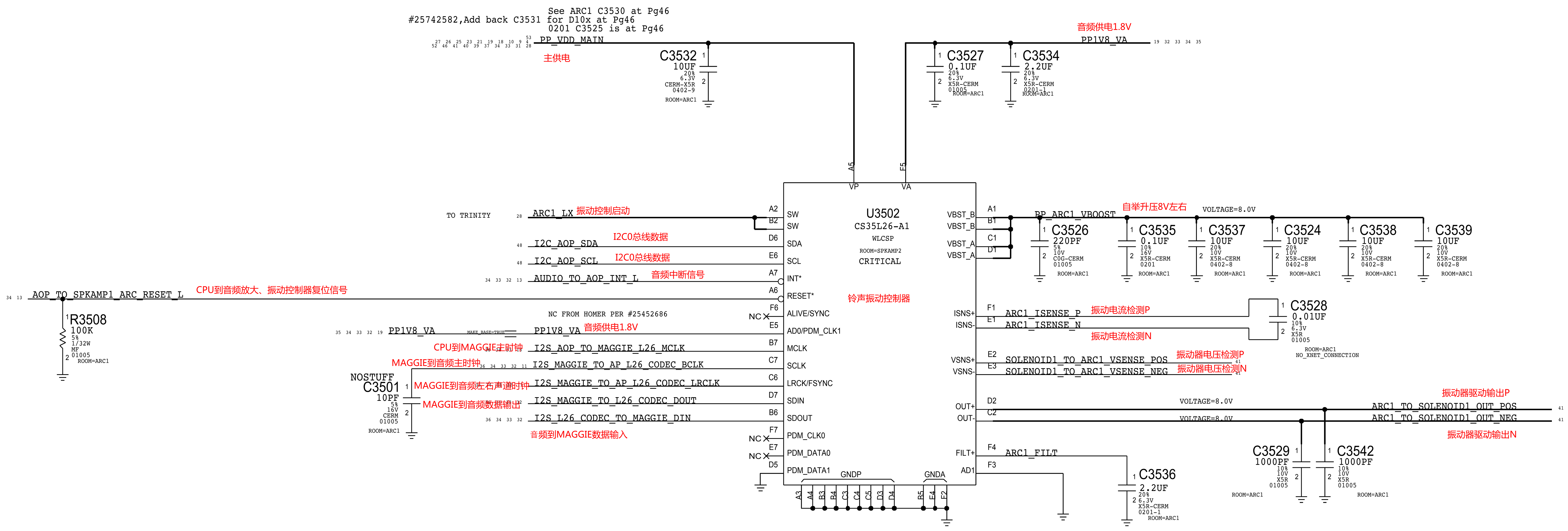


SPEAKER AMPLIFIER 1 (South)



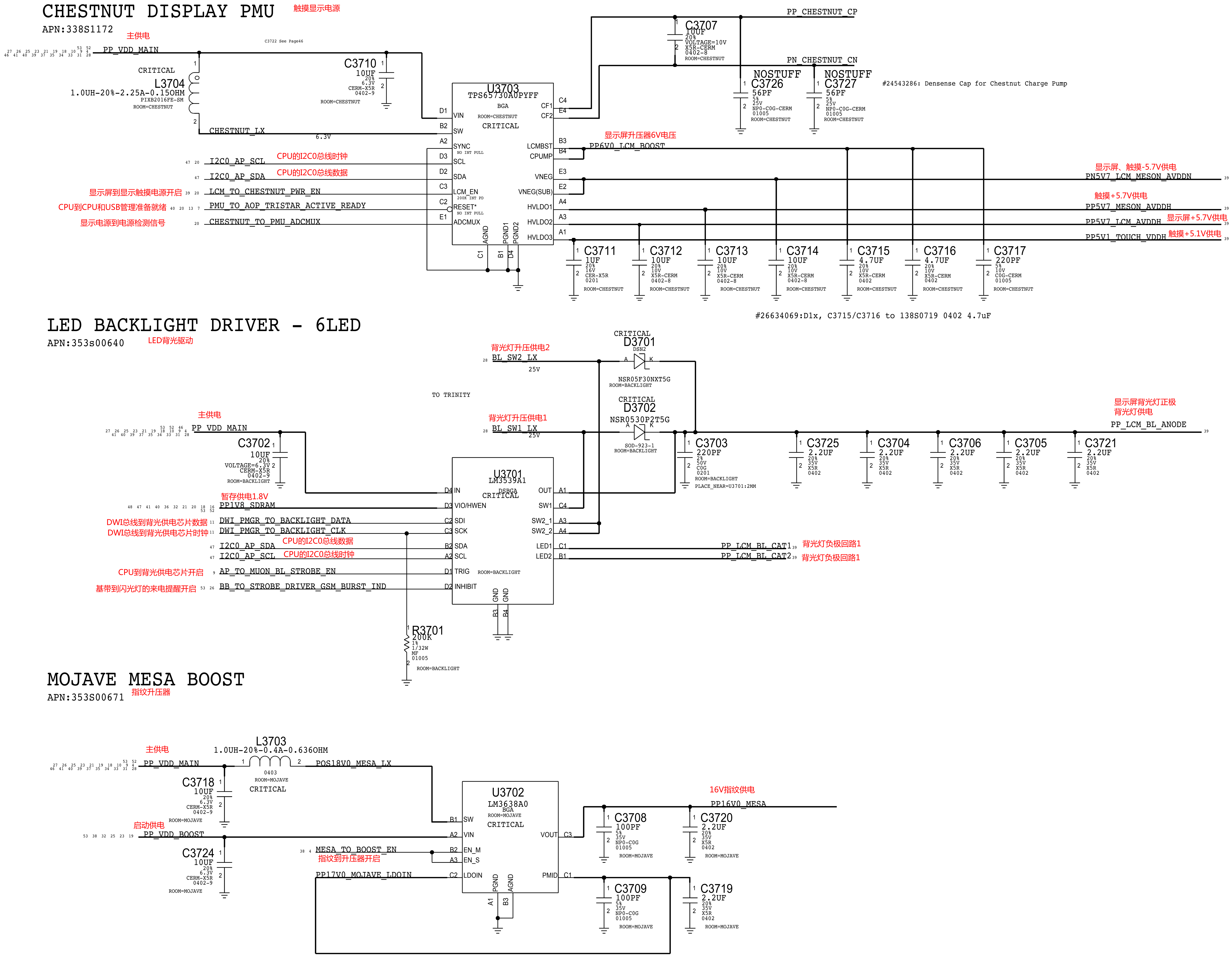
ARC DRIVER

振动控制器



MAGGIE LDO
APN: 353S00842
稳压器
U3603
LD39130S-1.2V/AP
暂存供电1.8V
PP1V8_SDRAM
MAGGIE
APN: 336S00020
MAGGIE供电1.2V
PP1V2_MAGGIE
C3605
4UF
C3601
0.1UF
R3601
100
PP1V2_MAGGIE_PLL
C3602
0.1UF
MAGGIE供电1.8V
VPP_2V5 must be > 1.71V for SPI Slave programming
PP1V8_MAGGIE_IMU
C3604
2.2UF
C3606
0.1UF
ICE5LP4K-SWG36I
U3602
HOMER STM32L0 MICRO
STM32L03 APN: 337S00231
HOMER微处理器
MAGGIE到HOMER唤醒
MAGGIE_TO_HOMER_WAKE
MAGGIE到HOMER时钟
SPI_MAGGIE_TO_HOMER_POS_SCLK
HOMER到MAGGIE数据输出
SPI_HOMER_TO_MAGGIE_POS_MISO
AOP_TO_MAGGIE_EN
CPU到MAGGIE开启
MAGGIE_TO_AOP_INT
MAGGIE到CPU中断
MAGGIE到音频主时钟
I2S_MAGGIE_TO_AP_L26_CODEC_BCLK_R
R3604
33.2
I2S_MAGGIE_TO_AP_L26_CODEC_BCLK
MAGGIE DRIVES TO ARC, SPKRS, AP, CODEC
MAGGIE_TO_AP_CDONE
MAGGIE到CPU数据控制
UART_AOP_TO_MAGGIE_TXD
CPU到MAGGIE数据发送
I2S_MAGGIE_TO_AP_DIN
MAGGIE到CPU数据输入
I2S_AOP_TO_MAGGIE_L26_MCLK
CPU到MAGGIE主时钟
I2S_MAGGIE_TO_L26_CODEC_DOUT
MAGGIE到音频数据输出
I2S_L26_CODEC_TO_MAGGIE_DIN
音频到MAGGIE数据输入
I2S_AP_TO_MAGGIE_DOUT
CPU到MAGGIE数据输出
MAGGIE到音频左右声道时钟
I2S_MAGGIE_TO_AP_L26_CODEC_LRCLK
SPI_CODEC_MAGGIE_TO_AP_MISO
音频MAGGIE到CPU数据输出
SPI_AP_TO_CODEC_MAGGIE_MOSI
CPU到音频MAGGIE数据输入
SPI_AP_TO_CODEC_MAGGIE_SCLK
CPU到音频MAGGIE时钟
SPI_AP_TO_MAGGIE_CS_L
CPU到MAGGIE片选
AP_TO_MAGGIE_CRESETB_L
CPU到MAGGIE复位
R3605
10K
R3602
10K
R3603
511K
ROOM=ARC_CTRL
CRITICAL
PP3601
PP3602
U3601
STM32L031E6Y6D
HOMER到CPU数据接收
UART_HOMER_TO_AP_RXD
UART_AP_TO_HOMER_TXD
CPU到HOMER数据发送
MAGGIE_TO_HOMER_WAKE
MAGGIE到HOMER唤醒
CPU和HOMER引导程序激活信号
AP_BI_HOMER_BOOTLOADER_ALIVE_R
R3607
100
HOMER到CPU唤醒中断
HOMER_TO_AOP_WAKE_INT
I2C_HOMER_SCL
HOMER的I2C总线时钟
I2C_HOMER_SDA
HOMER的I2C总线数据
CPU和HOMER的SWD总线数据
SWD_AP_BI_HOMER_SWDIO
CPU许多位置的SWD总线时钟
SWD_AP_TO_MANY_SWCLK
SPI_MAGGIE_TO_HOMER_POS_MOSI
MAGGIE到HOMER数据输入
NOTE: RESET HAS INTERNAL 65K PULLUP
PMU_TO_HOMER_RESET_L
电源到HOMER复位信号
CPU和HOMER引导程序激活信号
AP_BI_HOMER_BOOTLOADER_ALIVE
R3611
27K
C3607
0.1UF
C3603
2.2UF
#24543115: Scrub Value

DISPLAY & TOUCH - POWER SUPPLIES



MESA POWER
PP3V0_MESA
FL3803
80-OHM-25%-0.52A-0.17OHM
0201
ROOM=MAMBA_MESA
PP3V0_MESA_CONN
C3813
2.2UF
C3815
2.2UF
C3821
2.2UF
C3822
0.1UF
C3804
220PF
PP1V8_MESA
FL3801
150OHM-25%-200MA-0.7DCR
PP1V8_MESA_CONN
C3814
2.2UF
C3802
220PF
PP16V0_MESA
FL3802
150OHM-25%-200MA-0.7DCR
PP16V0_MESA_CONN
C3803
100PF
MESA DIGITAL I/O
FL3807
150OHM-25%-200MA-0.7DCR
SPI_AP_TO_MESA_MOSI
SPI_AP_TO_MESA_MOSI_CONN
R3807
511K
C3816
56PF
R3809
0.00
SPI_AP_TO_MESA_SCLK
SPI_AP_TO_MESA_SCLK_CONN
R3808
511K
C3817
56PF
R3811
33.2
SPI_MESA_TO_AP_MISO
SPI_MESA_TO_AP_MISO_CONN
C3818
56PF
R3801
681
MESA_TO_AP_INT
MESA_TO_AP_INT_CONN
C3819
100PF
R3802
681
MESA_TO_BOOST_EN
MESA_TO_BOOST_EN_CONN
C3801
100PF
FL3811
150OHM-25%-200MA-0.7DCR
AOP_TO_MESA_BLANKING_EN
AOP_TO_MESA_BLANKING_EN_CONN
C3825
56PF
MAMBA AND MESA CONNECTOR
MLB: 516S00141 (RCPT)
FLEX: 516S00142 (PLUG)
CRITICAL
J3801
BB35C-RA24-3A
F-ST-SM
Matches flex_x452_acf, schematic revision 1.5.0 pinout
PP3V0_MESA_CONN
PP16V0_MESA_CONN
MAMBA_TO_LCM_MDRIVE_CONN_MESA
显示屏到触摸同步信号
LCM_TO_MAMBA_MSYNC_CONN
AP_TO_TOUCH_MAMBA_RESET_CONN_L
TP_MAMBA_HINT_L
I2C_TOUCH_BI_MAMBA_SDA
I2C_TOUCH_TO_MAMBA_SCL
PP1V8_TOUCH_TO_MAMBA_CONN
GUARD
PP1V8_MESA_CONN
MESA_TO_AOP_FDINT_CONN
I2C_MESA_TURTLE_SDA_CONN
I2C_MESA_TURTLE_SCL_CONN
MESA_TO_BOOST_EN_CONN
MESA_TO_AP_INT_CONN
SPI_AP_TO_MESA_SCLK_CONN
SPI_AP_TO_MESA_MOSI_CONN
AOP_TO_MESA_BLANKING_EN_CONN
SPI_MESA_TO_AP_MISO_CONN
PP2V75_MAMBA_CONN
C3807
56PF
PP3801
P2MM-NSM
启动供电
PP_VDD_BOOST
MAMBA POWER
NOTE: OUTPUT IMPEDANCE MUST BE >0.005-OHM
IN ORDER TO MEET CAP ESR REQUIREMENT PER LDO SPEC.
VENDOR ALSO RECOMMENDS CIN = COUT FOR STABILITY
TI:353S00576
ST:353S00932
C3828
2.2UF
U3801
LP5907SNX-2.75
X2SON
VIN
VOUT
EN
GND EPAD
XW3801
SHORT-20L-0.05MM-SM
ROOM=PMU
OMIT
PP1V8_TOUCH
MAMBA_LDO_EN
PP2V75_MAMBA_CONN
C3823
10UF
C3812
220PF
R3805
0.00
PP1V8_TOUCH_TO_MAMBA_CONN
C3824
2.2UF
C3811
220PF
MAMBA DIGITAL I/O
FL3804
150OHM-25%-200MA-0.7DCR
MAMBA_TO_LCM_MDRIVE
MAMBA_TO_LCM_MDRIVE_CONN_MESA
C3805
56PF
NOSTUFF
C3806
56PF
FL3806
150OHM-25%-200MA-0.7DCR
MESA_TO_AOP_FDINT
MESA_TO_AOP_FDINT_CONN
C3826
100PF
#24543342: stuff 100pF

DISPLAY POWER
FL3912 240-OHM-25%-0.42A-0.31DCR 0201 ROOM=DISPLAY_B2B
PP5V7_LCM_AVDDH 显示屏+5.7V供电
PP5V7_LCM_AVDDH_CONN
C3901 10UF 20% 10V X5R-CERM 0402-8 ROOM=DISPLAY_B2B
C3929 10UF 20% 10V X5R-CERM 0402-8 ROOM=DISPLAY_B2B
C3933 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=DISPLAY_B2B
C3913 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3906 240-OHM-25%-0.42A-0.31DCR 0201 ROOM=DISPLAY_B2B
1.8V供电
PP1V8
PP1V8_LCM_CONN
C3934 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=DISPLAY_B2B
C3914 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
DISPLAY MIPI
L3901 65OHM-0.7-2GHZ-3.4OHM TAM0605 CRITICAL ROOM=DISPLAY_B2B
90_MIPI_AP_TO_LCM_CLK_P
90_MIPI_AP_TO_LCM_CLK_N
90_MIPI_AP_TO_LCM_CLK_CONN_P
90_MIPI_AP_TO_LCM_CLK_CONN_N
L3902 65OHM-0.7-2GHZ-3.4OHM TAM0605 CRITICAL ROOM=DISPLAY_B2B
90_MIPI_AP_TO_LCM_DATA0_P
90_MIPI_AP_TO_LCM_DATA0_N
90_MIPI_AP_TO_LCM_DATA0_CONN_P
90_MIPI_AP_TO_LCM_DATA0_CONN_N
L3903 65OHM-0.7-2GHZ-3.4OHM TAM0605 CRITICAL ROOM=DISPLAY_B2B
90_MIPI_AP_TO_LCM_DATA1_P
90_MIPI_AP_TO_LCM_DATA1_N
90_MIPI_AP_TO_LCM_DATA1_CONN_P
90_MIPI_AP_TO_LCM_DATA1_CONN_N
FL3913 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
显示供电开启
LCM_TO_CHESTNUT_PWR_EN
LCM_TO_CHESTNUT_PWR_EN_CONN
C3916 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
AP/TOUCH INTERFACE
FL3904 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
AP_TO_TOUCH_MAMBA_RESET_L
AP_TO_TOUCH_MAMBA_RESET_CONN_L
C3915 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3902 220PF 5% 10V C0G-CERM 01005 ROOM=MAMBA_MESA
To Display B2B
To Mamba/Mesa B2B
FL3908 240-OHM-25%-0.42A-0.31DCR 0201 ROOM=DISPLAY_B2B
显示屏、触摸-5.7V供电
PN5V7_LCM_MESON_AVDDN
PN5V7_LCM_MESON_AVDDN_CONN
C3909 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3909 240-OHM-25%-0.42A-0.31DCR 0201 ROOM=DISPLAY_B2B
PP5V7_MESON_AVDDH 触摸+5.7V供电
PP5V7_MESON_AVDDH_CONN
C3910 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3940 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=DISPLAY_B2B
FL3910 33-OHM-25%-1500MA 0201 ROOM=DISPLAY_B2B
PP1V8_TOUCH
PP1V8_TOUCH_CONN
C3932 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=DISPLAY_B2B
C3911 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3911 240-OHM-25%-0.42A-0.31DCR 0201 ROOM=DISPLAY_B2B
PP5V1_TOUCH_VDDH 触摸+5.1V供电
PP5V1_TOUCH_VDDH_CONN
C3912 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3915 150OHM-25%-200MA-0.7DCR 01005
CPU到显示屏复位
AP_TO_LCM_RESET_L
AP_TO_LCM_RESET_CONN_L
R3901 100K 5% 1/32W MF 01005 ROOM=DISPLAY_B2B
C3917 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
R3915 10 5% 1/32W MF 01005 ROOM=DISPLAY_B2B
PMU_TO_LCM_PANICB
PMU_TO_LCM_PANICB_CONN
C3918 100PF 5% 16V NP0-C0G 01005 ROOM=DISPLAY_B2B
AOP/TOUCH INTERFACE
FL3922 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
CPU到触摸SPI总线片选信号
SPI_AP_TO_TOUCH_CS_L
SPI_AP_TO_TOUCH_CS_CONN_L
C3924 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
R3923 0.00 0% 1/32W MF 01005 ROOM=DISPLAY_B2B
SPI_AP_TO_TOUCH_SCLK
SPI_AP_TO_TOUCH_SCLK_CONN
NOSTUFF C3922 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
C3923 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3924 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
SPI_AP_TO_TOUCH_MOSI
SPI_AP_TO_TOUCH_MOSI_CONN
C3925 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3919 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
SPI_TOUCH_TO_AP_MISO
SPI_TOUCH_TO_AP_MISO_CONN
C3926 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3920 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
TOUCH_TO_AP_INT_L 触摸到CPU中断信号
TOUCH_TO_AP_INT_L_CONN
C3927 100PF 5% 16V NP0-C0G 01005 ROOM=DISPLAY_B2B
R3908 33.2 1% 1/32W MF 01005 ROOM=DISPLAY_B2B
AP_TO_CUMULUS_CLK32K
AP_TO_CUMULUS_CLK_32K_CONN
C3928 100PF 5% 16V NP0-C0G 01005 ROOM=DISPLAY_B2B
BACKLIGHT
FL3901 33-OHM-25%-1500MA 0201 ROOM=DISPLAY_B2B
PP_LCM_BL_ANODE
PP_LCM_BL_ANODE_CONN VOLTAGE=25.0V
NOSTUFF C3903 220PF 5% 50V C0G 0201 ROOM=DISPLAY_B2B
#26634069: nostuff C3903 to help desense
FL3902 33-OHM-25%-1500MA 0201 ROOM=DISPLAY_B2B
PP_LCM_BL_CAT1
PP_LCM_BL_CAT1_CONN
C3904 100PF 5% 35V NP0-C0G 01005 ROOM=DISPLAY_B2B
FL3903 33-OHM-25%-1500MA 0201 ROOM=DISPLAY_B2B
PP_LCM_BL_CAT2
PP_LCM_BL_CAT2_CONN
C3905 100PF 5% 35V NP0-C0G 01005 ROOM=DISPLAY_B2B
FL3916 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
LCM_TO_MANY_BSYNC
LCM_TO_MANY_BSYNC_CONN
C3919 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3917 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
触摸到CPU数据接收
UART_TOUCH_TO_AOP_RXD
UART_TOUCH_TO_AOP_RXD_CONN
C3920 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
FL3918 150OHM-25%-200MA-0.7DCR 01005 ROOM=DISPLAY_B2B
CPU到触摸数据发送
UART_AOP_TO_TOUCH_TXD
UART_AOP_TO_TOUCH_TXD_CONN
C3921 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DISPLAY_B2B
AC Coupling Caps
PP_VDD_MAIN
C3930 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3931 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3935 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3936 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
AC return path for LCM MIPI which is referenced to GND and VDD_MAIN
C3937 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3938 220PF 5% 10V C0G-CERM 01005 ROOM=DISPLAY_B2B
C3939 220PF 5% 10V C0G-CERM 01005
For placement "along the way" as we route from SOC to B2B.

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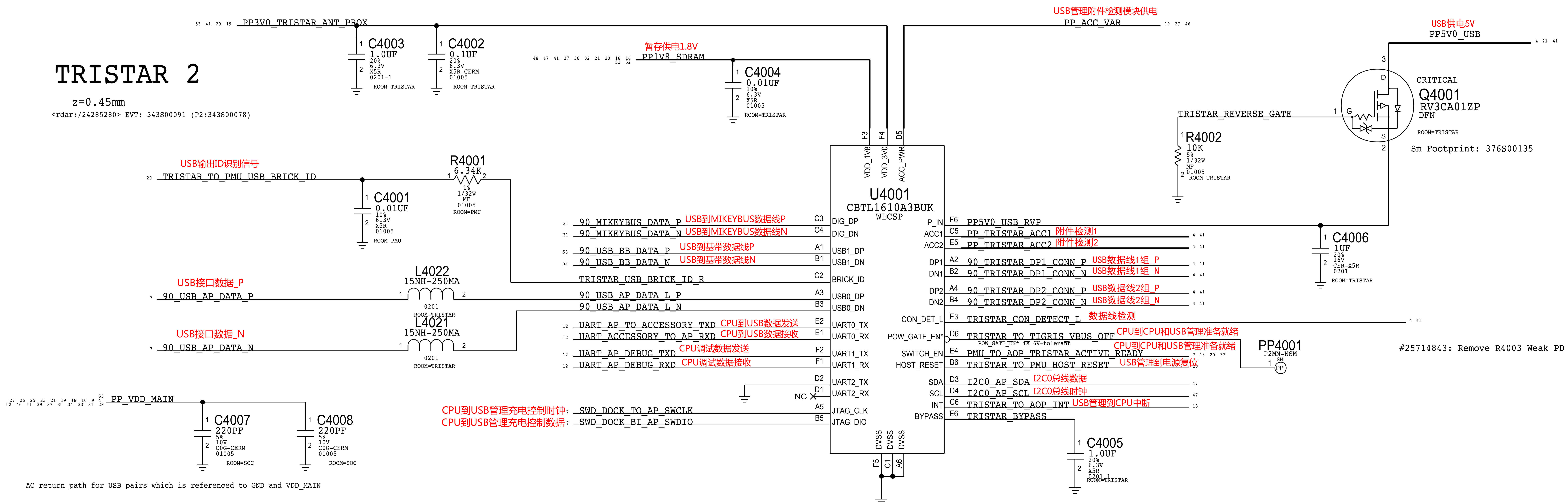
TRISTAR 2
z=0.45mm
<rdar://24285280> EVT: 343S00091 (P2:343S00078)
PP3V0_TRISTAR_ANT_PROX
C4003
1.0UF
C4002
0.1UF
暂存供电1.8V
PP1V8_SDRAM
C4004
0.01UF
USB管理附件检测模块供电
PP_ACC_VAR
USB供电5V
PP5V0_USB
CRITICAL
Q4001
RV3CA01ZP
DFN
TRISTAR_REVERSE_GATE
R4002
10K
Sm Footprint: 376S00135
USB输出ID识别信号
TRISTAR_TO_PMU_USB_BRICK_ID
R4001
6.34K
C4001
0.01UF
U4001
CBTL1610A3BUK
WLCSP
VDD_1V8
VDD_3V0
ACC_PWR
90_MIKEYBUS_DATA_P USB到MIKEYBUS数据线P
90_MIKEYBUS_DATA_N USB到MIKEYBUS数据线N
90_USB_BB_DATA_P USB到基带数据线P
90_USB_BB_DATA_N USB到基带数据线N
TRISTAR_USB_BRICK_ID_R
90_USB_AP_DATA_L_P
90_USB_AP_DATA_L_N
UART_AP_TO_ACCESSORY_TXD CPU到USB数据发送
UART_ACCESSORY_TO_AP_RXD CPU到USB数据接收
UART_AP_DEBUG_TXD CPU调试数据发送
UART_AP_DEBUG_RXD CPU调试数据接收
NC
CPU到USB管理充电控制时钟 SWD_DOCK_TO_AP_SWCLK
CPU到USB管理充电控制数据 SWD_DOCK_BI_AP_SWDIO
DIG_DP
DIG_DN
USB1_DP
USB1_DN
BRICK_ID
USB0_DP
USB0_DN
UART0_TX
UART0_RX
UART1_TX
UART1_RX
UART2_TX
UART2_RX
JTAG_CLK
JTAG_DIO
DVSS DVSS DVSS
P_IN
ACC1
ACC2
DP1
DN1
DP2
DN2
CON_DET_L
POW_GATE_EN*
SWITCH_EN
HOST_RESET
SDA
SCL
INT
BYPASS
PP5V0_USB_RVP
PP_TRISTAR_ACC1 附件检测1
PP_TRISTAR_ACC2 附件检测2
90_TRISTAR_DP1_CONN_P USB数据线1组_P
90_TRISTAR_DP1_CONN_N USB数据线1组_N
90_TRISTAR_DP2_CONN_P USB数据线2组_P
90_TRISTAR_DP2_CONN_N USB数据线2组_N
TRISTAR_CON_DETECT_L 数据线检测
TRISTAR_TO_TIGRIS_VBUS_OFF CPU到CPU和USB管理准备就绪
POW_GATE_EN* IS 6V-tolerant
PMU_TO_AOP_TRISTAR_ACTIVE_READY CPU到CPU和USB管理准备就绪
TRISTAR_TO_PMU_HOST_RESET USB管理到电源复位
I2C0_AP_SDA I2C0总线数据
I2C0_AP_SCL I2C0总线时钟
TRISTAR_TO_AOP_INT USB管理到CPU中断
TRISTAR_BYPASS
C4005
1.0UF
C4006
1UF
PP4001
P2MM-NSM
#25714843: Remove R4003 Weak PD
USB接口数据_P
90_USB_AP_DATA_P
L4022
15NH-250MA
USB接口数据_N
90_USB_AP_DATA_N
L4021
15NH-250MA
PP_VDD_MAIN
C4007
220PF
C4008
220PF
AC return path for USB pairs which is referenced to GND and VDD_MAIN


ANTENNA
Please loop in Matt Mow (Antenna Team) when changing these components!
FL4103 150OHM-25%-200MA-0.7DCR
PP3V0_LAT_CONN
C4134 220PF
FL4107 150OHM-25%-200MA-0.7DCR
PP3V0_TRISTAR_ANT_PROX
PP3V0_LAT1_CONN
VOLTAGE=3.0V
C4119 220PF
暂存供电1.8V
FL4108 150OHM-25%-200MA-0.7DCR
PP1V8_SDRAM
PP1V8_LAT_ARC_CONN
C4120 220PF
基带到天线时钟
R4109 0.00
BB_TO_LAT_ANT_SCLK
BB_TO_LAT_ANT_SCLK_CONN
C4121 33PF
基带到天线数据
R4110 0.00
BB_TO_LAT_ANT_DATA
BB_TO_LAT_ANT_DATA_CONN
C4122 33PF
ARC CONTROL
#26118161: Update FL4104 to 49.9ohm
R4104 49.9
I2C_HOMER_SCL
I2C_HOMER_SCL_CONN
C4135 56PF
FL4112 150OHM-25%-200MA-0.7DCR
I2C_HOMER_SDA
I2C_HOMER_SDA_CONN
C4136 56PF
ARC1
ARC1_TO_SOLENOID1_OUT_POS
ARC1_TO_SOLENOID1_OUT_NEG
C4101 220PF
C4102 220PF
Antenna GPIO
基带到天线接口1
FL4101 150OHM-25%-200MA-0.7DCR
BB_TO_LAT_GPO1
BB_TO_LAT_GPO1_CONN
C4110 56PF
基带到天线接口2
FL4102 150OHM-25%-200MA-0.7DCR
BB_TO_LAT_GPO2
BB_TO_LAT_GPO2_CONN
C4111 56PF
Per ANT Erdinc, change to cap
LOWER MIC1/4
FL4116 150OHM-25%-200MA-0.7DCR
LOWERMIC1_TO_CODEC_AIN1_P
LOWERMIC1_TO_CODEC_AIN1_CONN_POS
C4128 56PF
FL4117 150OHM-25%-200MA-0.7DCR
LOWERMIC1_TO_CODEC_AIN1_N
LOWERMIC1_TO_CODEC_AIN1_CONN_NEG
C4129 56PF
FL4118 150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_LOWERMIC1_BIAS
PP_CODEC_TO_LOWERMIC1_BIAS_CONN
C4130 220PF
FL4119 150OHM-25%-200MA-0.7DCR
LOWERMIC4_TO_CODEC_AIN2_P
LOWERMIC4_TO_CODEC_AIN2_CONN_POS
C4131 56PF
FL4120 150OHM-25%-200MA-0.7DCR
LOWERMIC4_TO_CODEC_AIN2_N
LOWERMIC4_TO_CODEC_AIN2_CONN_NEG
C4132 56PF
FL4121 150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_LOWERMIC4_BIAS
PP_CODEC_TO_LOWERMIC4_BIAS_CONN
C4133 220PF
SPEAKER1
FL4114 150OHM-25%-200MA-0.7DCR
SPEAKER_TO_SPEAKERAMP1_VSENSE_P
SPEAKER_TO_SPEAKERAMP1_VSENSE_CONN_POS
C4126 220PF
FL4115 150OHM-25%-200MA-0.7DCR
SPEAKER_TO_SPEAKERAMP1_VSENSE_N
SPEAKER_TO_SPEAKERAMP1_VSENSE_CONN_NEG
C4127 220PF
SPEAKERAMP1_TO_SPEAKER_OUT_POS
C4103 220PF
SPEAKERAMP1_TO_SPEAKER_OUT_NEG
C4104 220PF
VOLTAGE=14.0V
PP5V0_USB
C4109 0.1UF
C4105 0.1UF
C4106 0.1UF
C4107 330PF
C4108 330PF
J4101 245857 F-ST-SM
#25429221:Carrier Dock flex to add +1 ACC2 pin
SPEAKERAMP1_TO_SPEAKER_OUT_NEG
SPEAKER_TO_SPEAKERAMP1_VSENSE_CONN_NEG
BB_TO_LAT_GPO3_CONN
LOWERMIC4_TO_CODEC_AIN2_CONN_NEG
PP_CODEC_TO_LOWERMIC4_BIAS_CONN
PP_TRISTAR_ACC1_CONN
#22499940:Change Net Name to POS/NEG
90_TRISTAR_DP2_CONN_P
90_TRISTAR_DP2_CONN_N
90_TRISTAR_DP1_CONN_N
90_TRISTAR_DP1_CONN_P
MIKEYBUS_REFERENCE
LOWERMIC1_TO_CODEC_AIN1_CONN_POS
LOWERMIC1_TO_CODEC_BIAS_FILT_RET
I2C_MIC1_SDA_CONN
I2C_MIC1_SCL_CONN
ARC1_TO_SOLENOID1_OUT_POS
SOLENOID1_TO_ARC1_VSENSE_POS
PP3V0_LAT_CONN
SPEAKER_TO_SPEAKERAMP1_VSENSE_CONN_POS
SPEAKERAMP1_TO_SPEAKER_OUT_POS
PP_TRISTAR_ACC2_CONN
BB_TO_LAT_GPO2_CONN
LOWERMIC4_TO_CODEC_AIN2_CONN_POS
LOWERMIC4_TO_CODEC_BIAS_FILT_RET
BB_TO_LAT_GPO1_CONN
BB_TO_LAT_ANT_SCLK_CONN
BB_TO_LAT_ANT_DATA_CONN
PP1V8_LAT_ARC_CONN
PP3V0_LAT1_CONN
LOWERMIC1_TO_CODEC_AIN1_CONN_NEG
PP_CODEC_TO_LOWERMIC1_BIAS_CONN
TRISTAR_CON_DETECT_CONN_L
ARC1_TO_SOLENOID1_OUT_NEG
ARC1_TO_SOLENOID1_OUT_POS
SOLENOID1_TO_ARC1_VSENSE_NEG
I2C_HOMER_SDA_CONN
I2C_HOMER_SCL_CONN
DOCK FLEX CONNECTOR
TRISTAR
R4102 1.00K
TRISTAR_CON_DETECT_L
TRISTAR_CON_DETECT_CONN_L
C4116 27PF
FL4105 10-OHM-1.1A
PP_TRISTAR_ACC1
PP_TRISTAR_ACC1_CONN
VOLTAGE=4.3V
C4117 100PF
FL4106 22-OHM-25%-1800MA
PP_TRISTAR_ACC2
PP_TRISTAR_ACC2_CONN
VOLTAGE=4.3V
#25098110: Decrease DCR
C4118 100PF
USB AC Coupling
AC return path for USB pairs which is referenced to GND and VDD_MAIN
PP_VDD_MAIN
C4143 220PF

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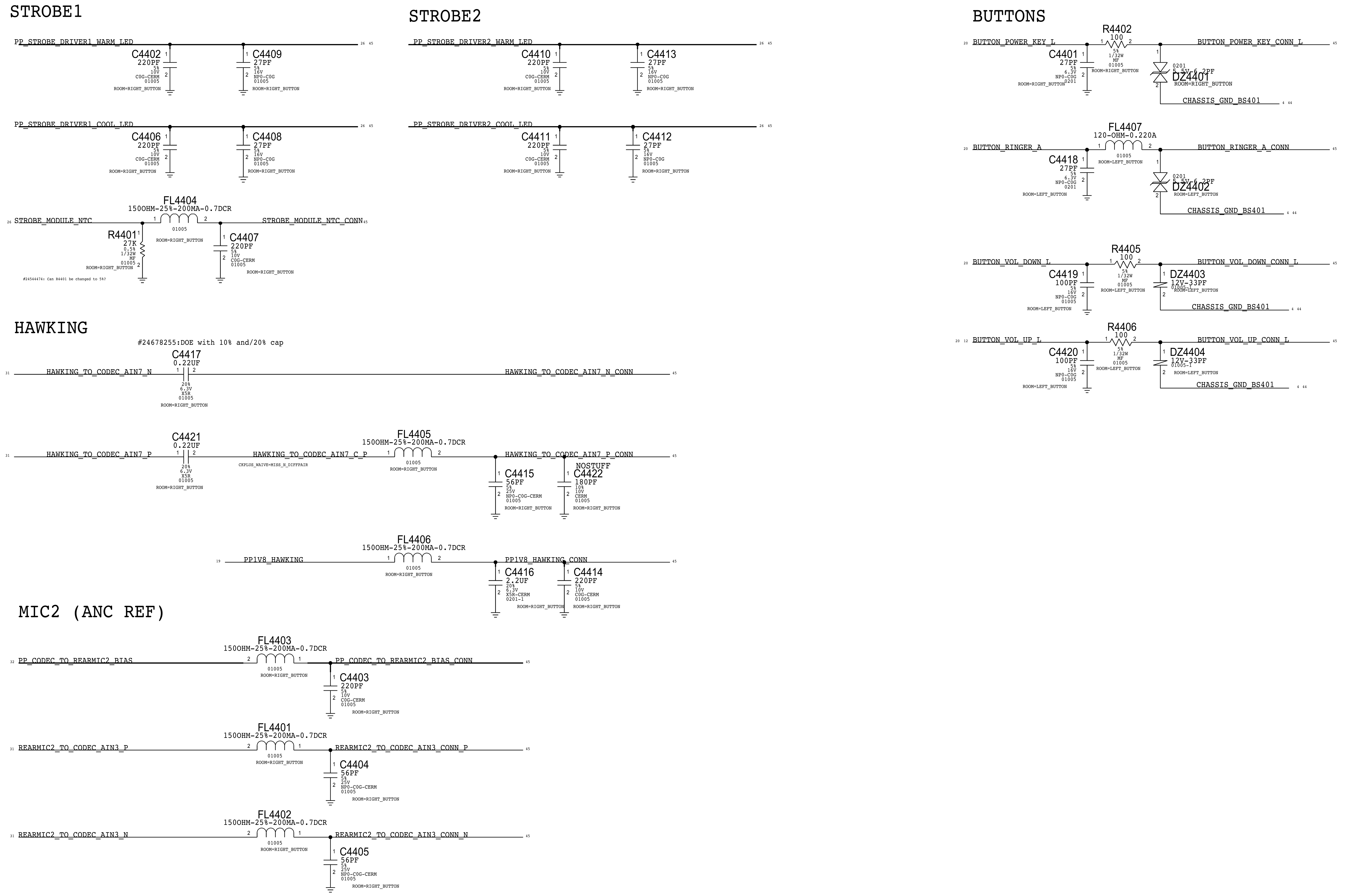
8 7 6 5 4 3 2 1


STROBE1
PP_STROBE_DRIVER1_WARM_LED
C4402 220PF
C4409 27PF
PP_STROBE_DRIVER1_COOL_LED
C4406 220PF
C4408 27PF
FL4404 150OHM-25%-200MA-0.7DCR
STROBE_MODULE_NTC
STROBE_MODULE_NTC_CONN
R4401 27K
C4407 220PF
STROBE2
PP_STROBE_DRIVER2_WARM_LED
C4410 220PF
C4413 27PF
PP_STROBE_DRIVER2_COOL_LED
C4411 220PF
C4412 27PF
HAWKING
#24678255:DOE with 10% and/20% cap
C4417 0.22UF
HAWKING_TO_CODEC_AIN7_N
HAWKING_TO_CODEC_AIN7_N_CONN
C4421 0.22UF
HAWKING_TO_CODEC_AIN7_P
HAWKING_TO_CODEC_AIN7_C_P
FL4405 150OHM-25%-200MA-0.7DCR
HAWKING_TO_CODEC_AIN7_P_CONN
C4415 56PF
NOSTUFF
C4422 180PF
FL4406 150OHM-25%-200MA-0.7DCR
PP1V8_HAWKING
PP1V8_HAWKING_CONN
C4416 2.2UF
C4414 220PF
MIC2 (ANC REF)
FL4403 150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_REARMIC2_BIAS
PP_CODEC_TO_REARMIC2_BIAS_CONN
C4403 220PF
FL4401 150OHM-25%-200MA-0.7DCR
REARMIC2_TO_CODEC_AIN3_P
REARMIC2_TO_CODEC_AIN3_CONN_P
C4404 56PF
FL4402 150OHM-25%-200MA-0.7DCR
REARMIC2_TO_CODEC_AIN3_N
REARMIC2_TO_CODEC_AIN3_CONN_N
C4405 56PF
BUTTONS
R4402 100
BUTTON_POWER_KEY_L
BUTTON_POWER_KEY_CONN_L
C4401 27PF
DZ4401
CHASSIS_GND_BS401
FL4407 120-OHM-0.220A
BUTTON_RINGER_A
BUTTON_RINGER_A_CONN
C4418 27PF
DZ4402
CHASSIS_GND_BS401
R4405 100
BUTTON_VOL_DOWN_L
BUTTON_VOL_DOWN_CONN_L
C4419 100PF
DZ4403 12V-33PF
CHASSIS_GND_BS401
R4406 100
BUTTON_VOL_UP_L
BUTTON_VOL_UP_CONN_L
C4420 100PF
DZ4404 12V-33PF
CHASSIS_GND_BS401

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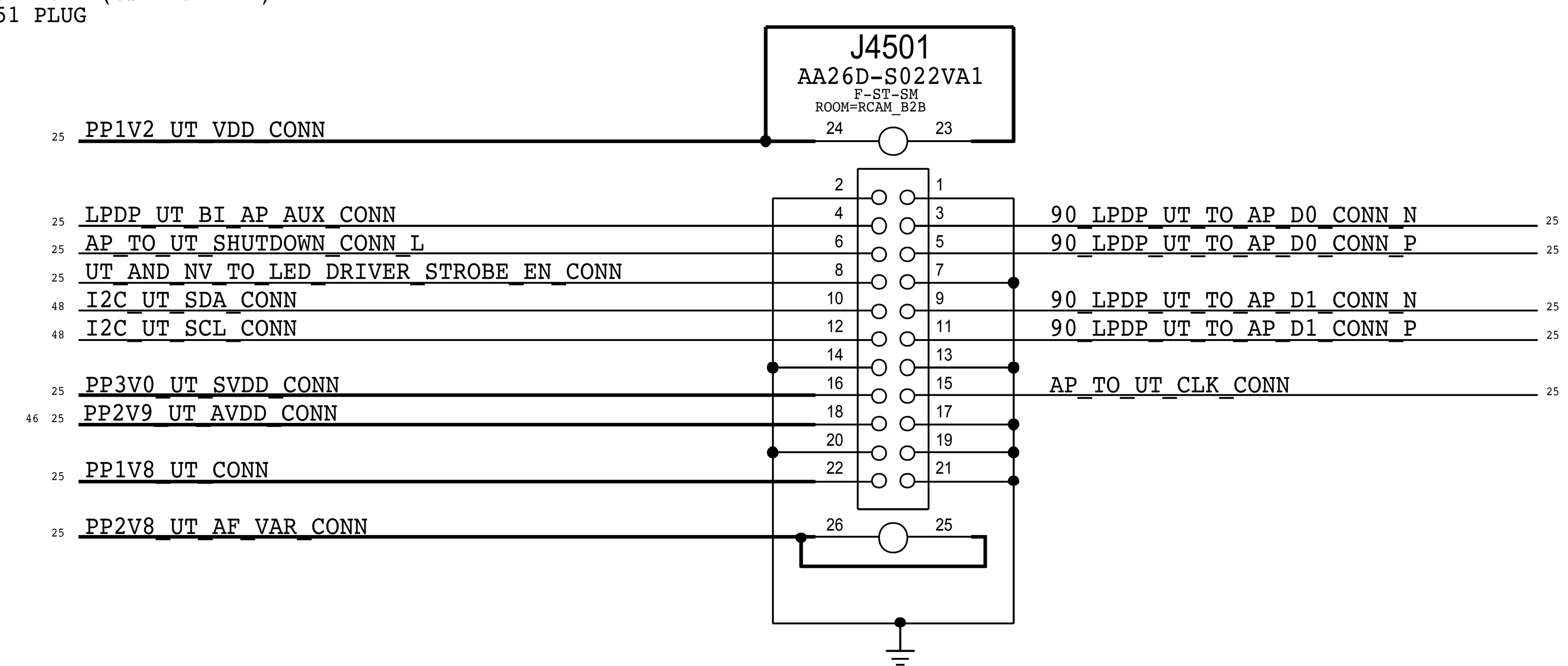
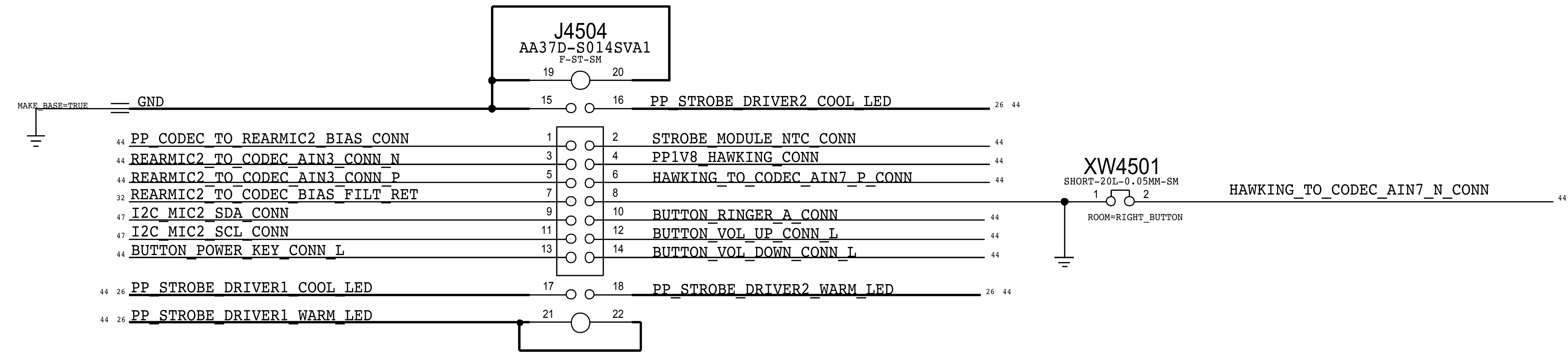
UTAH-C FLEX CONNECTOR

THIS ONE ---> 516S00152 RCPT (USED ON MLB)
516S00151 PLUG



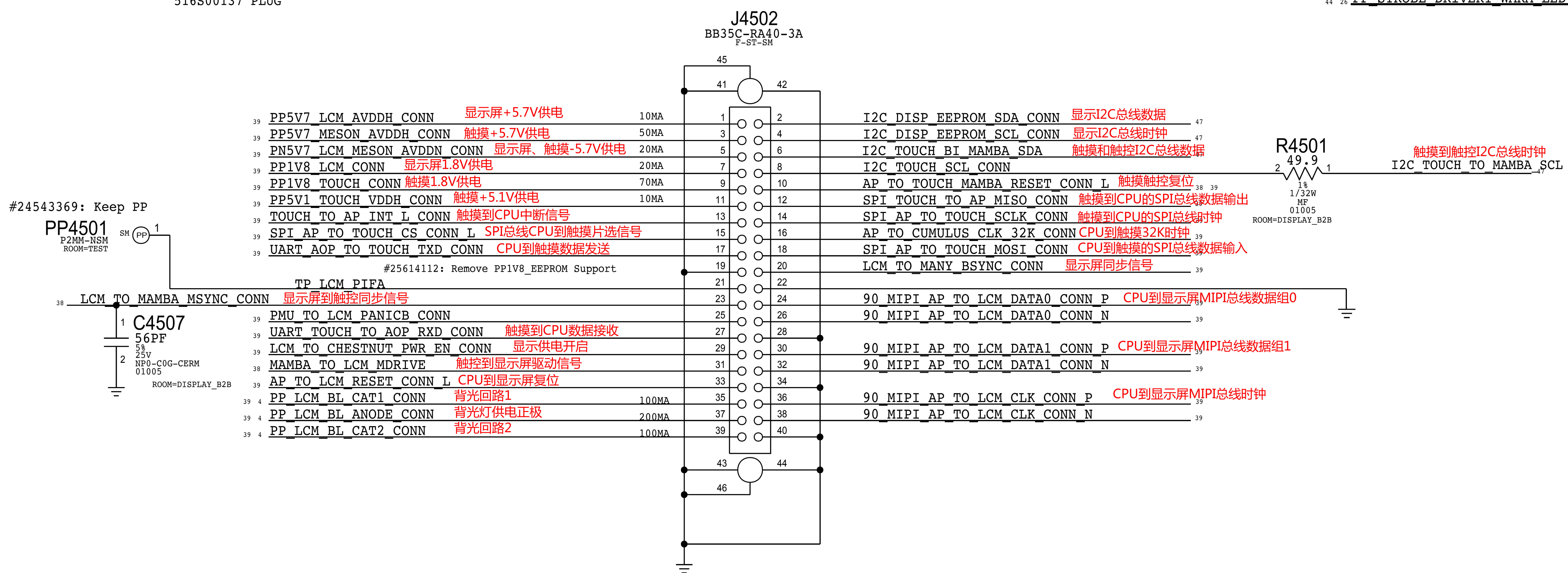
COMBINED BUTTON FLEX CONNECTOR

MLB APN: 516S00150
FLEX APN: 516S00149



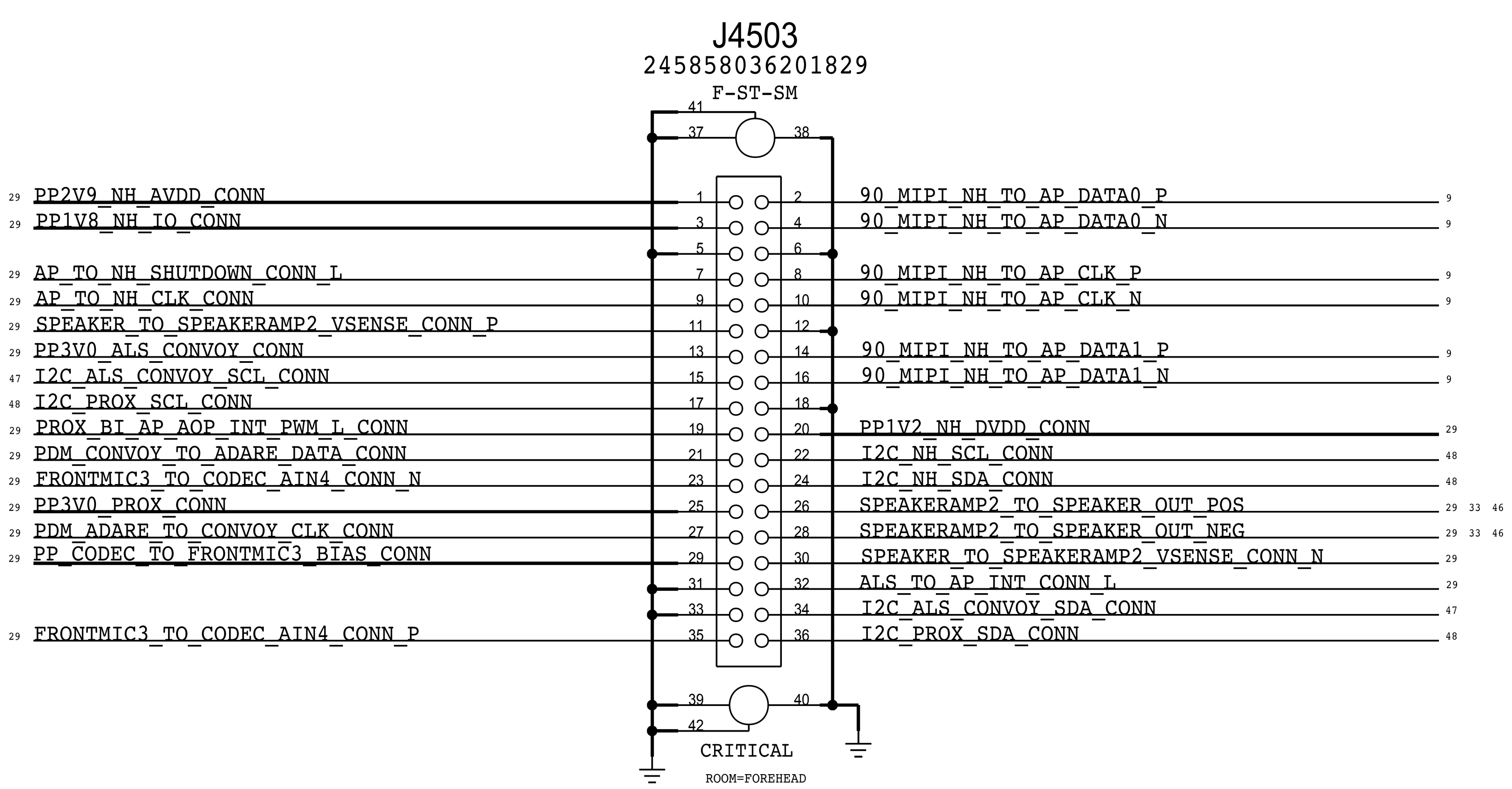
DISPLAY / TOUCH FLEX CONNECTOR

THIS ONE ---> 516S00138 RCPT (USED ON MLB)
516S00137 PLUG



FOREHEAD FLEX CONNECTOR

THIS ONE ---> 516S00146 RCPT (USED ON MLB)
516S00145 PLUG

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OTHER SMALL FORM FACTOR SPECIFIC PAGES:
4 - MECHANICAL
ISP I2C1
1.8V供电
PP1V8
R4603 2.2K 5% 1/32W MF 01005 ROOM=SOC
R4604 2.2K 5% 1/32W MF 01005 ROOM=SOC
I2C_ISP_NV_SCL
I2C_ISP_NV_SDA
MAKE_BASE=TRUE
#26682438:Move to Page 46
TOUCH I2C
PP1V8_TOUCH
NOSTUFF
R4711 10K 5% 1/32W MF 01005 ROOM=MAMBA_MESA
I2C_TOUCH_TO_MAMBA_SCL
NC_AP_LPDP_AUX2
NC Nets in Small FF
NC_90_LPDP_NV_TO_AP_D2_P
NC_90_LPDP_NV_TO_AP_D2_N
NC_90_LPDP_NV_TO_AP_D3_P
NC_90_LPDP_NV_TO_AP_D3_N
UT B2B
PP2V9_UT_AVDD_CONN
C2531 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=RCAM_B2B
C2507 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=RCAM_B2B
VDD_MAIN Cap
#26634069:D10x Only, 5x VDD_MAIN CAPS Change to 10UF/10V
PP_VDD_MAIN
Pg21 Pg34 Pg35 Pg37
C4601 10UF 20% VOLTAGE=10V X5R-CERM 0402-8 ROOM=BACKLIGHT
C2113 10UF 20% 10V X5R-CERM 0402-8 ROOM=CHARGER
C3424 10UF 20% 10V X5R-CERM 0402-8 ROOM=SPKAMP1
C3530 10UF 20% 10V X5R-CERM 0402-8 ROOM=ARC1
C3722 10UF 20% 10V X5R-CERM 0402-8 ROOM=CHESTNUT
Top Speaker Compass Coil
SPEAKERAMP2_TO_SPEAKER_OUT_POS
SPEAKERAMP2_TO_SPEAKER_OUT_NEG
R3332 910 1% 1/32W TK 01005 ROOM=SPKAMP2
R3333 910 1% 1/32W TK 01005 ROOM=SPKAMP2
NEG_COMPASS_COIL_COMP
POS_COMPASS_COIL_COMP
XW3333 SHORT-20L-0.05MM-SM NO_XNET_CONNECTION ROOM=MAMBA_MESA OMIT
C3332 220PF 5% 10V C0G-CERM 01005 ROOM=SPKAMP2
C3333 220PF 5% 10V C0G-CERM 01005 ROOM=SPKAMP2
#25742582,Add back C3531 in layout at ARC
PP_VDD_MAIN
C3531 10UF 20% 10V X5R-CERM 0402-8 ROOM=ARC1
#26104509:C3525 Change to 1UF 0201 in DVT
PP_VDD_MAIN
C3525 1.0UF 20% 6.3V X5R 0201-1 ROOM=ARC1
Dock B2B (Pg 41)
基带到天线接口3
BB_TO_LAT_GPO3
FL4604 150OHM-25%-200MA-0.7DCR 01005 ROOM=DOCK_B2B
BB_TO_LAT_GPO3_CONN
C4608 56PF 5% 25V NP0-C0G-CERM 01005 ROOM=DOCK_B2B
ACC Buck Caps
PP_ACC_VAR
USB管理附件检测模块供电
C2707 2.2UF 20% 6.3V X5R-CERM 0201-1 ROOM=TRISTAR

AP
I2C0
1.8V供电
PP1V8
R4701
4.02K
R4702
4.02K
I2C0总线时钟
I2C0_AP_SCL
I2C0总线数据
I2C0_AP_SDA
MAKE_BASE=TRUE
#24544434
NOSTUFF
C4701
56PF
NOSTUFF
C4702
56PF
NP0-C0G-CERM
01005
ROOM=SOC
I2C0_AP_SCL
I2C0_AP_SDA
I2C1
1.8V供电
PP1V8
R4703
4.02K
R4704
4.02K
I2C1_AP_SCL
I2C1_AP_SDA
D11/111 ONLY
#24544426
NOSTUFF
C4703
56PF
NOSTUFF
C4704
56PF
ROOM=PMU
I2C1_AP_SCL
I2C1_AP_SDA
I2C2
1.8V供电
PP1V8
R4705
2.2K
R4706
2.2K
I2C2_AP_SCL
I2C2_AP_SDA
ROOM=FOREHEAD
R4707
0.00
I2C_ALS_CONVOY_SCL_CONN
R4708
0.00
I2C_ALS_CONVOY_SDA_CONN
C4707
56PF
C4708
56PF
TO FOREHEAD FLEX
TOUCH
PP1V8_TOUCH
NOSTUFF
R4712
10K
ROOM=MAMBA_MESA
#26682438:Move to Page 46
NOTE:MAMBA I2C 2.2K PULL-UPS TO PP1V8_TOUCH INSIDE GALILEO
ADDING R3803, R3804 AS OPTION FOR TWEAKING VALUE
I2C_TOUCH_TO_MAMBA_SCL
I2C_TOUCH_BI_MAMBA_SDA
TO MAMBA / MESA FLEX
NOSTUFF
C4709
56PF
NOSTUFF
C4710
56PF
I2C_TOUCH_TO_MAMBA_SCL
I2C_TOUCH_BI_MAMBA_SDA
TO DISPLAY / TOUCH FLEX
C4711
56PF
C4712
56PF
ROOM=DISPLAY_B2B
HOMER
暂存供电1.8V
PP1V8_SDRAM
R4713
1.00K
R4714
1.00K
ROOM=HOMER
I2C_HOMER_SCL
I2C_HOMER_SDA
I2C5
1.8V供电
PP1V8
R4715
2.2K
R4716
2.2K
I2C5_SCL
I2C5_SDA
I2C3
1.8V供电
PP1V8
R4709
2.2K
R4710
2.2K
I2C3_AP_SCL
I2C3_AP_SDA
ROOM=DOCK
FL4730
150OHM-25%-200MA-0.7DCR
ckplus_waive=I2C_PULLUP
I2C_MIC1_SCL_CONN
#26633265:mitigate MIC1 undershoot
FL4729
100
CKPLUS_WAIVE=I2C_PULLUP
I2C_MIC1_SDA_CONN
TO DOCK FLEX
C4730
56PF
C4729
56PF
ROOM=DOCK_B2B
ROOM=RIGHT_BUTTON
FL4732
150OHM-25%-200MA-0.7DCR
I2C_MIC2_SCL_CONN
FL4731
150OHM-25%-200MA-0.7DCR
I2C_MIC2_SDA_CONN
TO COMBINED BUTTON FLEX
C4732
56PF
C4731
56PF
FL4742
150OHM-25%-200MA-0.7DCR
I2C_DISP_EEPROM_SCL_CONN
FL4741
150OHM-25%-200MA-0.7DCR
I2C_DISP_EEPROM_SDA_CONN
TO DISPLAY FLEX
C4742
56PF
C4741
56PF
ROOM=DISPLAY_B2B

AOP
I2C
暂存供电1.8V
PP1V8_SDRAM
R4801
2.2K
R4802
2.2K
C4807
0.1UF
U4802
MAX20312
WLP
VCC
IOVCC1
IOVCC2
GND
ROOM=SOC
#24544699: Support 1MHz
I2C_AOP_SCL
I2C_AOP_SDA
MAKE_BASE=TRUE
FL4815
150OHM-25%-200MA-0.7DCR
01005
ROOM=FOREHEAD
TO FCAM FLEX
I2C_PROX_SCL_CONN
I2C_PROX_SDA_CONN
CKPLUS_WAIVE=I2C_PULLUP
#24958320:Intentional R4815 Change
Reduce undershoot when Prox Driving
R4815
33.2
C4803
56PF
C4804
56PF
NP0-C0G-CERM
Intentional R4815 Change
PP1V8_MESA
R4803
4.7K
R4804
4.7K
ROOM=MAMBA_MESA
U4805
74LVC1G3157GX
X2SON6
NC
I2C_AOP_SCL_ISO
AOP_TO_MESA_I2C_ISO_EN
R4805
511K
U4806
74LVC1G3157GX
X2SON6
I2C_AOP_SDA_ISO
R4806
0.00
R4807
0.00
I2C_MESA_TURTLE_SCL_CONN
I2C_MESA_TURTLE_SDA_CONN
C4809
56PF
C4810
56PF
TO MAMBA/MESA FLEX
ISP
I2C0
1.8V供电
PP1V8
#24550735: ISP I2C0 PU
R4808
1.00K
R4809
1.00K
I2C_ISP_UT_SCL
I2C_ISP_UT_SDA
I2C1
See page 46
R4816
0.00
C4817
56PF
I2C_UT_SCL_CONN
R4817
0.00
C4816
56PF
I2C_UT_SDA_CONN
ROOM=RCAM_B2B
I2C2
R4810
2.2K
R4811
2.2K
I2C_ISP_NH_SCL
I2C_ISP_NH_SDA
R4812
0.00
C4812
56PF
I2C_NH_SCL_CONN
R4813
0.00
C4813
56PF
I2C_NH_SDA_CONN

D1x I2C Table

Bus	Device	Binary	7-bit Address	8-bit Address	Max Speed
ISP I2C0 1MHz	SALT LAKE (PRIMARY)	0010000X	0x10	0x20	1 MHz
	SALT LAKE (SECONDARY)	0110000X	0x30	0x60	1 MHz
	GRUNBERG (STANDARD)	0011100X	0x1C	0x38	1 MHz
	GRUNBERG (SPECIAL ABSOLUTE R/W)	0011101X	0x1D	0x3A	1 MHz
	GRUNBERG (SPECIAL DELTA READ)	0011110X	0x1E	0x3C	1 MHz
	STROBE 1	1100011X	0x63	0xC6	1 MHz
ISP I2C1 1MHz	LAS VEGAS (PRIMARY)	0100000X	0x20	0x40	1 MHz
	LAS VEGAS (SECONDARY)	0110000X	0x30	0x60	1 MHz
	EDWIN	1100011X	0x63	0xC6	400 kHz
	STROBE 2	0000110X	0x06	0x0C	1 MHz
ISP I2C2 400kHz	CONCORD	0010000X	0x10	0x20	1 MHz
AOP I2C 1MHz	ARC DRIVER 1	100000YX	0x41	0x82	1 MHz
	PROX	1011000X	0x58	0x80	1 MHz
	SPKAMP1 (L26)	100000YX	0x40	0x80	1 MHz
	TURTLE	0101100X	0x2c	0x58	400 kHz
	MESA	1100001X	0x61	0xC2	400 kHz
Touch I2C 400kHz	MAMBA	1100000X	0x60	0xC0	1 MHz
	MESON	1000000X	0x40	0x80	510 kHz
EEPROM I2C	EEPROM	1010001X	0x51	0xA2	400 kHz
AP I2C0 400kHz	BOOST	1110101X	0x75	0xEA	3.4 MHz
	TRISTAR	0011010X	0x1A	0x34	400 kHz
	BACKLIGHT 1	1100010X	0x62	0xC4	1 MHz
	CHESTNUT	0100111X	0x27	0x4E	400 kHz
AP I2C1 400kHz	ADELYN	1110100X	0x74	0xE8	400 kHz
	TIGRIS	1110101X	0x75	0xEA	400 kHz
	BACKLIGHT 2	1100010X	0x62	0xC4	1 MHz
AP I2C2 400kHz	ALS	0101001X	0x29	0x52	400 kHz
	CONVOY	0100001X	0x21	0x42	3.4 MHz
		0100010X	0x22	0x44	400 kHz
	SPKAMP 2 (L26)	10000ZYX	0x40	0x80	400 MHz
AP I2C3 700kHz	DOCK FLEX (MIC1)	1010100X	0x54	0xA8	1 MHz
	COMBINED BUTTON FLEX (MIC2)	1010100X	0x56	0xAC	1 MHz
	DISP EEPROM	1010001X	0x51	0xA2	400kHz
Homer I2C 1MHz	SCHRODINGER (STANDARD)	0001110X	0xE	0x1C	1 MHz
	SCHRODINGER (CAL REG'S)	0001111X	0xF	0x1D	1 MHz
	EEPROM (STANDARD)	1010000X	0x50	0xA0	
	EEPROM (CONFIG)	1010000X	0x50	0xA0	

8 7 6 5 4 3 2 1

D C B A

This page contains items which differ accross all MLB designs

PCIe lanes

Wifi/BT
Cellular
FF SPECIFIC
NFC
WIFI_MLB
RADIO_MLB
RADIO_MLB_FF
STOCKHOLM_MLB
I15
I2
I16
PP_VDD_MAIN 主供电
PP1V8_SDRAM 暂存供电1.8V
PMUGPIO_TO_WLAN_CLK32K 电源到WiFi时钟32.768KHz
PMU_TO_WLAN_REG_ON 电源到WiFi激活信号
PMU_TO_BT_REG_ON 电源到蓝牙激活信号
BT_TO_PMU_HOST_WAKE 蓝牙到电源唤醒信号
AP_TO_BT_WAKE CPU到蓝牙唤醒
CPU到WiFi的PCIE总线时钟
CPU到WiFi的PCIE总线数据发送
WiFi到CPU的PCIE总线数据接收
CPU到WiFi复位信号
WiFi到CPU的时钟请求
90_PCIE_AP_TO_WLAN_REFCLK_P
90_PCIE_AP_TO_WLAN_REFCLK_N
90_PCIE_AP_TO_WLAN_TXD_P
90_PCIE_AP_TO_WLAN_TXD_N
90_PCIE_WLAN_TO_AP_RXD_P
90_PCIE_WLAN_TO_AP_RXD_N
PCIE_AP_TO_WLAN_RESET_L
PCIE_WLAN_BI_AP_CLKREQ_L
WLAN_TO_PMU_HOST_WAKE WiFi到电源唤醒
AP_TO_WLAN_DEVICE_WAKE CPU到WiFi唤醒
UART_AP_TO_WLAN_TXD CPU到WiFi数据发送
UART_WLAN_TO_AP_RXD WiFi到CPU数据接收
UART_AP_TO_WLAN_RTS_L CPU到WiFi数据发送请求
UART_WLAN_TO_AP_CTS_L WiFi到CPU数据发送请求
UART_AP_TO_BT_TXD CPU到蓝牙数据发送
UART_BT_TO_AP_RXD 蓝牙到CPU数据接收
UART_AP_TO_BT_RTS_L CPU到蓝牙数据发送请求
UART_BT_TO_AP_CTS_L 蓝牙到CPU数据发送请求
AOP_TO_WLAN_CONTEXT_A CPU到WiFi数据接口A
AOP_TO_WLAN_CONTEXT_B CPU到WiFi数据接口B
I2S_AP_TO_BT_BCLK CPU到蓝牙位时钟
I2S_AP_TO_BT_LRCLK CPU到蓝牙左右声道时钟
I2S_BT_TO_AP_DIN 蓝牙到CPU数据输入
I2S_AP_TO_BT_DOUT CPU到蓝牙数据出
UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX
50_UAT_WLAN_5G_WEST
50_UAT_WLAN_2G_EAST
50_UAT_WLAN_5G_EAST
50_LAT_WLAN_G_1
50_LAT_WLAN_A_1
50_UAT2_M
PP_VDD_BOOST 启动供电
BBPMU_TO_PMU_AMUX1 基带电源到主电源开关信号1
BBPMU_TO_PMU_AMUX2
BBPMU_TO_PMU_AMUX3
AP_TO_BBPMU_RADIO_ON_L
Opposite polarity on Karoo -->
PMU_TO_BBPMU_RESET_L
AP_TO_BB_RESET_L
BB_TO_AP_RESET_DETECT_L
基带到闪光灯的来电提醒开启
BB_TO_STROBE_DRIVER_GSM_BURST_IND
AP_TO_BB_MESA_ON
AP_TO_BB_TIME_MARK CPU到基带时间标志
AP_TO_BB_COREDUMP
LCM_TO_MANY_BSYNC 显示屏同步信号
AP_TO_BB_IPC_GPIO1
CPU到基带参考时钟_P
CPU到基带参考时钟_N
90_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_AP_TO_BB_REFCLK_N
90_PCIE_AP_TO_BB_TXD_P
90_PCIE_AP_TO_BB_TXD_N
90_PCIE_BB_TO_AP_RXD_P
90_PCIE_BB_TO_AP_RXD_N
PCIE_AP_TO_BB_RESET_L
PCIE_BB_BI_AP_CLKREQ_L
BB_TO_PMU_PCIE_HOST_WAKE_L
UART_AOP_TO_BB_TXD
UART_BB_TO_AOP_RXD
I2S_BB_TO_AP_BCLK
I2S_BB_TO_AP_LRCLK
I2S_AP_TO_BB_DOUT
I2S_BB_TO_AP_DIN
UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX
NFC_SWP
NFC_SWP_MUX
AP_TO_ICEFALL_FW_DWLD_REQ
SE2_PRESENT
ICEFALL_LDO_ENABLE
CPU许多位置的SWD总线时钟
SWD_AP_TO_MANY_SWCLK
SWD_AOP_BI_BB_SWDIO CPU和基带的SWD总线数据
PMU_TO_BB_USB_VBUS_DETECT
90_USB_BB_DATA_P USB到基带数据线P
90_USB_BB_DATA_N USB到基带数据线N
50_UAT_WLAN_2G_WEST_PLEXER
50_UAT_LB_MLB_SOUTH
50_UAT_MB_HB_SOUTH
50_UUAT_LB_MLB_NORTH
50_UAT1_WEST
BB_TO_UAT_SCLK 基带到上天线时钟
BB_TO_UAT_DATA 基带到上天线数据
50_UAT1_TUNER
BB_BUFFER_GPO1
BB_BUFFER_GPO2
BB_TO_LAT_GPO1 基带到天线接口1
BB_TO_LAT_GPO2 基带到天线接口2
BB_TO_LAT_GPO3 基带到天线接口3
To UAT
To LAT
BB_TO_LAT_ANT_SCLK 基带到天线时钟
BB_TO_LAT_ANT_DATA 基带到天线数据
PP3V0_TRISTAR_ANT_PROX
PP1V8_SDRAM
BB_TO_LAT_ANT_SCLK
BB_TO_LAT_ANT_DATA
PP5304
PP5301
PP5302
PP5303
PMU_TO_GNSS_EN
UART_AP_TO_GNSS_TXD
UART_GNSS_TO_AP_RXD
UART_AP_TO_GNSS_RTS_L
UART_GNSS_TO_AP_CTS_L
GNSS_TO_PMU_HOST_WAKE
AP_TO_GNSS_TIME_MARK
PMU_TO_NFC_EN 电源到NFC开启
BB_TO_NFC_CLK 基带电源到NFC时钟
NFC_TO_BB_CLK_REQ NFC到基带电源时钟请求
AP_TO_NFC_FW_DWLD_REQ CPU到NFC中断请求
AP_TO_NFC_DEV_WAKE CPU到NFC唤醒
NFC_TO_PMU_HOST_WAKE NFC到电源唤醒
UART_AP_TO_NFC_TXD CPU到NFC数据发送
UART_NFC_TO_AP_RXD NFC到CPU数据接收
UART_AP_TO_NFC_RTS_L CPU到NFC数据发送请求
UART_NFC_TO_AP_CTS_L NFC到CPU数据发送请求
NFC_SWP
SE2_READY
SE2_PWR_REQ
SE2_PRESENT
NFC_SWP_MUX
ICEFALL_LDO_ENABLE

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
8	0006400877	ENGINEERING RELEASED	2016-06-14

D1X WIFI_MLB (PERENNIAL) FEBRUARY 1, 2016

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2	76	PERENNIAL
3	77	WIFI FRONT-END

BOM OPTIONS:

D10_JP:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0648	1	CAP,CER,0.3PF,+/-0.05,01005	C7705_RF	CRITICAL	D10_JP
152S00029	1	IND,1.1NH,UH-Q,01005	R7703_RF	CRITICAL	D10_JP
131S0893	1	CAP,CER,0.2PF,+/-0.05,01005	C7706_RF	CRITICAL	D10_JP
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7711_RF	CRITICAL	D10_JP
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7702_RF	CRITICAL	D10_JP
152S1980	1	IND,1.0NH,UH-Q,01005	R7704_RF	CRITICAL	D10_JP
131S0404	1	CAP,3.9PF,+/-1.0PF,01005	R6711_RF	CRITICAL	D10_JP
152S2061	1	IND,7.5NH,UH-Q,01005	C6729_RF	CRITICAL	D10_JP
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7700_RF	CRITICAL	D10_JP
152S2043	1	IND,6.2NH,UH-Q,01005	C7702_RF	CRITICAL	D10_JP
152S1998	1	IND, 0.8NH,UH-Q,01005	L7700_RF	CRITICAL	D10_JP
152S2043	1	IND,6.2NH,UH-Q,01005	L7701_RF	CRITICAL	D10_JP
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7701_RF	CRITICAL	D10_JP

NOSTUFF:C7729_RF,C7711_RF,C7709_RF,C7710_RF, C7707_RF, C7708_RF
C7700_RF, C7703_RF,C7704_RF

D10_ROW:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0648	1	CAP,CER,0.3PF,+/-0.05,01005	C7705_RF	CRITICAL	D10_ROW
152S00029	1	IND,1.1NH,UH-Q,01005	R7703_RF	CRITICAL	D10_ROW
131S0893	1	CAP,CER,0.2PF,+/-0.05,01005	C7706_RF	CRITICAL	D10_ROW
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	R7711_RF	CRITICAL	D10_ROW
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	R7702_RF	CRITICAL	D10_ROW
152S1988	TRUE 1	IND,2.4NH,UH-Q,01005	R7704_RF	CRITICAL	D10_ROW
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	R6711_RF	CRITICAL	D10_ROW
152S1853	TRUE 1	IND,9.1NH,UH-Q,01005	C6729_RF	CRITICAL	D10_ROW
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	R7700_RF	CRITICAL	D10_ROW
152S2043	1	IND,6.2NH,UH-Q,01005	C7702_RF	CRITICAL	D10_ROW
152S1998	1	IND, 0.8NH,UH-Q,01005	L7700_RF	CRITICAL	D10_ROW
152S2043	TRUE 1	IND,6.2NH,UH-Q,01005	L7701_RF	CRITICAL	D10_ROW
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7701_RF	CRITICAL	D10_ROW

NOSTUFF:C7729_RF,C7711_RF,C7709_RF,C7710_RF, C7707_RF, C7708_RF
C7700_RF, C7703_RF,C7704_RF

D101_WIFI:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0648	1	CAP,CER,0.3PF,+/-0.05,01005	C7705_RF	CRITICAL	D101
152S00029	1	IND,1.1NH,UH-Q,01005	R7703_RF	CRITICAL	D101
131S0893	1	CAP,CER,0.2PF,+/-0.05,01005	C7706_RF	CRITICAL	D101
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7711_RF	CRITICAL	D101
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7702_RF	CRITICAL	D101
152S1988	TRUE 1	IND,2.4NH,UH-Q,01005	R7704_RF	CRITICAL	D101
131S0648	TRUE 1	CAP,CER,0.3PF,+/-0.05PF,01005	C7708_RF	CRITICAL	D101
118S0724	TRUE 1	RES,MF, 0 OHM,1/20W, 0201	R6711_RF	CRITICAL	D101
152S2054	TRUE 1	IND,9.1NH,UH-Q,0201	C6729_RF	CRITICAL	D101
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7700_RF	CRITICAL	D101
152S2043	TRUE 1	IND,6.2NH,UH-Q,01005	C7702_RF	CRITICAL	D101
152S1998	1	IND, 0.8NH,UH-Q,01005	L7700_RF	CRITICAL	D101
152S2043	1	IND,6.2NH,UH-Q,01005	L7701_RF	CRITICAL	D101
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7701_RF	CRITICAL	D101

NOSTUFF: C7729_RF,C7711_RF,C7709_RF,C7710_RF, C7707_RF
C7700_RF, C7703_RF,C7704_RF

D11_JP:

PART#	TRUE QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00273	TRUE 1	IND,0.6NH,UH-Q,01005	R7703_RF	CRITICAL	D11_JP
152S1976	TRUE 1	IND,0.7NH,UH-Q,01005	R7711_RF	CRITICAL	D11_JP
131S0400	TRUE 1	CAP,CER,3.5PF+/-0.1,01005	R7702_RF	CRITICAL	D11_JP
152S1986	TRUE 1	IND,FILM,2.2NH,UH-Q,01005	R7704_RF	CRITICAL	D11_JP
131S0648	TRUE 1	CAP,CER,0.3PF,+/-0.05PF,01005	C7708_RF	CRITICAL	D11_JP
131S0593	TRUE 1	CAP,3.9PF,+/-1.0PF,0201,HI-Q	R6711_RF	CRITICAL	D11_JP
152S2055	TRUE 1	IND,7.5NH,UH-Q,0201	C6729_RF	CRITICAL	D11_JP
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	R7700_RF	CRITICAL	D11_JP
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	L7700_RF	CRITICAL	D11_JP
152S1853	TRUE 1	IND,9.1NH,UH-Q,01005	L7701_RF	CRITICAL	D11_JP
117S0161	TRUE 1	RES,MF,0 OHM,1/32W,01005	R7701_RF	CRITICAL	D11_JP
131S0893	TRUE 1	CAP,CER,0.2PF,+/-0.05PF,01005	C7705_RF	CRITICAL	D11_JP
131S0893	1	CAP,CER,0.2PF,+/-0.05PF,01005	C7729_RF	CRITICAL	D11_JP

NOSTUFF:C7706_RF, C7711_RF,C7709_RF,C7710_RF, C7707_RF
C7700_RF,C7702_RF, C7703_RF,C7704_RF

D11_ROW:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00273	1	IND,0.6NH,UH-Q,01005	R7703_RF	CRITICAL	D11_ROW
152S1976	1	IND,0.7NH,UH-Q,01005	R7711_RF	CRITICAL	D11_ROW
131S0400	1	CAP,CER,3.5PF+/-0.1,01005	R7702_RF	CRITICAL	D11_ROW
152S1986	TRUE 1	IND,FILM,2.2NH,UH-Q,01005	R7704_RF	CRITICAL	D11_ROW
118S0724	TRUE 1	RES,MF,0 OHM,1/20W,0201	R6711_RF	CRITICAL	D11_ROW
152S2054	TRUE 1	IND,9.1NH,UH-Q,0201	C6729_RF	CRITICAL	D11_ROW
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7700_RF	CRITICAL	D11_ROW
117S0161	1	RES,MF,0 OHM,1/32W,01005	L7700_RF	CRITICAL	D11_ROW
152S1853	1	IND,9.1NH,UH-Q,01005	L7701_RF	CRITICAL	D11_ROW
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7701_RF	CRITICAL	D11_ROW
131S0893	1	CAP,CER,0.2PF,+/-0.05PF,01005	C7705_RF	CRITICAL	D11_ROW
131S0893	1	CAP,CER,0.2PF,+/-0.05PF,01005	C7729_RF	CRITICAL	D11_ROW

NOSTUFF:C7706_RF,C7711_RF,C7709_RF,C7710_RF, C7707_RF, C7708_RF
C7700_RF,C7702_RF, C7703_RF,C7704_RF

D111_WIFI:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S00273	1	IND,0.6NH,UH-Q,01005	R7703_RF	CRITICAL	D111
152S1976	1	IND,0.7NH,UH-Q,01005	R7711_RF	CRITICAL	D111
131S0400	1	CAP,CER,3.5PF+/-0.1,01005	R7702_RF	CRITICAL	D111
152S1986	TRUE 1	IND,FILM,2.2NH,UH-Q,01005	R7704_RF	CRITICAL	D111
118S0724	TRUE 1	RES,MF, 0 OHM,1/20W, 0201	R6711_RF	CRITICAL	D111
152S2054	TRUE 1	IND,9.1NH,UH-Q,0201	C6729_RF	CRITICAL	D111
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7700_RF	CRITICAL	D111
117S0161	1	RES,MF,0 OHM,1/32W,01005	L7700_RF	CRITICAL	D111
152S1853	1	IND,9.1NH,UH-Q,01005	L7701_RF	CRITICAL	D111
117S0161	1	RES,MF,0 OHM,1/32W,01005	R7701_RF	CRITICAL	D111
131S0893	1	CAP,CER,0.2PF,+/-0.05PF,01005	C7705_RF	CRITICAL	D111
131S0893	1	CAP,CER,0.2PF,+/-0.05PF,01005	C7729_RF	CRITICAL	D111

NOSTUFF:C7706_RF,C7711_RF,C7709_RF,C7710_RF, C7707_RF, C7708_RF
C7700_RF,C7702_RF, C7703_RF,C7704_RF

POWER

55 53 IN PP_VDD_MAIN 主供电
55 53 IN PP1V8_SDRAM 暂存供电1.8V

CLOCKS

55 53 IN PMUGPIO_TO_WLAN_CLK32K

CONTROL

55 53 IN PMU_TO_WLAN_REG_ON
55 53 IN PMU_TO_BT_REG_ON
55 53 OUT BT_TO_PMU_HOST_WAKE
55 53 IN AP_TO_BT_WAKE

WLAN PCIE

55 53 IN 90_PCIE_AP_TO_WLAN_REFCLK_P
55 53 IN 90_PCIE_AP_TO_WLAN_REFCLK_N
55 53 IN 90_PCIE_AP_TO_WLAN_TXD_P
55 53 IN 90_PCIE_AP_TO_WLAN_TXD_N
55 53 OUT 90_PCIE_WLAN_TO_AP_RXD_P
55 53 OUT 90_PCIE_WLAN_TO_AP_RXD_N
55 53 IN PCIE_AP_TO_WLAN_RESET_L
2 IO PCIE_WLAN_BI_AP_CLKREQ_L
55 53 OUT WLAN_TO_PMU_HOST_WAKE
55 53 IN AP_TO_WLAN_DEVICE_WAKE

WLAN UART

55 53 OUT UART_WLAN_TO_AP_RXD
55 53 IN UART_AP_TO_WLAN_TXD
55 53 OUT UART_WLAN_TO_AP_CTS_L
55 53 IN UART_AP_TO_WLAN_RTS_L

BLUETOOTH UART

55 53 IN UART_AP_TO_BT_TXD
55 53 OUT UART_BT_TO_AP_RXD
55 53 IN UART_AP_TO_BT_RTS_L
55 53 OUT UART_BT_TO_AP_CTS_L

AOP

55 53 IN AOP_TO_WLAN_CONTEXT_A
55 53 IN AOP_TO_WLAN_CONTEXT_B

AUDIO

55 53 IN I2S_AP_TO_BT_BCLK
55 53 IN I2S_AP_TO_BT_LRCLK
55 53 OUT I2S_BT_TO_AP_DIN
55 53 IN I2S_AP_TO_BT_DOUT

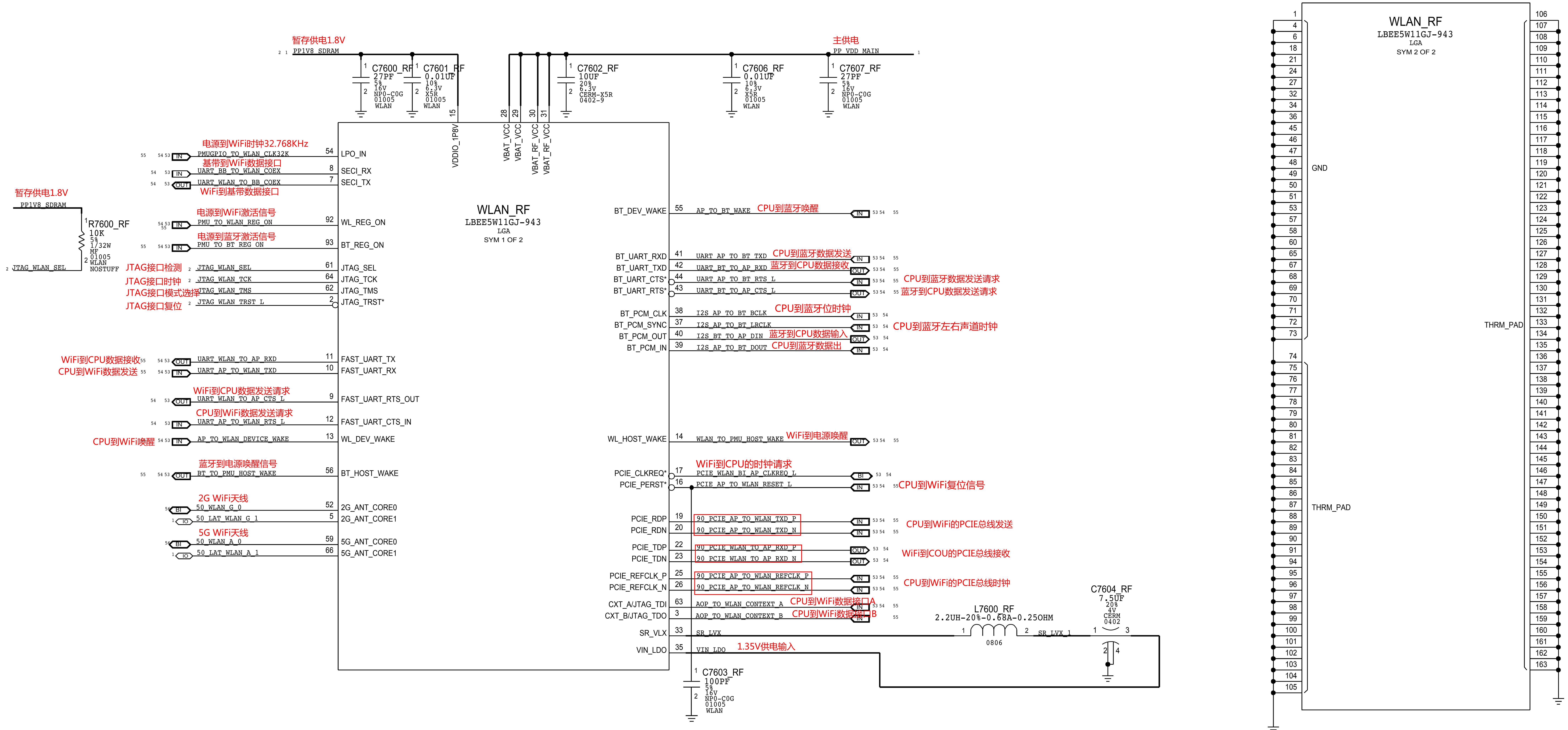
COEX

55 53 IN UART_BB_TO_WLAN_COEX
55 53 OUT UART_WLAN_TO_BB_COEX

ANTENNA

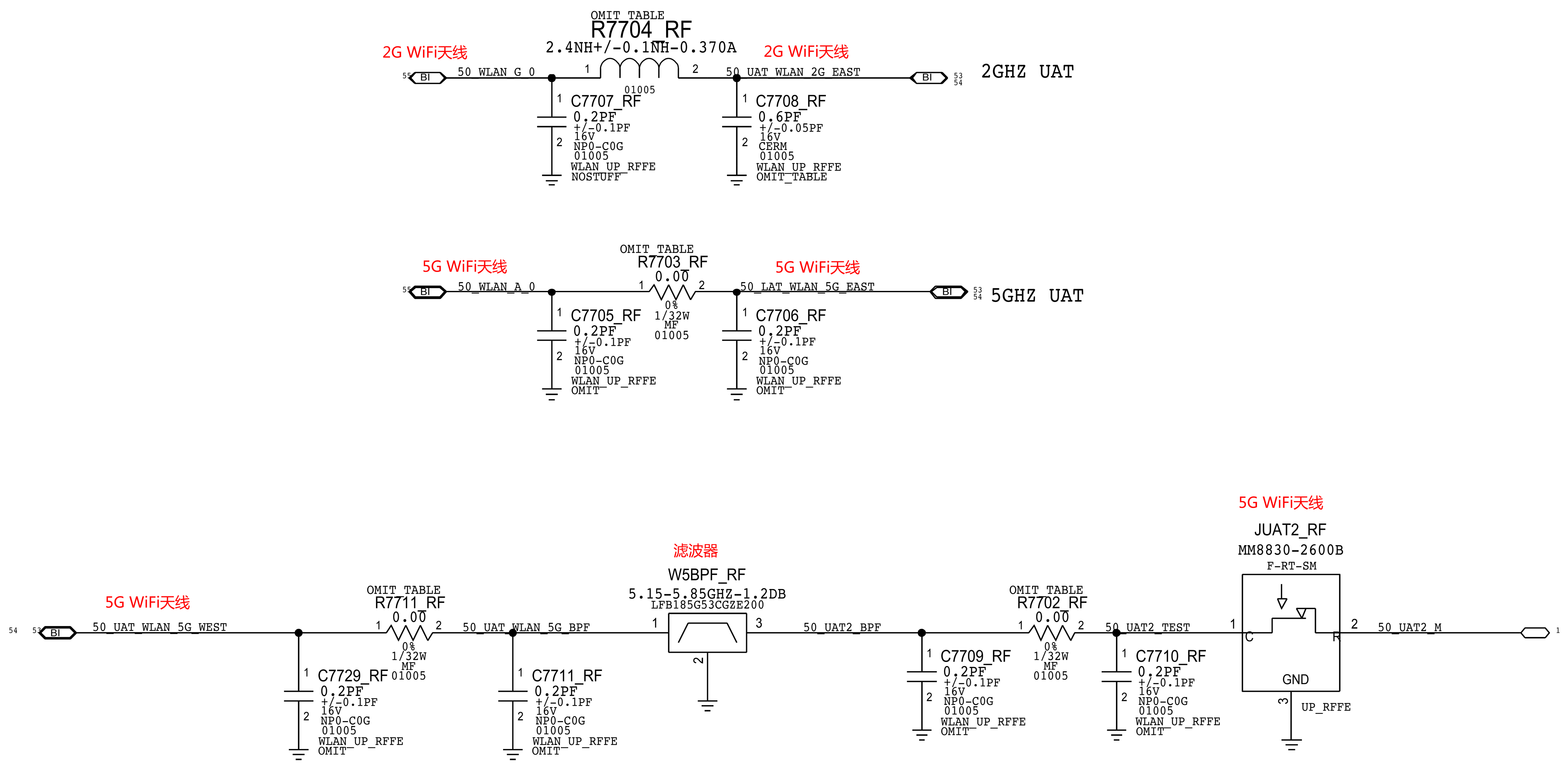
3 IO 50_UAT_WLAN_5G_WEST
3 IO 50_UAT_WLAN_2G_EAST
3 IO 50_LAT_WLAN_5G_EAST
2 IO 50_LAT_WLAN_G_1
2 IO 50_LAT_WLAN_A_1
3 IO 50_UAT2_M

WIFI/BT



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S00201	339S00199	ALTERNATE	WLAN_RF	ALT WIFI/BT MODULE

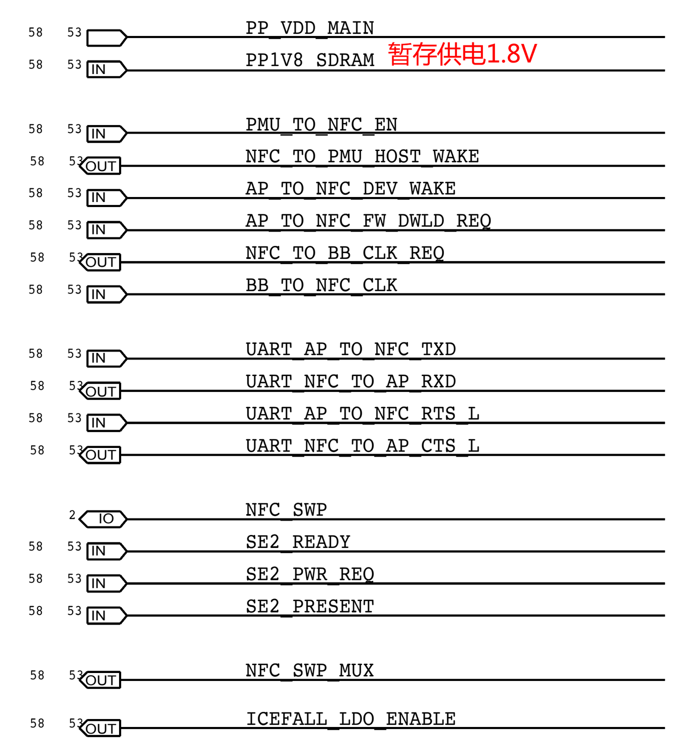
WIFI UPPER ANTENNA FEEDS



1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD
8	0006400877	ENGINEERING RELEASED	

STOCKHOLM MLB
JUNE 9, 2016



ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	REFERENCE DESIGNATOR(S)	DESCRIPTION	BOM OPTION
132S0436	132S0400	C7504_RF	0.22UF 20% 6.3V 01005	?

BOM OPTIONS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S0883	1	220PF, 0201 2% 50V	C7514_RF	D101
131S00055	1	22PF, 0201 2% 50V	C7512_RF	D101
131S00117	1	120PF, 0201 2% 50V	C7518_RF	D101
131S00026	1	820PF, 0201 2% 50V	C7516_RF	D101
131S0883	1	220PF, 0201 2% 50V	C7514_RF	D10_JP
131S00055	1	22PF, 0201 2% 50V	C7512_RF	D10_JP
131S00019	1	150PF, 0201 2% 50V	C7518_RF	D10_JP
131S0825	1	560PF, 0201 2% 50V	C7516_RF	D10_JP
131S0883	1	220PF, 0201 2% 50V	C7514_RF	D10_ROW
131S00055	1	22PF, 0201 2% 50V	C7512_RF	D10_ROW
131S00117	1	120PF, 0201 2% 50V	C7518_RF	D10_ROW
131S00026	1	820PF, 0201 2% 50V	C7516_RF	D10_ROW

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
131S00081	1	270PF, 0201 2% 25V	C7514_RF	D111
131S00118	1	180PF, 0201 2% 50V	C7518_RF	D111
131S00033	1	680PF, 0201 2% 50V	C7516_RF	D111
131S00081	1	270PF, 0201 2% 25V	C7514_RF	D11_JP
131S0731	1	100PF, 0201 2% 50V	C7518_RF	D11_JP
131S00033	1	680PF, 0201 2% 50V	C7516_RF	D11_JP
131S00081	1	270PF, 0201 2% 25V	C7514_RF	D11_ROW
131S00118	1	180PF, 0201 2% 50V	C7518_RF	D11_ROW
131S00033	1	680PF, 0201 2% 50V	C7516_RF	D11_ROW

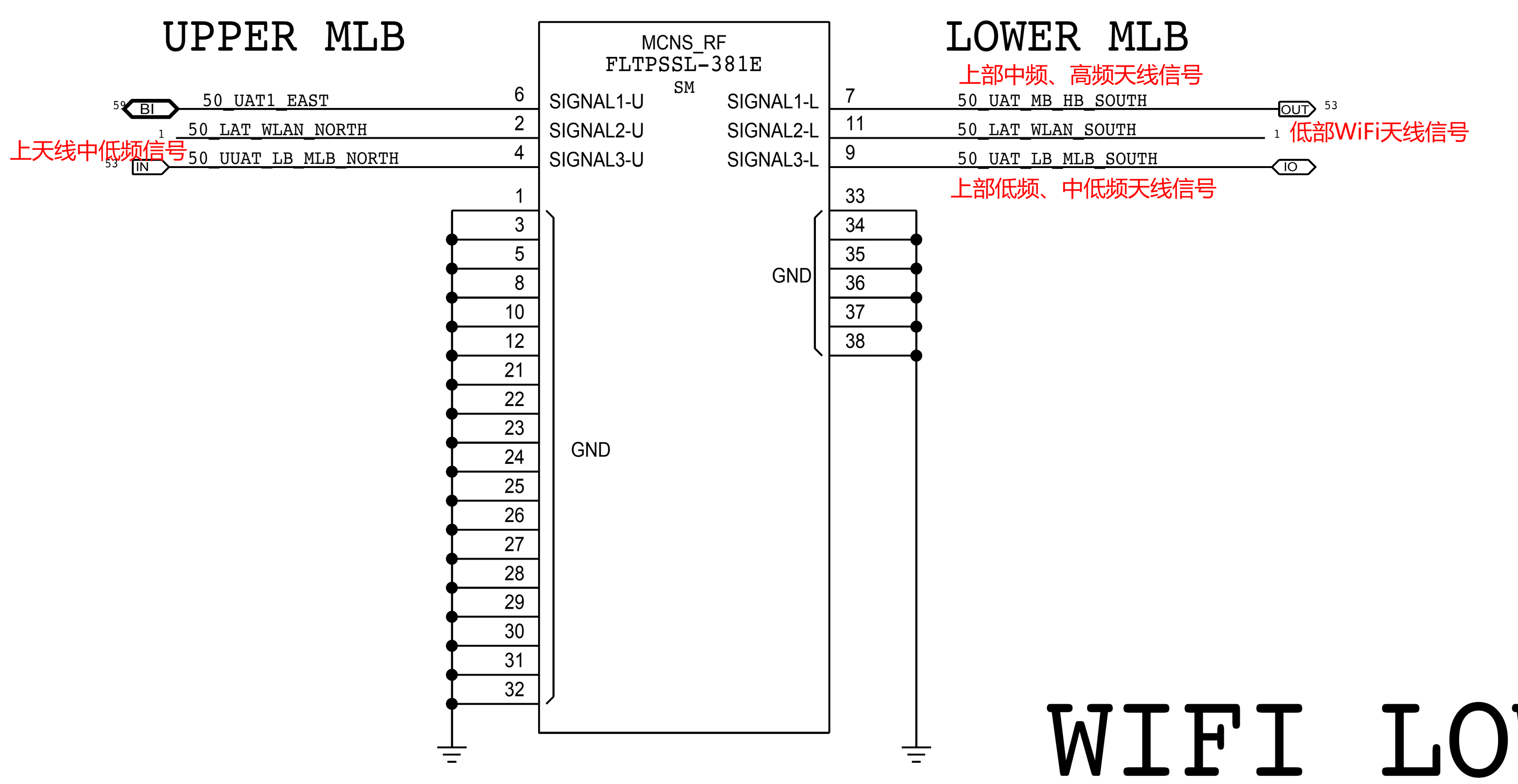
STOCKHOLM
5V BOOSTER
NFBST_RF
WLCSP
FAN48614BUC50X
NFC主供电
PP_VDD_MAIN_NFC
C7521_RF
C7511_RF
L7502_RF
NFC_BOOST_SW
NFC_BOOST_EN
NFC升压使能
VDD_NFC_5V
C7517_RF
C7522_RF
NFC CONTROLLER
NFC供电5V
VDD_NFC_5V
C7520_RF
暂存供电1.8V
PP1V8_SDRAM
PP_VDD_MAIN_NFC
R7502_RF
VDD_NFC_AVDD
VDD_NFC_DVDD
C7500_RF
C7502_RF
VDD_NFC_TVDD
C7506_RF
C7526_RF
C7527_RF
C7504_RF
C7505_RF
NFC_RF
PN67VEU3-B001D004
UFLGA
NFC到电源唤醒
NFC_TO_PMU_HOST_WAKE
SE2_PRESENT
AP_TO_NFC_FW_DWLD_REQ
CPU到NFC中断请求
NFC_TO_BB_CLK_REQ
NFC到基带电源时钟请求
BB_TO_NFC_CLK
基带电源到NFC时钟
UART_AP_TO_NFC_TXD
CPU到NFC数据发送
UART_NFC_TO_AP_RXD
NFC到CPU数据接收
UART_AP_TO_NFC_RTS_L
CPU到NFC数据发送请求
UART_NFC_TO_AP_CTS_L
NFC到CPU数据发送请求
PMU_TO_NFC_EN
电源到NFC开启
VDD_NFC_ESE
SE2_READY
ICEFALL_LDO_ENABLE
NFC_SWP_MUX
NFC_SWP
NFC_TEST_OUT
NFC测试点
NFC_BOOST_EN
NFC升压使能
NFC_RXP
NFC接收信号P
NFC_RXN
NFC接收信号N
NFC_TXP
NFC发送信号P
NFC_TXN
NFC发送信号N
AP_TO_NFC_DEV_WAKE
CPU到NFC唤醒
NFC_VMID
SE2_PWR_REQ
PP1V8_SDRAM
暂存供电1.8V
C7503_RF
R7599_RF
NFC FRONT END
NFC接收信号P
C7507_RF
NFC_RXP_CAP
R7508_RF
NFC发送信号P
L7500_RF
NFC_BALP
C7509_RF
BALUN_RF
180 PHASE SHIFT INTRODUCED BY BALUN
DONE FOR BEST ROUTING
NFC发送信号N
L7501_RF
NFC_BALN
C7510_RF
NFC_ANT_MATCH
C7514_RF
C7512_RF
C7515_RF
C7516_RF
C7518_RF
NFC天线
NFC_ANT
TP7500_RF
NFC接收信号N
C7508_RF
NFC_RXN_CAP
R7509_RF
NFC_TEST_OUT
TP7505_RF
NFC LOAD SWITCH
NFCSW_RF
FPF1204UCX
WLCSP-COMBO
主供电
PP_VDD_MAIN
暂存供电1.8V
PP1V8_SDRAM
NFC主供电
PP_VDD_MAIN_NFC
R7520_RF
NOSTUFF
PP7503_RF
UART_AP_TO_NFC_TXD
PP7504_RF
UART_NFC_TO_AP_RXD
PP7505_RF
UART_AP_TO_NFC_RTS_L
PP7506_RF
UART_NFC_TO_AP_CTS_L
PP7507_RF
NFC_TO_PMU_HOST_WAKE
PP7508_RF
PMU_TO_NFC_EN
PP7509_RF
AP_TO_NFC_DEV_WAKE

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

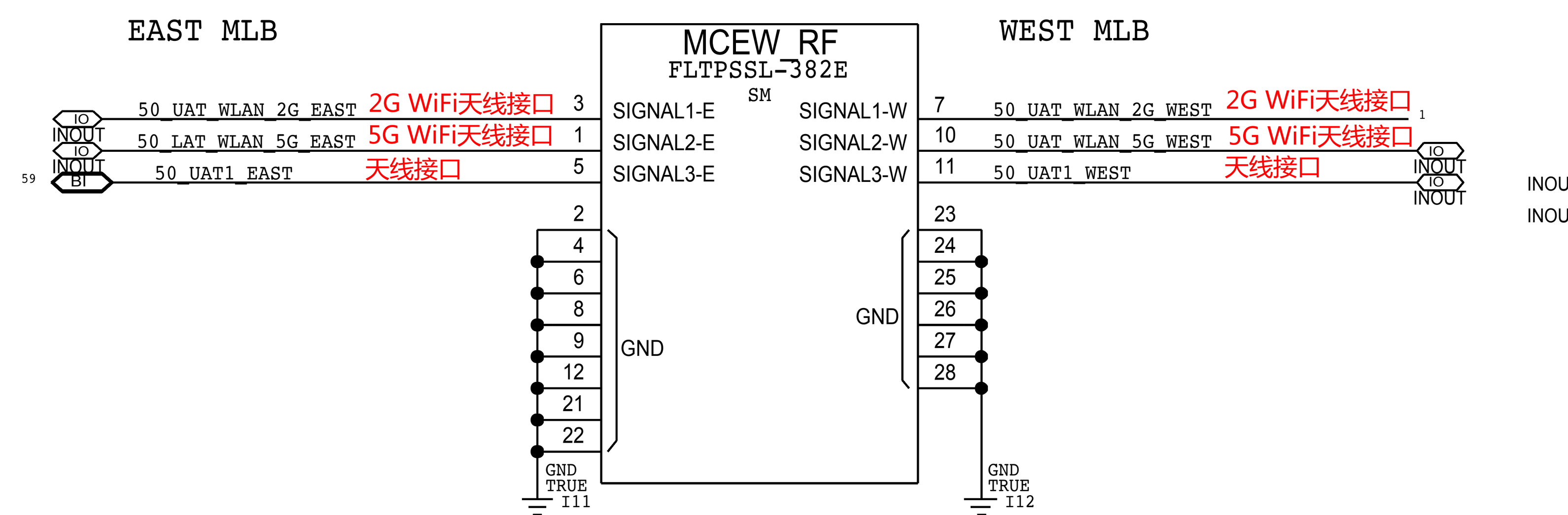
REV	ECN	DESCRIPTION OF REVISION	DATE
8	0006400877	ENGINEERING RELEASED	2016-06-14

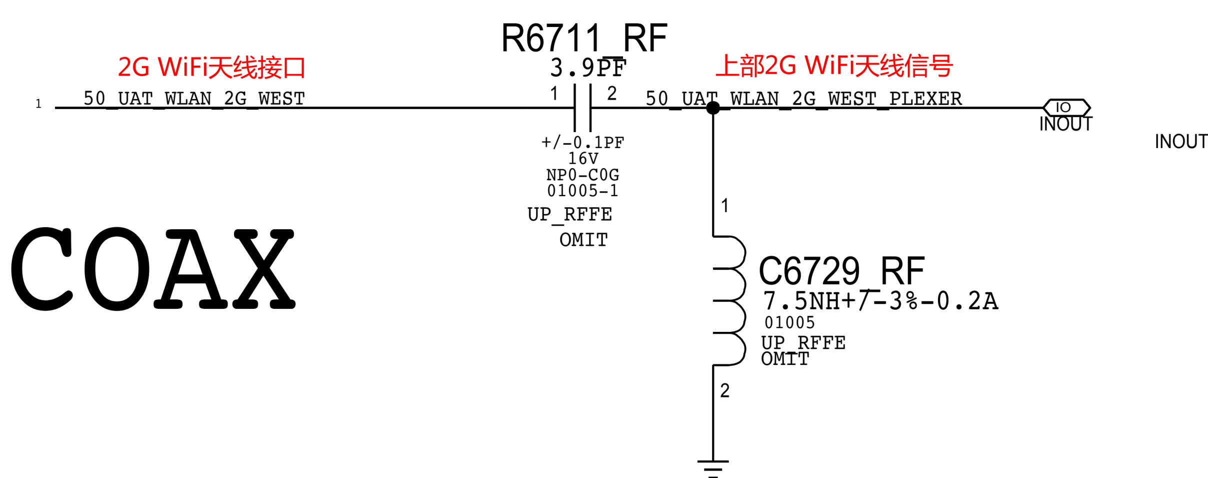
D10 RADIO_MLB_FF
FEB 19, 2016

D10 NORTH-SOUTH METROCIRC
339S00086



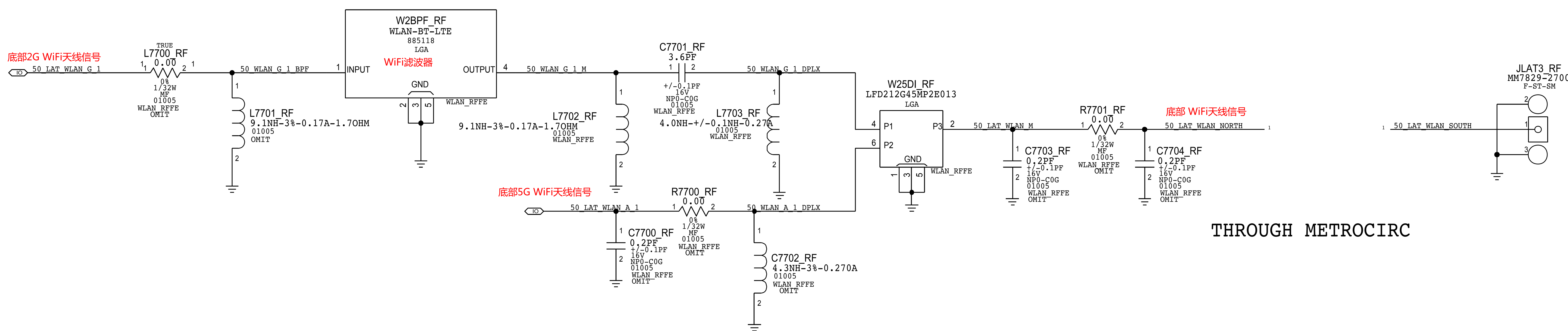
D10 EAST-WEST METROCIRC
339S00110





COAX

WIFI LOWER ANTENNA FEED



THROUGH METROCIRC

ADD REV ID FOR D10/D11 HERE


UAT TUNER FLEX
PLEASE CONTACT ANTENNA (MATT MOW)
FOR ANY COMPONENT CHANGE.
BB_TO_UAT_SCLK
C5904_RF
120PF
BB_TO_UAT_DATA
C5905_RF
68PF
BB_TO_LAT_ANT_SCLK
C5908_RF
33PF
BB_TO_LAT_ANT_DATA
C5909_RF
33PF
暂存供电1.8V
PP1V8_SDRAM
FL6700_RF
150OHM-25%-200MA-0.7DCR
VDD_TUNER_RFFE_VIO_1V8_FILT
C6701_RF
27PF
基带到上天线数据
BB_TO_UAT_DATA
FL6701_RF
0.00
UAT_TUNER_RFFE_DATA_FILT
C6702_RF
27PF
TUNFX_RF
505066-0620
PP3V0_TRISTAR_UAT_TUNER_B2B_FILT
FL6703_RF
150OHM-25%-200MA-0.7DCR
PP3V0_TRISTAR_ANT_PROX
C6705_RF
27PF
UAT_TUNER_RFFE_CLK_FILT
FL6702_RF
0.00
基带到上天线时钟
BB_TO_UAT_SCLK
C6703_RF
27PF
LB/MLB/GNSS/MB/HB
STANDOFF
PP3V0_TRISTAR_ANT_PROX
C6720_RF
0.01UF
50_UAT1_TUNER
C6704_RF
0.00
50_UAT1_TUNER_M
R6706_RF
2.2NH+/-0.1NH-0.6A
50_UAT1_NOTCH
C6716_RF
18PF
50_UAT1_FEED
SUAT1_RF
STDOFF-2.560D1.4ID.99H-SM
UP_RFFE
C6714_RF
5.6NH+/-3%-0.4A
C6700_RF
1.2PF
L6707_RF
9.1NH+/-0.3%-0.3A
BB_BUFFER_GPO2
USPDT2_RF
RF1341
WLCSP
C6721_RF
33PF
ALT UAT1 GND
L8007_RF
560NH-5%-2.8OHM
USPDT_VDD
PP3V0_TRISTAR_ANT_PROX
C8007_RF
0.01UF
USPDT_RF
RF1341
L8008_RF
3.0NH+/-0.1NH-0.6A
DC_BLOCK
BB_BUFFER_GPO1
C8005_RF
33PF
C8008_RF
18PF
ALT_GND
TP8000_RF
SM-TP1P25-TOP
OMIT
OMIT_TABLE
L8009_RF
9.1NH-0.4A
PP8000_RF
CHASSIS_GND
UAT GROUND RING
CHASSIS_GND
C6734_RF
100PF
C6735_RF
18PF
C6730_RF
220PF
C6731_RF
56PF
C6732_RF
4.0PF
C6733_RF
220PF
5G WIFI
STANDOFF
50_UAT2_M
C7731_RF
1.2PF
50_UAT2_FEED
SUAT2_RF
STDOFF-2.560D1.4ID.99H-SM
NOSTUFF
L6710_RF
4.3NH+/-3%-0.5A
C7730_RF
1.8NH+/-0.1NH-0.8A
NOSTUFF
L7709_RF
3.0NH+/-0.1NH-0.6A
SGND_RF
STDOFF-2.560D1.4ID.99H-SM

8 7 6 5 4 3 2 1

D

BOM LIST

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1860	1	LONGER PATH INDUCTOR	L8009_RF	CRITICAL	LMBRF
152S0570	1	3.0 NH,03015	L8009_RF	CRITICAL	NOLMBRF

C

B

A

8 7 6 5 4 3 2 1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
8	0006400877	ENGINEERING RELEASED	2016-06-14

MAV16 RADIO_MLB

LAST_MODIFICATION=Wed Jun 8 12:54:09 2016

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64	<CSA_PAGE3>	PMU: CONTROL AND CLOCKS	<SYNC_MASTER3>	<SYNC_DATE3>
65	<CSA_PAGE4>	PMU: SWITCHERS AND LDOS	<SYNC_MASTER4>	<SYNC_DATE4>
66	<CSA_PAGE5>	BASEBAND: POWER2	<SYNC_MASTER5>	<SYNC_DATE5>
67	<CSA_PAGE6>	BASEBAND: CONTROL	<SYNC_MASTER6>	<SYNC_DATE6>
68	<CSA_PAGE7>	BASEBAND GPIOS	<SYNC_MASTER7>	<SYNC_DATE7>
69	<CSA_PAGE8>	TRANSCEIVER0/1: POWER	<SYNC_MASTER8>	<SYNC_DATE8>
70	<CSA_PAGE9>	TRANSCEIVER0/1: TX PORTS	<SYNC_MASTER9>	<SYNC_DATE9>
71	<CSA_PAGE10>	TRANSCEIVER0/1: PRX PORTS	<SYNC_MASTER10>	<SYNC_DATE10>
72	<CSA_PAGE11>	RECEIVE MATCHING	<SYNC_MASTER11>	<SYNC_DATE11>
73	<CSA_PAGE12>	LOWER ANTENNA & COUPLERS	<SYNC_MASTER12>	<SYNC_DATE12>
74	<CSA_PAGE13>	DIVERSITY RECEIVE ASM'S	<SYNC_MASTER13>	<SYNC_DATE13>
75	<CSA_PAGE14>	DIVERSITY RECEIVE LNA'S	<SYNC_MASTER14>	<SYNC_DATE14>
76	<CSA_PAGE15>	UPPER ANTENNA FEEDS	<SYNC_MASTER15>	<SYNC_DATE15>
77	<CSA_PAGE16>	PMU: ET MODULATOR	<SYNC_MASTER16>	<SYNC_DATE16>
78	<CSA_PAGE17>	TEST POINTS & BOOT CONFIG	<SYNC_MASTER17>	<SYNC_DATE17>
79	<CSA_PAGE18>	TDD TRANSMIT	<SYNC_MASTER18>	<SYNC_DATE18>
80	<CSA_PAGE19>	FDD TRANSMIT	<SYNC_MASTER19>	<SYNC_DATE19>
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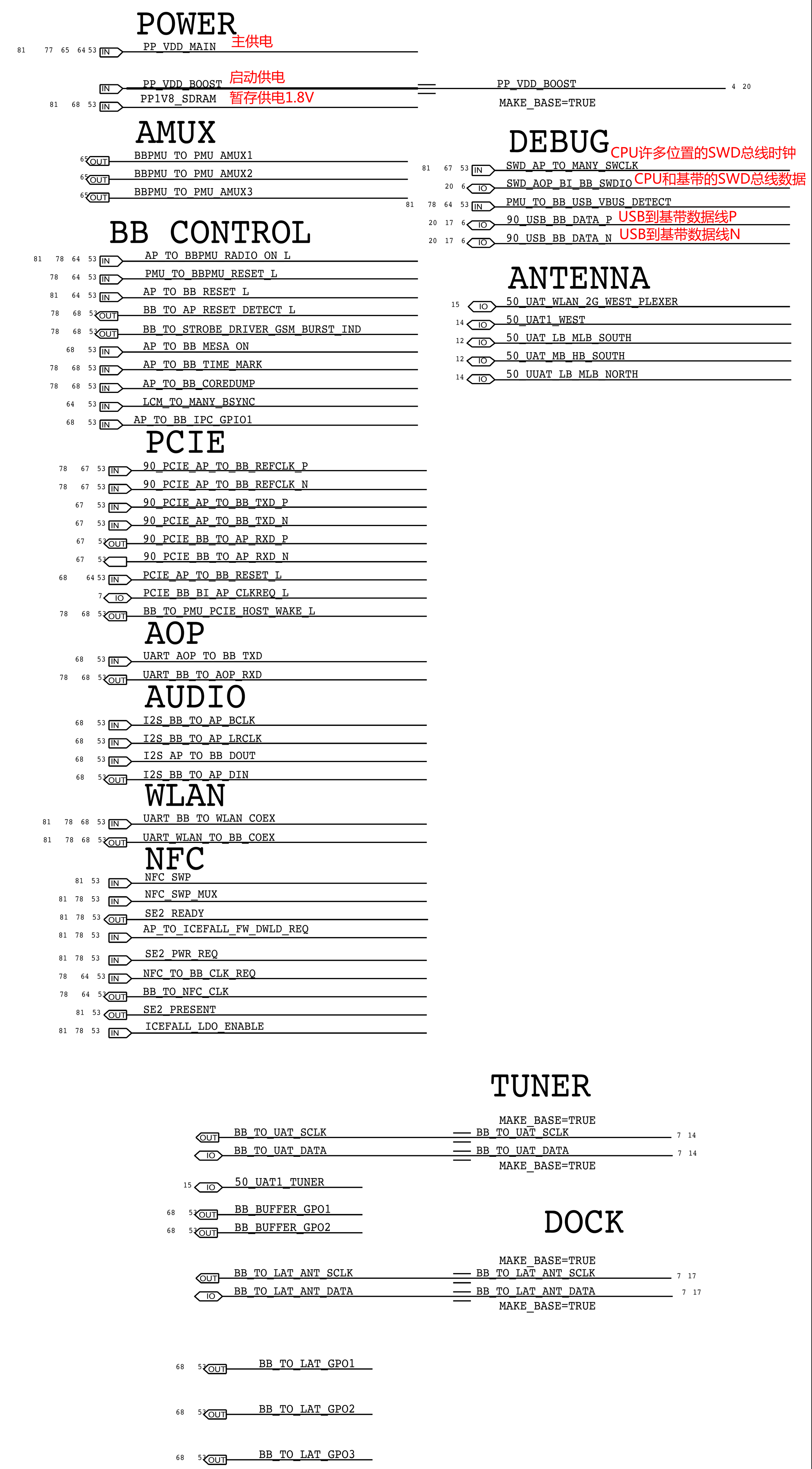
AP CONNECTIONS



SCH: 951-00964
BOM: 939-00826

ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0565	197S0593	ALTERNATE	Y5501_RF	19P2 MHZ XTAL
197S0598	197S0593	ALTERNATE	Y5501_RF	19P2 MHZ XTAL
138S00101	138S00095	ALTERNATE	ALL	TRANSITION CAP
335S00013	335S0894	ALTERNATE	EPROM_RF	IC EEPROM
353S00253	353S00321	ALTERNATE	SWPMX_RF	IC SWITCH SPDT
155S00235	155S00234	ALTERNATE	UPPDI_RF	NOLMBRF

ALL:C5616_RF-C5618_RF,C5632_RF-C5634_RF

BOM OPTIONS:

LMBRF:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S00723	1	MLB PAD	MLBPA_RF	CRITICAL	LMBRF
117S0002	1	MLB PAD LAT OUTPUT MATCH	R7105_RF	CRITICAL	LMBRF
152S2042	1	MLB PAD UAT OUTPUT MATCH	R7106_RF	CRITICAL	LMBRF
353S00627	1	MLB LNA	MLBLN_RF	CRITICAL	LMBRF
117S0002	1	MLB LNA OUTPUT MATCH	R6710_RF	CRITICAL	LMBRF
155S00139	1	PENTAPLEXER	PPLXR_RF	CRITICAL	LMBRF
118S0643	1	RES,MF,4.99K OHM,01005	R7506_RF	CRITICAL	LMBRF
155S00193	1	FLTR,DIPLEXER,LB-MB-HB,DPX,SHIELD,0805	UATDI_RF	CRITICAL	LMBRF
155S00158	1	FLTR,DIPLEXER,LB-MB/HB,MIRROR,0805	UPPDI_RF	CRITICAL	LMBRF
337S00284	1	IC,SECURE ELEMENT,BCM20211,WLBGA25	SE2_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7201_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7528_RF	CRITICAL	LMBRF
132S0316	1	CAP,0.1UF,01005	C7501_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7523_RF	CRITICAL	LMBRF
138S0739	1	CAP,1UF,0201	C7524_RF	CRITICAL	LMBRF
138S00032	1	CAP,2.2UF,0201	C7529_RF	CRITICAL	LMBRF
138S00032	1	CAP,2.2UF,0201	C7530_RF	CRITICAL	LMBRF
131S0217	1	CAP,100PF,01005	C7531_RF	CRITICAL	LMBRF
118S0627	1	RES,10KOHM,01005	R7512_RF	CRITICAL	LMBRF
353S00026	1	LDO,BGA,2X2	SE2LDO_RF	CRITICAL	LMBRF
131S0630	1	CAP,CER,NPO/COG,27PF,2%,16V,01005	C6345_RF	CRITICAL	LMBRF
131S0341	1	CAP,CER,COG,3.0PF,+/-0.05,25V,0201,HI-Q	C6348_RF	CRITICAL	LMBRF
152S2006	1	IND,FILM,6.2NH,3%,400MA,UH-Q,0201	L6322_RF	CRITICAL	LMBRF
131S0630	1	CAP,CER,NPO/COG,27PF,2%,16V,01005	C6315_RF	CRITICAL	LMBRF
152S2006	1	IND,FILM,6.2NH,3%,400MA,UH-Q,0201	L6305_RF	CRITICAL	LMBRF
131S0341	1	CAP,CER,COG,3.0PF,+/-0.05PF,25V,0201,HI-Q	C6306_RF	CRITICAL	LMBRF
131S0630	1	CAP,CER,NPO/COG,27PF,2%,16V,01005	C6106_RF	CRITICAL	LMBRF
152S2051	1	IND,1.3NH,1.1A,0201	R6404_RF	CRITICAL	LMBRF
152S2000	1	IND,2.0NH,600MA,0201	R7104_RF	CRITICAL	LMBRF
131S00001	1	CAP,CER,0.1PF,25V,0201	C7119_RF	CRITICAL	LMBRF
118S0724	1	RES,MF,0OHM,1/20W,0201	R6703_RF	CRITICAL	LMBRF
118S0724	1	RES,MF,0OHM,1/20W,0201	R6708_RF	CRITICAL	LMBRF
131S0363	1	CAP,CER,C0G,HQ,0.6PF,+/-0.05PF,25V,0201	C7123_RF	CRITICAL	LMBRF
152S2002	1	IND,2.7NH,+/-0.1NH,600MA,0201,UH-Q	R6605_RF	CRITICAL	LMBRF
131S0337	1	CAP,1.5PF,+/-0.05PF,25V,0201,HQ	C6610_RF	CRITICAL	LMBRF
118S0724	1	RES,MF,0OHM,1/20W,0201	R6606_RF	CRITICAL	LMBRF
131S0275	1	CAP,CER,C0G,0.3PF,+/-0.1PF,25V,0201,HQ	C6416_RF	CRITICAL	LMBRF
118S0608	1	RES,MF,1K OHM,1%,1/32W,01005	R5911_RF	CRITICAL	LMBRF
152S1356	1	IND,10NH,3%,250MA,HI-Q,0201	C6613_RF	CRITICAL	LMBRF

NOLMBRF:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0444	1	DCS/PCS RX FILTER MATCH	R6301_RF	CRITICAL	NOLMBRF
152S2020	1	DCS/PCS RX FILTER MATCH	L6320_RF	CRITICAL	NOLMBRF
155S00149	1	DCS/PCS RX FILTER	GSMRX_RF	CRITICAL	NOLMBRF
152S2005	1	DCS/PCS RX MATCH (DCS)	L6318_RF	CRITICAL	NOLMBRF
131S0319	1	DCS/PCS RX MATCH (DCS)	C6332_RF	CRITICAL	NOLMBRF
131S0630	1	DCS/PCS RX MATCH (DCS)	C6340_RF	CRITICAL	NOLMBRF
152S2005	1	DCS/PCS RX MATCH (DCS)	L6319_RF	CRITICAL	NOLMBRF
131S0385	1	DCS/PCS RX MATCH (DCS)	C6333_RF	CRITICAL	NOLMBRF
131S0630	1	DCS/PCS RX MATCH (DCS)	C6341_RF	CRITICAL	NOLMBRF
155S00163	1	QUADPLEXER	PPLXR_RF	CRITICAL	NOLMBRF
155S00199	1	FLTR,DPX,PASS THRU,LB-MB-HB,SHLD,0805	UATDI_RF	CRITICAL	NOLMBRF
155S00234	1	FLTR,DPX,PASS THRU,LB-MB-HB,UNSHLD,0805	UPPDI_RF	CRITICAL	NOLMBRF
118S0724	1	RES,MF,0OHM,1/20W,0201	R6404_RF	CRITICAL	NOLMBRF
152S2044	1	IND,2.2NH,+/-0.1NH,600MA,0201	R7104_RF	CRITICAL	NOLMBRF
152S00157	1	1.2NH,+/-0.05NH,1.1A,UH-Q,0201	R6703_RF	CRITICAL	NOLMBRF
131S0549	1	18PF,0201,25V	R6708_RF	CRITICAL	NOLMBRF
131S0445	1	CAP,CER,12PF,5%,25V,0201,HI-Q	R6606_RF	CRITICAL	NOLMBRF

D10 SPECIFIC:

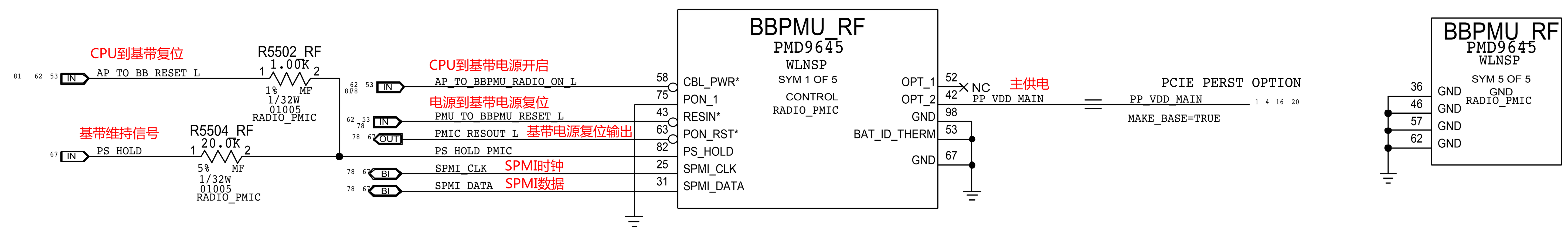
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0278	1	CAP,CER,COG,1PF,+/-0.1PF,25V,0201,HI-Q	C7120_RF	CRITICAL	D10

D11 SPECIFIC:

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0425	1	CAP,CER,COG,0.5PF,+/-0.05PF,25V,0201,HI-Q	C7120_RF	CRITICAL	D11

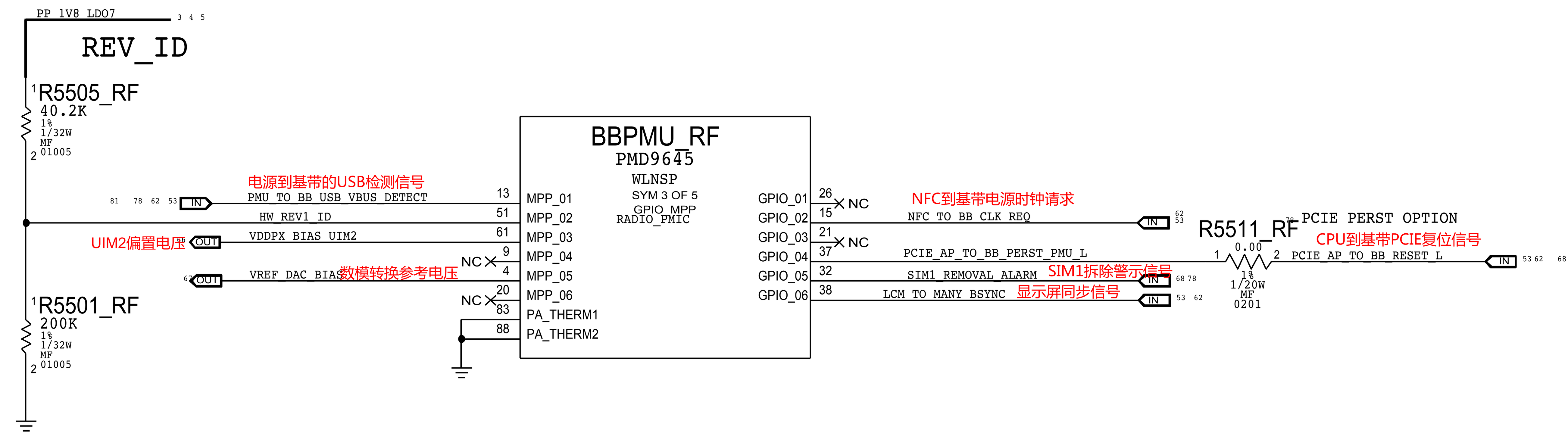
PMU: CONTROL AND CLOCKS

RESET AND CONTROL: PMU

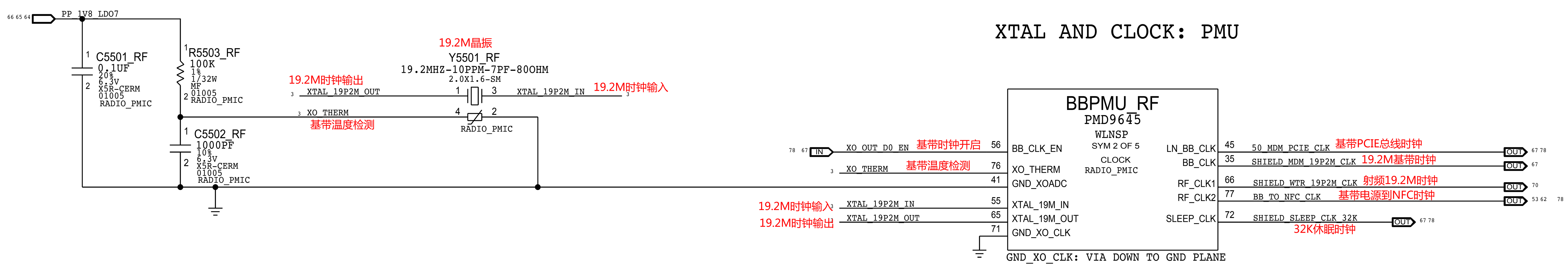


HW_REV_ID	R5505	R5501	REVISION
0.10V	887K	51.1K	DEV1
0.20V	422K	51.1K	DEV2
0.30V	255K	51.1K	DEV 2.1
0.40V	180K	51.1K	DEV 3
0.50V	124K	51.1K	T181
0.60V	102K	51.1K	PP/P1
0.70V	82.5K	51.1K	DEV4
0.80V	63.4K	51.1K	P2
0.90V	51.1K	51.1K	DEV5
1.00V	51.1K	63.4K	EVT
1.10V	51.1K	82.5K	EVT DOE
1.20V	50.0K	100K	EVT ALT/CARBON
1.30V	39.0K	100K	DEV6
1.40V	14.7K	51.1K	CARRIER
1.50V	40.2K	200K	DVT
1.60V	6.34K	51.1K	PVT

MPPS AND GPIOS: PMU



XTAL AND CLOCK: PMU



PMU: SWITCHERS AND LDOS

BASEBAND: POWER

BASEBAND: CONTROL AND INTERFACES

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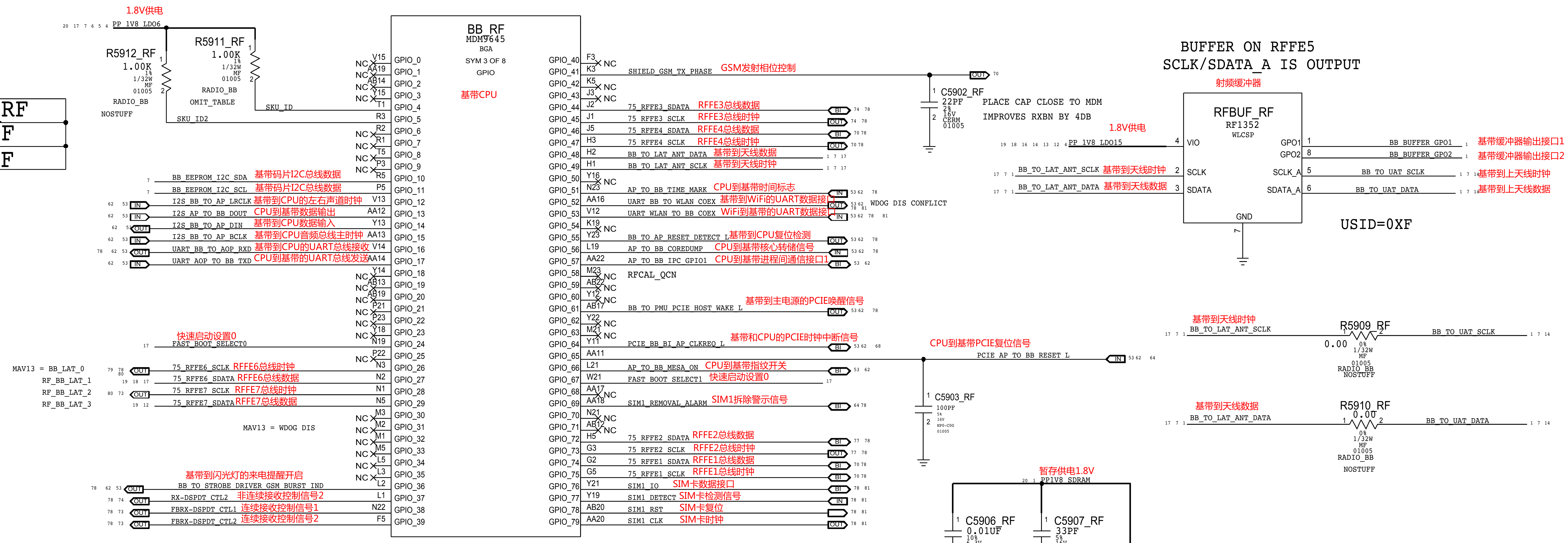
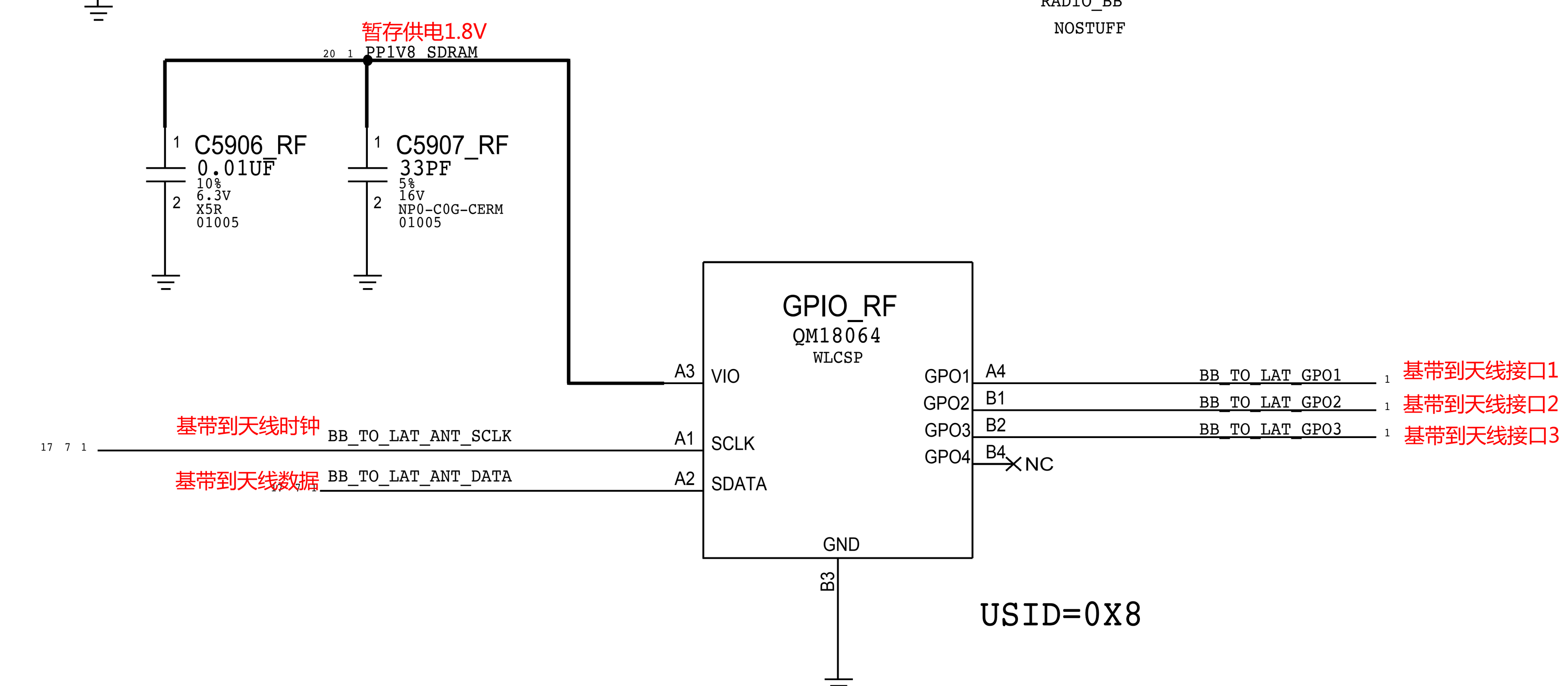
BASEBAND: GPIOS

SKU	R5911_RF	R5912_RF
ROW	NOSTUFF	NOSTUFF
JP	1.0 KOHM	NOSTUFF





RFFE USAGE TABLE

RFFE1 WTR3925

RFFE2 QFE3100

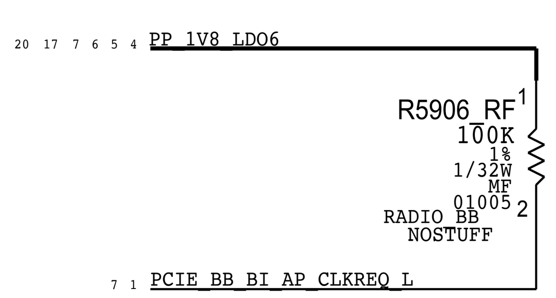
RFFE3 DIV MODULES

RFFE4 WTR4905

RFFE5 TUNERS + ELNAS

RFFE6 2G PA,MLB PA,MB/HB TDD PA,MB/HB FDD PA

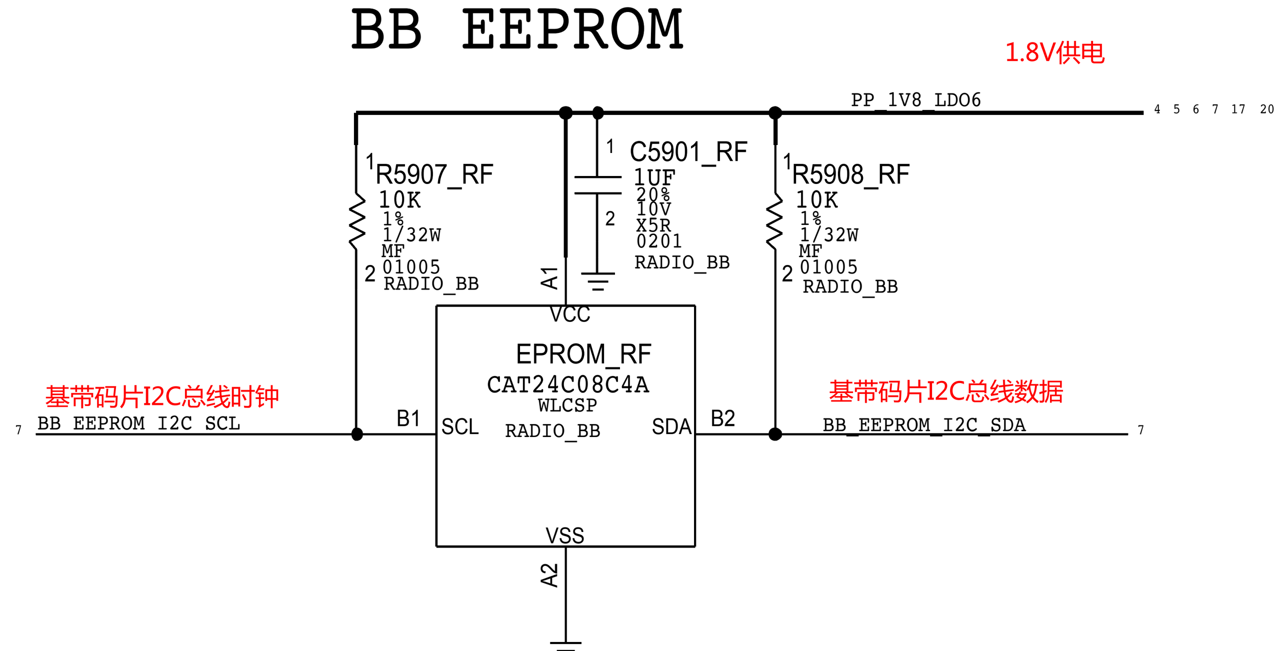
RFFE7 LB PA, COUPLERS



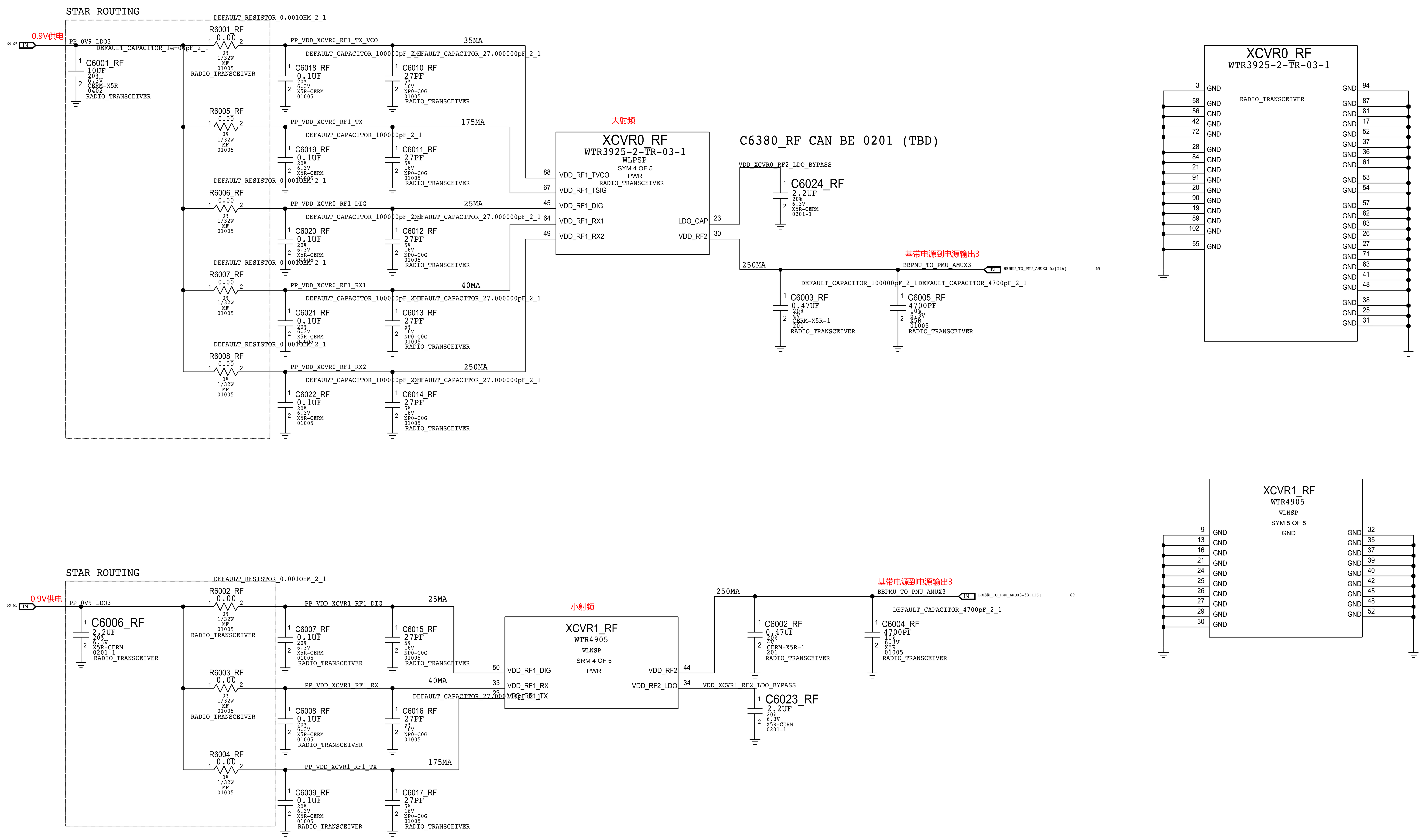
PCIE PULL-UPS TO BB RAIL

基带码片

BB EEPROM



TRANSCEIVER: POWER



TRANSCEIVER: TX PORTS

TRANSCEIVER: PRX, DRX, & GPS PORTS

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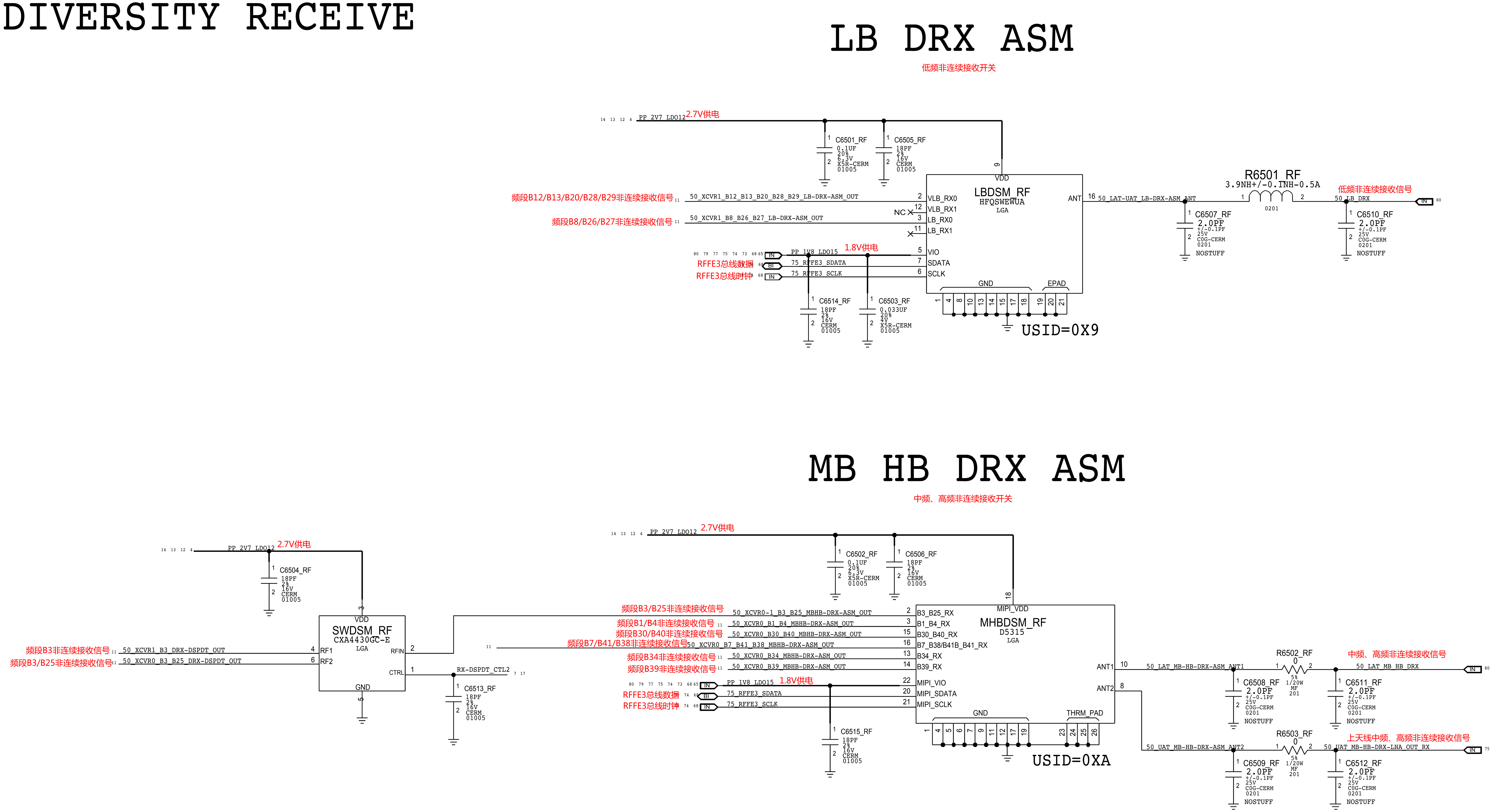
PRIMARY & DIVERSITY RECEIVE MATCHING

DC BLOCKING CAP VALUES CANNOT BE MORE THAN 33PF

LOWER ANTENNA AND COUPLER

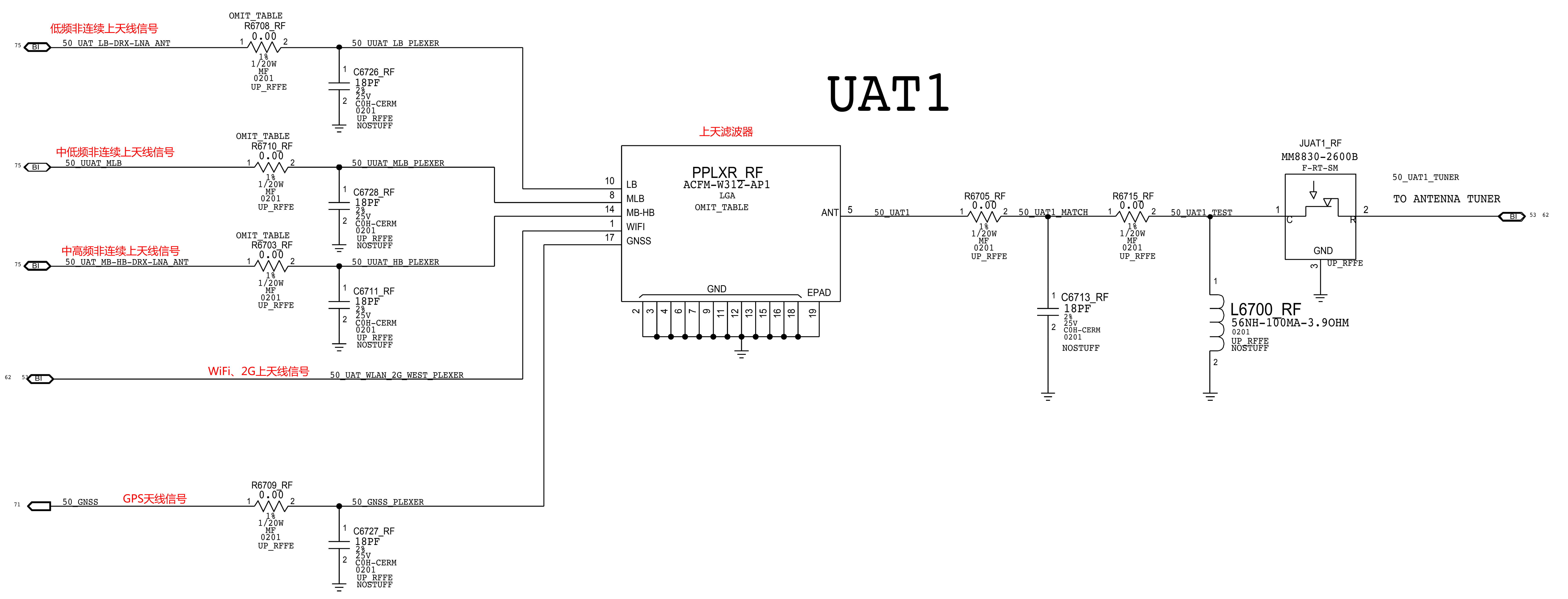
UPPER ANTENNA COUPLER

DIVERSITY RECEIVE

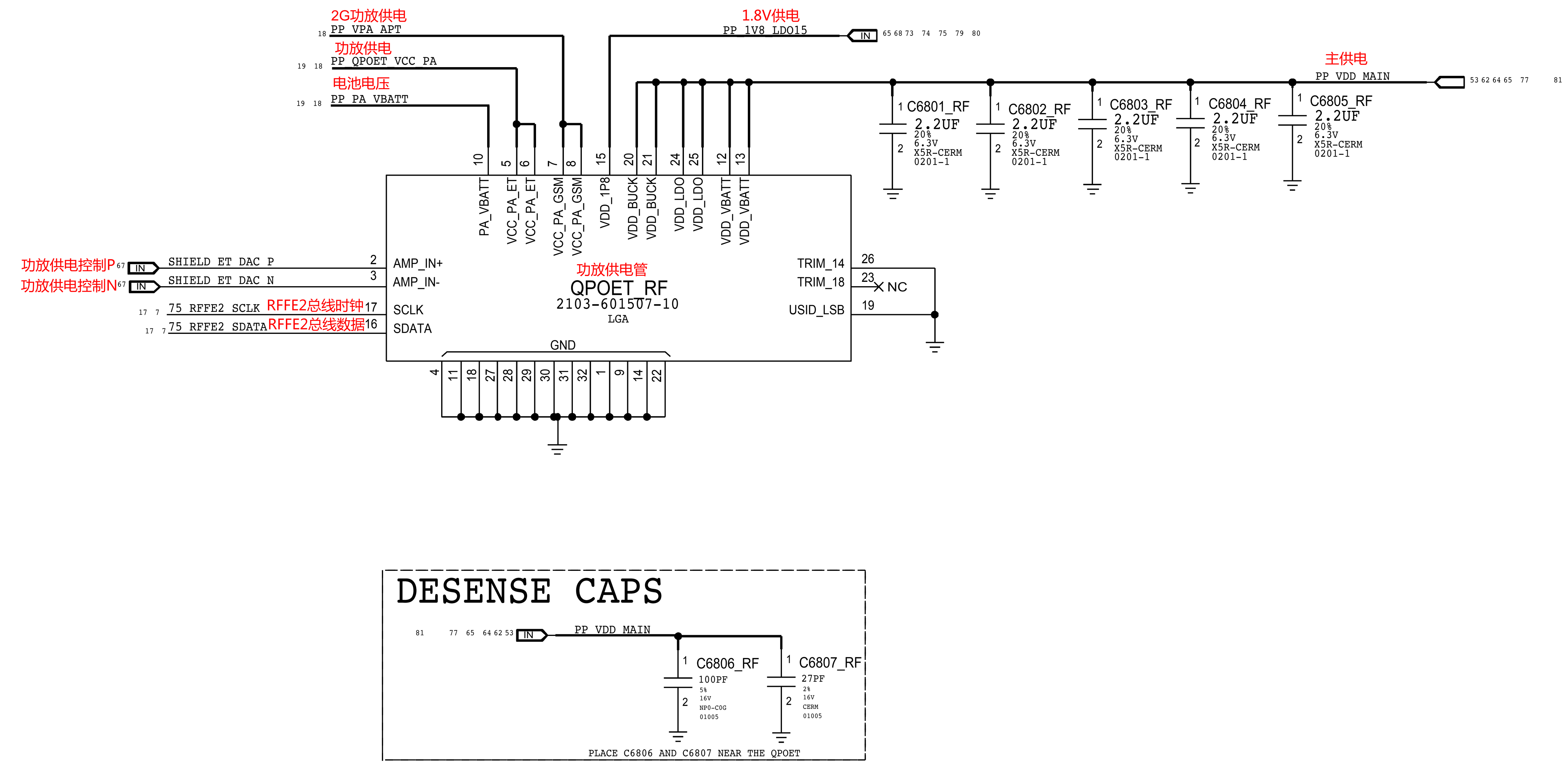


DIVERSITY RECEIVE LNAS

UPPER ANTENNA FEEDS



PMU: ET MODULATOR

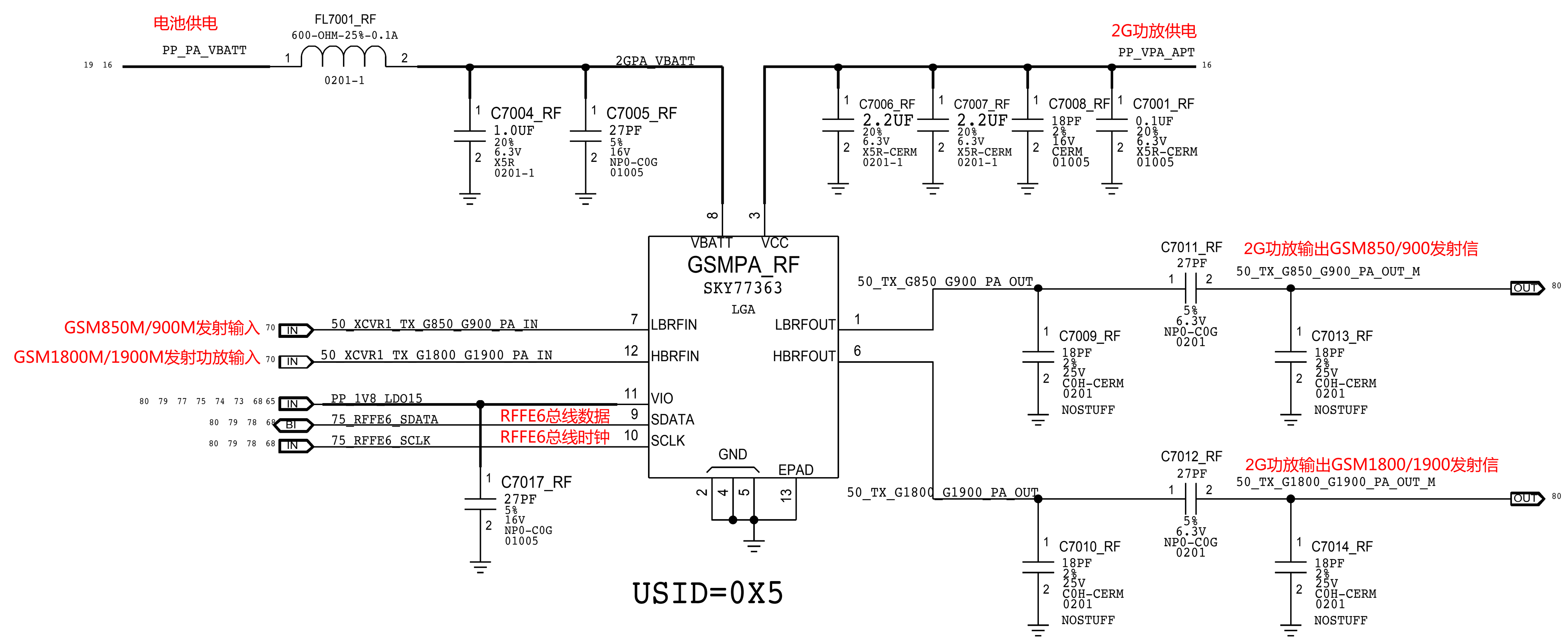


MLB TEST POINTS

TDD TRANSMIT

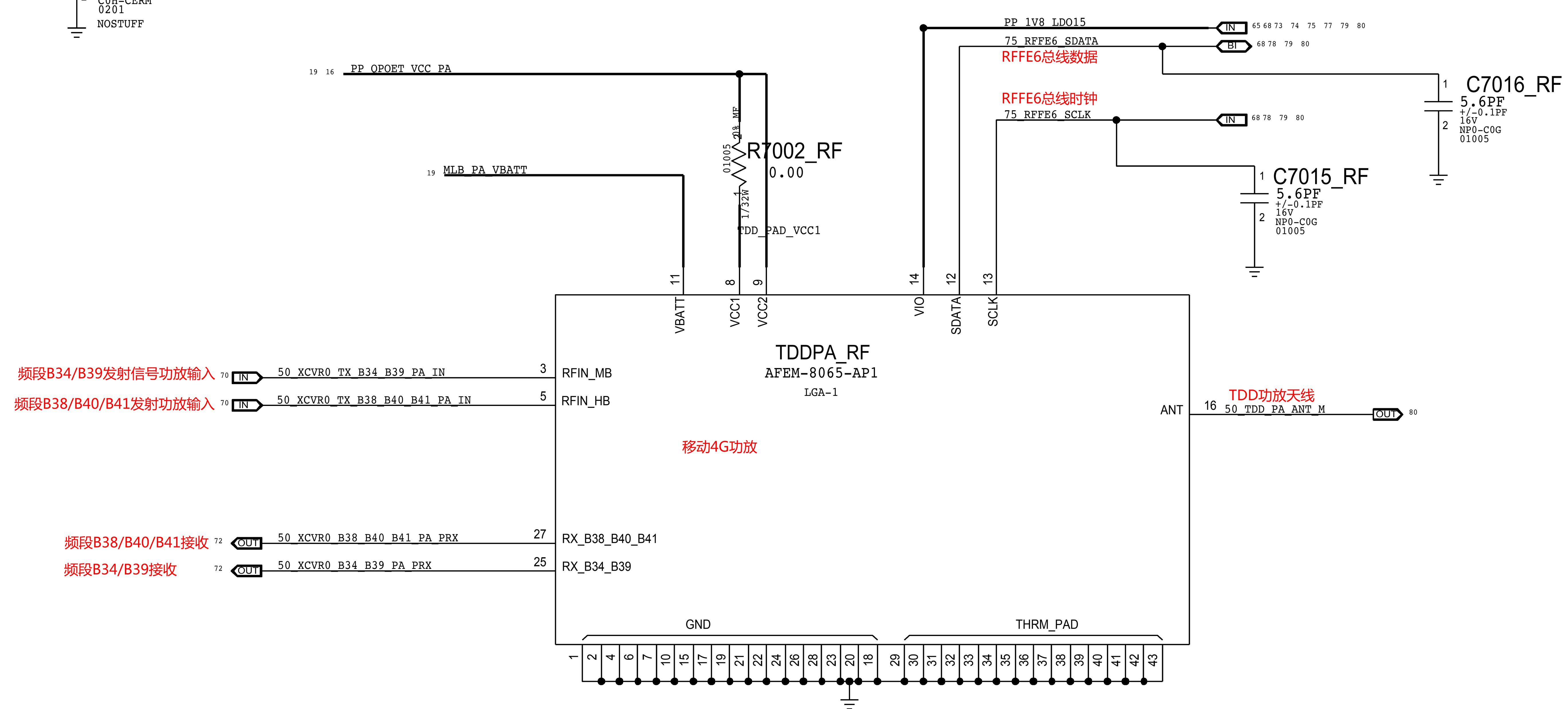
2G PA

2G功放



USID=0X5

MB HB TDD PA



USID=0XF

FDD TRANSMIT


ICEFALL
SWP MUX
ICEFALL LDO
SIM CARD CONNECTOR
DEBUG CONNECTOR
PP1V8_ICEFALL_LDO
C7201_RF
NOSTUFF
C7528_RF
OMIT_TABLE
C7501_RF
VDD_SE2_1V8
VOLTAGE=1.80V
VDD_SE2_1V2
VOLTAGE=1.20V
C7523_RF
C7524_RF
PP1V8_SDRAM
R7506_RF
SE2_PRESENT
SE2_READY
R7512_RF
10K
SE2_RF
BCM20211CP
WLBGA
SPI_CLK
SPI_CS
SPI_INT
SPI_MISO
SPI_MOSI
GPIO_0
GPIO_1
DB_RX
DB_TX
DWP
SWP
REG_PU
TCAL_CLK
AP_TO_ICEFALL_FW_DWLD_REQ
SE2_SWP
SE2_PWR_REQ
暂存供电1.8V
R7511_RF
NFC_SWP_MUX
C7525_RF
SWPMX_RF
NLAS3257CMX2TCG
DFN
SIM1_SWP
NFC通信数据
NFC_SWP
R7510_RF
NFC_SWP_R
主供电
PP_VDD_MAIN
SE2LDO_RF
LP5907UVX-1.825-S
DSBGA
ICEFALL_LDO_ENABLE
C7530_RF
C7529_RF
C7531_RF
100PF
调试接口
SIM卡数据接口
SIM1_IO
DZ6905_RF
DZ6902_RF
ESD202-B1-CSP01005
SIM卡复位
SIM1_RST
DZ6903_RF
SIM卡时钟
SIM1_CLK
DZ6904_RF
J_DEBUG
20-5857-036-001-829
F-ST-SM
PMU_TO_BB_USB_VBUS_DETECT
90_USB_BB_DATA_P
90_USB_BB_DATA_N
AP_TO_BBPMU_RADIO_ON_L
VDD_SIM1
SIM1_DETECT
AP_TO_BB_RESET_L
SWD_AP_TO_MANY_SWCLK
SWD_AOP_BI_BB_SWDIO
CPU许多位置的SWD总线时钟
BB_JTAG_RST_L
UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX
启动供电
PP_VDD_BOOST
SIM卡检测信号
J_SIM_RF
RCPT-WIDE-HSG-THICK-PIVOT
SIM_DETECT_GND
SIM_DETECT
SIM供电
VDD_SIM1
C6900_RF
DZ6901_RF
12V-33PF
DZ6900_RF
5.5V-6.2PF
C6901_RF
100PF
R6900_RF
15.00K
1.8V供电
PP_1V8_LDO6
R6904_RF
100K